

The most probable first-passage time of the lazy walk on a segment: a rigorous one-dimensional theory Companion note R1

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Note. This document is the companion note R1 of the article “Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices” (`paper.pdf`, Section 7; Supplementary Material, Section S8), in the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>. In the repository, its scripts are in `code/msc_rigorous_1d/` and its data in `data/msc_rigorous_1d/`; paths written `scripts/...` and `data/...` below refer to these folders; the paths in the `% src` comments of the L^AT_EX source are relative to the root of the repository, and the run logs they cite are in `data/msc_rigorous_*/logs/`. Other run logs, the code of the internal checks and working notes mentioned below are not part of the repository; `notes/VERIFICATION.md` summarises how this note was checked.

Abstract

For the lazy nearest-neighbour walk on $\{1, \dots, N\}$ with move probability $q \in (0, 1]$, a reflecting wall behind site 1 and an absorbing target at site N , we replace the formal asymptotics of the first-passage mode by theorems. With $L = N - \frac{1}{2}$ and T the first-passage time from the reflecting end we prove: the exact spectral law for every start; unimodality (continuous time: every start, every q ; discrete time: every start when the spectrum is nonnegative, i.e. $q \leq q_N$; from the reflecting end, log-concavity for $q \leq \frac{2}{3}$ and unimodality for *all* N whenever $q \leq \frac{4}{5}$, which is sharp, by an “eventual total positivity” argument: Q^k is TP_2 for all $k \geq 10$ when $q = \frac{4}{5}$; also $q \leq \frac{9}{10}$ for $N \geq 9$), and exactly how it fails for q near 1 (from the reflecting end) and for q above an explicit threshold depending only on $N - x_0$ (from any other start); that the limiting density $G(u) = \frac{\pi}{2} \sum_{k \text{ odd}} (-1)^{(k-1)/2} k e^{-k^2 \pi^2 u/8}$ is log-concave with a unique maximiser

$$c = 0.33328426549494789874852691654344242109703655907113188526100593708624 \dots$$

(enclosed to 70 digits); and explicit finite- N laws for the mode t^* :

- continuous time, all $N \geq 3$: $|q t_c^*/L^2 - c + \kappa/L^2| < 1.78/L^4$, $\kappa = 0.0862737793 \dots$, and $L^4(q t_c^*/L^2 - c + \kappa/L^2) \rightarrow \kappa_2 = -1.5308900198 \dots$, given in closed form by derivatives of G at c ;
- discrete time, $q \leq \frac{4}{5}$, all $N \geq 3$ (and $q \leq \frac{9}{10}$, all $N \geq 10$): $\tau - E < t^* < \tau + 1 + E$ with $\tau = (cL^2 - \kappa)/q + 1 - \rho$, $\rho = 1.9911786618 \dots$, $E = 1.78/(qL^2)$; hence $|q t^*/L^2 - c| < (0.0863 + 0.9912q)/L^2 + 1.78/L^4$, and $t^* = \lceil \tau \rceil$ exactly unless τ is within E of an integer;
- discrete time, $\frac{9}{10} < q < 1$: the same (with 1.56) for $N \geq 601$ and $(1-q)L^2 \gtrsim 27 \ln L + 100$ (precisely: condition (33); e.g. all $N \geq 601$ if $q \leq 0.999$);
- $q = 1$ (simple random walk), all $N \geq 3$: $t^* \equiv N - 1 \pmod{2}$ and $\tau_1 - 10.3/L < t^* < \tau_1 + 2 + 10.3/L$ with $\tau_1 = cL^2 - bL - 1.2503 \dots$, $b = 0.3325085829 \dots$; so here the error of $t^*/L^2 \rightarrow c$ is of exact order N^{-1} ;
- general start x_0 , $\xi_N = (x_0 - \frac{1}{2})/L$: the limit density G_ξ has exactly one critical point $c(\xi)$, which is non-degenerate, and $\text{mode-}q/L^2 \rightarrow c(\xi)$ (continuous time, and discrete

time for $q \leq \frac{1}{2}$); for starts in the half next to the wall ($\xi_N \leq \frac{1}{2}$) and all $N \geq 3$: $|qt_c^*/L^2 - c(\xi_N) + \kappa(\xi_N)/L^2| < 3.5/L^4$ in continuous time and $\tau - E < t^* < \tau + 1 + E$, $\tau = (c(\xi_N)L^2 - \kappa(\xi_N))/q + 1 - \rho(\xi_N)$, $E = 3.5/(qL^2)$, in discrete time with $q \leq \frac{1}{2}$; on the other hand, for every start $x_0 \geq 2$ the discrete-time PMF is *not* unimodal as soon as $q > \frac{4}{5}$, $q > \frac{4+\sqrt{2}}{7}$ or $q > 1 - \frac{1}{d}$, according as $d = N - x_0$ is 1, 2 or ≥ 3 .

In particular $\text{mode}/\mathbb{E}T \rightarrow c$ for every fixed $q \in (0, 1]$. Statements are labelled *Theorem* only when a complete proof is given; computer-assisted steps (CA) use ball arithmetic (Arb, via `python-flint`) or exact integer/rational arithmetic and are listed with their scripts in Section 9, together with what remains open.

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1 Setting and notation

Fix an integer $N \geq 2$ and $q \in (0, 1]$. The walk lives on $\{1, \dots, N\}$. At each step it attempts a move with probability q , to the left or to the right with probability $q/2$ each, and rests with probability $1 - q$; an attempted move from site 1 to the left is cancelled (the walker stays). Site N is absorbing. Write $n = N - 1$ for the number of transient sites and

$$L := N - \frac{1}{2}.$$

The transition matrix restricted to the transient sites $\{1, \dots, n\}$ is the symmetric tridiagonal matrix

$$Q_{11} = 1 - \frac{q}{2}, \quad Q_{jj} = 1 - q \quad (2 \leq j \leq n), \quad Q_{j,j\pm 1} = \frac{q}{2}. \quad (1)$$

For a start site $x_0 \in \{1, \dots, n\}$ let T be the absorption time, $d := N - x_0$ the distance to the target,

$$f(t) = f_{x_0}(t) := \mathbb{P}_{x_0}(T = t) = \frac{q}{2} (Q^{t-1})_{x_0, n}, \quad t \geq 1,$$

because $T = t$ means that the walker is at site n at time $t - 1$ without earlier absorption and then steps to the right. The *continuous-time walk* has generator $Q - I$ on the transient sites (each neighbour is attempted at rate $q/2$); its absorption time has density $g(t) = g_{x_0}(t) = \frac{q}{2} (e^{t(Q-I)})_{x_0, n}$. A *mode* is a maximiser of f over $t \in \{1, 2, \dots\}$ (resp. of g over $t \geq 0$).

Throughout,

$$\theta_m := \frac{(2m-1)\pi}{2N-1} = \frac{(2m-1)\pi}{2L}, \quad \nu_m := q(1 - \cos \theta_m), \quad \lambda_m := 1 - \nu_m, \quad m = 1, \dots, n.$$

For odd $k \geq 1$ we write $y_k := k\pi/4$, $\sigma_k := (-1)^{(k-1)/2}$, so that with $k = 2m - 1$ we have $\theta_m/2 = y_k/L =: x_k$ and $\sigma_k = (-1)^{m+1}$.

A sequence $(a_t)_{t \geq 1}$ of nonnegative numbers is called *unimodal* if there is t^* with $a_1 \leq \dots \leq a_{t^*} \geq a_{t^*+1} \geq \dots$, and *log-concave* if $a_t^2 \geq a_{t-1}a_{t+1}$ for all $t \geq 2$ and its support is an interval of integers.

2 Exact formulas

Lemma 2.1 (Spectrum of Q). *For $m = 1, \dots, n$ the vector $v_m(j) = \cos(\theta_m(j - \frac{1}{2}))$, $j = 1, \dots, n$, satisfies $Qv_m = \lambda_m v_m$, and $\|v_m\|^2 = \sum_{j=1}^n v_m(j)^2 = (2N-1)/4$. The λ_m are pairwise distinct, $1 > \lambda_1 > \dots > \lambda_n > 1 - 2q$, and the v_m form an orthogonal basis of $\mathbb{R}^n$.*

Proof. Put $v(j) = \cos(\theta(j - \frac{1}{2}))$ for all integers j , $\theta = \theta_m$. From $\cos(A - \theta) + \cos(A + \theta) = 2 \cos A \cos \theta$, $(1-q)v(j) + \frac{q}{2}(v(j-1) + v(j+1)) = (1-q + q \cos \theta)v(j)$ for every j . For $2 \leq j \leq n-1$ this is $(Qv)(j) = \lambda v(j)$. For $j = 1$: $v(0) = \cos(-\theta/2) = v(1)$, hence $(Qv)(1) = (1 - \frac{q}{2})v(1) + \frac{q}{2}v(2) = (1-q)v(1) + \frac{q}{2}(v(0) + v(2)) = \lambda v(1)$. For $j = n$: $v(n+1) = \cos(\theta_m(N - \frac{1}{2})) = \cos((2m-1)\pi/2) = 0$, hence $(Qv)(n) = (1-q)v(n) + \frac{q}{2}v(n-1) = (1-q)v(n) + \frac{q}{2}(v(n-1) + v(n+1)) = \lambda v(n)$. Since $0 < \theta_1 < \dots < \theta_n < \pi$ and $\cos$ is strictly decreasing on $[0, \pi]$, the eigenvalues are distinct and lie in $(1 - 2q, 1)$; eigenvectors of a symmetric matrix for distinct eigenvalues are orthogonal. Finally $\sum_{j=1}^n \cos^2(\theta(j - \frac{1}{2})) = \frac{n}{2} + \frac{1}{2} \sum_{j=1}^n \cos((2j-1)\theta) = \frac{n}{2} + \frac{\sin(2n\theta)}{4 \sin \theta}$, and $2n\theta_m = (2m-1)\pi - \theta_m$ gives $\sin(2n\theta_m) = \sin \theta_m$; so $\|v_m\|^2 = \frac{n}{2} + \frac{1}{4} = \frac{2N-1}{4}$. $\square$

Theorem 2.2 (Exact first-passage law, every start). *Let $x_0 \in \{1, \dots, N-1\}$ and $d = N - x_0$. Then for all $t \geq 1$ (with $0^0 := 1$) and all $s \geq 0$,*

$$f_{x_0}(t) = \frac{2q}{2N-1} \sum_{m=1}^{N-1} \sin \theta_m \sin(d\theta_m) \lambda_m^{t-1}, \quad (2)$$

$$g_{x_0}(s) = \frac{2q}{2N-1} \sum_{m=1}^{N-1} \sin \theta_m \sin(d\theta_m) e^{-\nu_m s}. \quad (3)$$

For the reflecting end, $x_0 = 1$: $\sin(d\theta_m) = (-1)^{m+1} \cos(\theta_m/2)$. Moreover $f_{x_0}(t) = 0$ for $t < d$, $f_{x_0}(d) = (q/2)^d$, and $g_{x_0}^{(i)}(0) = 0$ for $i < d-1$, $g_{x_0}^{(d-1)}(0) = (q/2)^d$.

Proof. By Lemma 2.1, $(Q^{t-1})_{x_0, n} = \sum_m \lambda_m^{t-1} v_m(x_0) v_m(n) / \|v_m\|^2$. With $\theta_m L = (2m-1)\pi/2$ and $\cos((2m-1)\frac{\pi}{2} - a) = \sin((2m-1)\frac{\pi}{2}) \sin a = (-1)^{m+1} \sin a$,

$$v_m(n) = \cos(\theta_m L - \theta_m) = (-1)^{m+1} \sin \theta_m, \quad v_m(x_0) = \cos(\theta_m L - d\theta_m) = (-1)^{m+1} \sin(d\theta_m).$$

Multiplying by $\frac{q}{2} \cdot \frac{4}{2N-1}$ gives (2); (3) is the same computation for $e^{s(Q-I)}$. For $x_0 = 1$, $d\theta_m = \theta_m L - \theta_m/2$. Since Q is tridiagonal, $(Q^i)_{x_0,n} = 0$ for $i < n - x_0 = d - 1$ and $(Q^{d-1})_{x_0,n} = (q/2)^{d-1}$ (a single path); the same holds for $(Q - I)^i$, which gives the statements on f near $t = d$ and on the derivatives of g at 0. $\square$

Theorem 2.3 (Product formula from the reflecting end). *Let $x_0 = 1$. Then $\mathbb{P}(T < \infty) = 1$ and, for $|z| \leq 1$ and $s \geq 0$,*

$$\mathbb{E} z^T = \prod_{m=1}^{N-1} \frac{(1 - \lambda_m) z}{1 - \lambda_m z}, \quad \mathbb{E} e^{-sT_{\text{cont}}} = \prod_{m=1}^{N-1} \frac{\nu_m}{\nu_m + s}, \quad \prod_{m=1}^{N-1} (1 - \cos \theta_m) = 2^{-(N-1)}. \quad (4)$$

Consequently: in continuous time T is distributed as a sum of $N - 1$ independent exponential variables with rates $\nu_1, \dots, \nu_{N-1}$; in discrete time, if all $\lambda_m \geq 0$, i.e. if

$$q \leq q_N := \frac{1}{2 \cos^2(\pi/(2N-1))} \quad (\text{note } q_N > \frac{1}{2}, \quad q_N \downarrow \frac{1}{2}), \quad (5)$$

T is distributed as a sum of $N - 1$ independent geometric variables on $\{1, 2, \dots\}$ with success probabilities $1 - \lambda_m$. In all cases $f(n+k) = (q/2)^n h_k(\lambda_1, \dots, \lambda_n)$, $k \geq 0$, with h_k the complete homogeneous symmetric polynomial of degree k .

Proof. For $|z| \leq 1$ the matrix $I - zQ$ is invertible (the eigenvalues of Q lie in $(-1, 1)$) and $\mathbb{E} z^T = \sum_{t \geq 1} z^t \frac{q}{2} (Q^{t-1})_{1,n} = \frac{qz}{2} [(I - zQ)^{-1}]_{1,n}$. By Cramer's rule $[(I - zQ)^{-1}]_{1,n} = (-1)^{1+n} M_{n,1} / \det(I - zQ)$, where $M_{n,1}$ is the minor of $I - zQ$ obtained by deleting row n and column 1. That minor is the determinant of a lower triangular matrix (its (j, k) entry is $(I - zQ)_{j,k+1}$, which vanishes for $k > j$) with diagonal entries $(I - zQ)_{j,j+1} = -zq/2$, $j = 1, \dots, n - 1$, so $M_{n,1} = (-zq/2)^{n-1}$ and

$$\mathbb{E} z^T = \frac{(qz/2)^n}{\det(I - zQ)} = \frac{(qz/2)^n}{\prod_m (1 - \lambda_m z)}.$$

The row sums of Q are 1 except the last, which is $1 - q/2$; hence $(I - Q)\mathbf{1} = \frac{q}{2} e_n$, i.e. $\frac{q}{2}(I - Q)^{-1} e_n = \mathbf{1}$, which says $\mathbb{P}_x(T < \infty) = \sum_t f_x(t) = 1$ for every x . Putting $z = 1$ gives $(q/2)^n = \prod_m (1 - \lambda_m)$, which is the third identity in (4) and turns the display into the first. The second follows in the same way from $\mathbb{E} e^{-sT_{\text{cont}}} = \frac{q}{2} [((s+1)I - Q)^{-1}]_{1,n}$. The factors in (4) are the generating function of a geometric variable (if $0 \leq \lambda_m < 1$) and the Laplace transform of an exponential variable, and $\lambda_n = 1 - q(1 + \cos \frac{2\pi}{2N-1}) = 1 - 2q \cos^2 \frac{\pi}{2N-1}$ is the smallest eigenvalue. Expanding $\prod_m (1 - \lambda_m z)^{-1} = \sum_k h_k(\lambda) z^k$ gives the last statement. $\square$

Remark 2.4. The distributional statements are the Karlin–McGregor/Keilson theorem for birth–death chains (S. Karlin, J. McGregor, *Coincidence properties of birth and death processes*, Pacific J. Math. **9** (1959) 1109–1140; J. Keilson, *Log-concavity and log-convexity in passage time densities of diffusion and birth-death processes*, J. Appl. Probab. **8** (1971) 391–398); in the form used here they are J. A. Fill, *The passage time distribution for a birth-and-death chain: strong stationary duality gives a first stochastic proof*, J. Theoret. Probab. **22** (2009) 543–557, Theorem 1.1 (continuous time) and Theorem 1.2 (discrete time; the product form of the generating function holds for all eigenvalues in $[-1, 1)$, the interpretation as a sum of geometric variables requires them to be nonnegative). The hypotheses of those theorems (birth–death chain on $\{0, \dots, d\}$, started at the reflecting state 0, d absorbing, positive birth and death probabilities) hold here with $d = N - 1$. The proof above is self-contained. For $q > q_N$ some eigenvalues are negative and T is *not* a sum of geometric variables; this is the origin of the parity effects of Section 3.

Corollary 2.5 (Moments). *For every start, $\mathbb{E}_{x_0} T = (N(N-1) - x_0(x_0-1))/q$ in discrete and in continuous time. From the reflecting end, $\mathbb{E} T = N(N-1)/q = (L^2 - \frac{1}{4})/q$, $\text{Var } T_{\text{cont}} = \sum_m \nu_m^{-2}$ and $\text{Var } T = \sum_m \nu_m^{-2} - \mathbb{E} T$.*

Proof. $m(x) := (N(N-1) - x(x-1))/q$ satisfies $m(N) = 0$, $\frac{q}{2}(m(1) - m(2)) = 1$ and, for $2 \leq x \leq N-1$, $q m(x) - \frac{q}{2}(m(x-1) + m(x+1)) = \frac{1}{2}[-2x(x-1) + (x-1)(x-2) + (x+1)x] = 1$; that is $(I - Q)m = \mathbf{1}$, the equation of the mean absorption time (the same linear system in continuous time). The variances follow from (4): cumulants add, an exponential of rate ν has variance ν^{-2} and the factor $(1-\lambda)z/(1-\lambda z)$ has second cumulant $\lambda/(1-\lambda)^2 = \nu^{-2} - \nu^{-1}$ (as a formal identity this holds also for negative λ), and $\sum_m \nu_m^{-1} = \mathbb{E}T$. $\square$

3 Unimodality and log-concavity

3.1 Descartes' rule for exponential sums

Lemma 3.1 (Laguerre's extension of Descartes' rule). *Let $\beta_1 < \dots < \beta_r$ be real, $c_1, \dots, c_r$ real and not all zero, and $F(t) = \sum_{i=1}^r c_i e^{\beta_i t}$. Then the number $Z(F)$ of real zeros of F , counted with multiplicity, is at most the number V of sign changes in the sequence $(c_1, \dots, c_r)$ (zero entries discarded).*

Proof. We may assume all $c_i \neq 0$. Induction on V . If $V = 0$ all terms have the same sign and F has no zero. Let $V \geq 1$, choose i with $c_i c_{i+1} < 0$ and $\beta \in (\beta_i, \beta_{i+1})$, and put $h(t) = e^{-\beta t} F(t)$, which has the same zeros as F with the same multiplicities. Then $h'(t) = \sum_j c_j (\beta_j - \beta) e^{(\beta_j - \beta)t}$; its coefficient sequence is obtained from (c_j) by reversing the signs of $c_1, \dots, c_i$, so it has exactly $V-1$ sign changes and no zero entry. By induction $Z(h') \leq V-1$. If h had at least $V+1$ real zeros counted with multiplicity, say distinct zeros $t_1 < \dots < t_p$ with multiplicities $\mu_1, \dots, \mu_p$, $\sum \mu_i \geq V+1$, then h' would have a zero of multiplicity $\mu_i - 1$ at each t_i and, by Rolle's theorem, at least one zero in each of the $p-1$ gaps: $Z(h') \geq \sum (\mu_i - 1) + p - 1 \geq V$, a contradiction. Hence $Z(F) = Z(h) \leq V$. $\square$

(E. Laguerre, J. Math. Pures Appl. (3) **9** (1883) 99–146; see also G. Pólya and G. Szegő, *Problems and Theorems in Analysis II*, Part V, Chapter 1.)

Lemma 3.2. *For integers $1 \leq D$ and reals $0 < \vartheta_1 < \dots < \vartheta_r < \Theta \leq \pi$, the sequence $(\sin(D\vartheta_i))_{i=1}^r$ has at most as many sign changes as $\sin(D\vartheta)$ has zeros in $(0, \Theta)$, i.e. at most $\#\{j \geq 1 : j\pi/D < \Theta\}$. For $\Theta = \pi$ this is $D-1$; for $\Theta = \pi/2$ it is $\lceil D/2 \rceil - 1 = \lfloor (D-1)/2 \rfloor$.*

Proof. Between two sample points at which the values have opposite signs the continuous function $\sin(D\vartheta)$ has a zero; disjoint index gaps give disjoint open intervals. $\square$

3.2 Continuous time, every start

Theorem 3.3 (Strict unimodality in continuous time). *Let $x_0 \in \{1, \dots, N-1\}$, $d = N - x_0$.*

1. *If $d = 1$, g_{x_0} is completely monotone; in particular it is strictly decreasing on $[0, \infty)$.*
2. *If $d \geq 2$, then $g_{x_0} > 0$ on $(0, \infty)$ and g'_{x_0} has exactly one zero t_c in $(0, \infty)$; $g'_{x_0} > 0$ on $(0, t_c)$, $g'_{x_0} < 0$ on (t_c, ∞) and $g''_{x_0}(t_c) < 0$.*

Proof. By (3), $g = \frac{2q}{2N-1} \sum_m c_m e^{-\nu_m t}$ with $c_m = \sin \theta_m \sin(d\theta_m)$, $\sin \theta_m > 0$, and exponents $-\nu_m$ strictly decreasing in m . If $d = 1$ all $c_m = \sin^2 \theta_m > 0$. Let $d \geq 2$. By Lemma 3.2 the coefficient sequence has $V \leq d-1$ sign changes, and the same is true for the coefficient sequence $(-\nu_m c_m)$ of g' . By Theorem 2.2, g (an entire function of t) has a zero of multiplicity exactly $d-1$ at $t=0$ and $g^{(d-1)}(0) > 0$. By Lemma 3.1, g has no other real zero; hence $g > 0$ on $(0, \infty)$. Likewise g' has a zero of multiplicity exactly $d-2$ at 0, so it has at most one other real zero, counted with multiplicity. Since $g(0) = 0 < g(t)$ for $t > 0$ and $g(t) \rightarrow 0$ as $t \rightarrow \infty$, g attains its maximum at some $t_c > 0$, where $g'(t_c) = 0$. Thus t_c is the only zero of g' in $(0, \infty)$, it is simple, so $g''(t_c) \neq 0$, and $g''(t_c) < 0$ because t_c is a maximum. The signs of g' on the two sides follow. $\square$

Remark 3.4. For birth–death processes the unimodality of all passage-time densities is due to J. Keilson, *On the unimodality of passage time densities in birth-death processes*, Statist. Neerlandica **35** (1981) 49–55 (and, for diffusions, U. Rösler, *Unimodality of passage times for one-dimensional strong Markov processes*, Ann. Probab. **8** (1980) 853–859). The argument above does not rely on these and gives in addition the non-degeneracy $g''(t_c) < 0$.

3.3 Discrete time, nonnegative spectrum, every start

Theorem 3.5 (Unimodality in discrete time when $q \leq q_N$). *Let $q \leq q_N$ (see (5)); in particular any $q \leq \frac{1}{2}$, $x_0 \in \{1, \dots, N-1\}$ and $d = N - x_0$. Then there is an integer $t^* \geq d$ such that*

$$0 = f(1) = \dots = f(d-1) < f(d) < \dots < f(t^*), \quad f(t^*) \geq f(t^*+1) > f(t^*+2) > \dots$$

Thus f_{x_0} is unimodal with at most two modes, which are then adjacent. If $q < q_N$ and $d \geq 2$, the real-variable increment Ψ defined in the proof has exactly one zero t_0 in $(d-1, \infty)$, and every mode lies in $[t_0, t_0+1]$.

Proof. The case $q < q_N$. All λ_m are positive. Put $\Phi(t) = \sum_m c_m \lambda_m^{t-1} = \sum_m (c_m/\lambda_m) e^{t \ln \lambda_m}$ for real t , with $c_m = \sin \theta_m \sin(d\theta_m)$; then $f(t) = \frac{2q}{2N-1} \Phi(t)$ for integers $t \geq 1$, the exponents $\ln \lambda_m$ are strictly decreasing, and $V \leq d-1$ (Lemma 3.2). The function $\Psi(t) = \Phi(t+1) - \Phi(t) = \sum_m (c_m/\lambda_m)(\lambda_m - 1) e^{t \ln \lambda_m}$ has a coefficient sequence with the same number of sign changes. If $d = 1$, $\Psi < 0$ everywhere and the claim holds with $t^* = 1$. Let $d \geq 2$. Ψ vanishes at $t = 1, \dots, d-2$ (where $f(t) = f(t+1) = 0$), so by Lemma 3.1 it has at most one further real zero, counted with multiplicity. $\Psi(d-1) = \Phi(d) > 0$, and Ψ is negative at some integer $> d-1$ because $f(t) \rightarrow 0$. Hence Ψ has exactly one zero t_0 in $(d-1, \infty)$, it is simple, $\Psi > 0$ on $[d-1, t_0)$ and $\Psi < 0$ on (t_0, ∞) . Take $t^* = \lceil t_0 \rceil$: the increments $f(t+1) - f(t)$ are positive for $d-1 \leq t < t_0$, nonpositive at $t = t^*$ and negative for $t > t_0$, which is the stated pattern (also when $t_0 \in \mathbb{Z}$, where $f(t^*) = f(t^*+1)$).

The case $q = q_N$ (relevant only for $N \geq 3$, since $q_2 = 2$). Now $\lambda_n = 1 - 2q_N \cos^2 \frac{\pi}{2N-1} = 0$ and $\lambda_1 > \dots > \lambda_{n-1} > 0$, so $\lambda_n^{t-1} = 0$ for $t \geq 2$ and, for integers $t \geq 2$, $f(t) = \frac{2q}{2N-1} \Phi'(t)$ with the $(n-1)$ -term exponential sum $\Phi'(t) := \sum_{m < n} c_m \lambda_m^{t-1}$; put $\Psi'(t) := \Phi'(t+1) - \Phi'(t) = \sum_{m < n} (c_m/\lambda_m)(\lambda_m - 1) e^{t \ln \lambda_m}$ (real t). Its coefficient sequence is, up to the factors $(\lambda_m - 1)/\lambda_m < 0$, a subsequence of $(c_m)_{m \leq n}$, so it has at most $d-1$ sign changes (Lemma 3.2), and by Lemma 3.1 Ψ' has at most $d-1$ real zeros, counted with multiplicity. Moreover $c_1 = \sin \theta_1 \sin(d\theta_1) > 0$ because $0 < d\theta_1 = d\pi/(2N-1) < \frac{\pi}{2}$, so the leading term gives $\Psi'(t) < 0$ for all large t . For integers $t \geq 2$ we have $f(t+1) - f(t) = \frac{2q}{2N-1} \Psi'(t)$. If $d = 1$, all $c_m = \sin^2 \theta_m > 0$, so $\Psi' < 0$ everywhere and f is strictly decreasing from $t = 2$ on; moreover $f(2) < f(1) = \frac{q}{2}$, because $f(2) = \frac{q}{2}(1-q)$ if $N \geq 3$ (one rest at the interior site $N-1$, then a step to N); so $t^* = 1$. If $d = 2$, Ψ' has at most one real zero, counted with multiplicity, and it is negative for large t ; hence on $[2, \infty)$ either $\Psi' < 0$, or Ψ' has exactly one zero $t_0 \geq 2$, which is simple, with $\Psi' > 0$ on $[2, t_0)$ and $\Psi' < 0$ on (t_0, ∞) ; together with $f(2) = (q/2)^2 > 0 = f(1)$ this gives the pattern with $t^* = 2$, resp. $t^* = \lceil t_0 \rceil$. If $d \geq 3$, Ψ' vanishes at the $d-3$ integers $t = 2, \dots, d-2$ (where $f(t) = f(t+1) = 0$) and $\Psi'(d-1) = \frac{2N-1}{2q} f(d) > 0$ (note $d-1 \geq 2$). Since $\Psi'(d-1) > 0$ and $\Psi' < 0$ for large t , the total multiplicity of the zeros of Ψ' in $(d-1, \infty)$ is odd; these zeros are different from the $d-3$ known ones, so the total multiplicity in $(d-1, \infty)$ is at most $(d-1) - (d-3) = 2$, hence it is 1: there is exactly one zero $t_0 > d-1$, it is simple, $\Psi' > 0$ on $[d-1, t_0)$ and $\Psi' < 0$ on (t_0, ∞) . With $t^* = \lceil t_0 \rceil$ the stated strict pattern follows exactly as in the first case. (Numerical sanity check at $q = q_N$, $3 \leq N \leq 25$, all starts, 60 digits: script 38.) $\square$

3.4 Log-concavity from the reflecting end for $q \leq 2/3$

Theorem 3.6 (Log-concavity). *Let $x_0 = 1$.*

1. If $q \leq \frac{2}{3}$, the PMF f is log-concave: $f(t)^2 \geq f(t-1)f(t+1)$ for all $t \geq 2$, and $f(t) > 0$ exactly for $t \geq N-1$. In particular f is unimodal.

2. In continuous time the density g is log-concave on $(0, \infty)$, for every $q \in (0, 1]$.

Proof. (1) For $N = 2$, f is geometric. Let $n \geq 2$ and $K_t(x) := (Q^t)_{x,n}$ for $x \in \{1, \dots, n\}$, $t \geq 0$, so that $f(t) = \frac{q}{2}K_{t-1}(1)$ and $K_t(x) = \sum_z Q_{xz}K_{t-1}(z)$. Define, for $t \geq 1$, $M_t(x, y) := K_t(x)K_{t-1}(y) - K_{t-1}(x)K_t(y)$.

Claim: $M_t(x, y) \geq 0$ for all $t \geq 1$ and $x < y$. For $t = 1$, $K_0 = e_n$ and $x < y \leq n$ give $M_1(x, y) = Q_{x,n}\delta_{y,n} \geq 0$. For the induction step,

$$M_{t+1}(x, y) = \sum_{z,w} Q_{xz}Q_{yw} M_t(z, w) = \sum_{z < w} (Q_{xz}Q_{yw} - Q_{xw}Q_{yz})M_t(z, w),$$

using $M_t(z, w) = -M_t(w, z)$. It remains to check that Q is totally positive of order 2: $Q_{xz}Q_{yw} \geq Q_{xw}Q_{yz}$ for $x < y, z < w$. If $Q_{xw}Q_{yz} \neq 0$ then $|x-w| \leq 1$ and $|y-z| \leq 1$; with $x < y$ and $z < w$ this forces $z \geq y-1 \geq x$ and $w \leq x+1$, hence $z = x, w = x+1, y = x+1$, and the minor is $Q_{xx}Q_{x+1,x+1} - \frac{q^2}{4} \geq (1-q)^2 - \frac{q^2}{4} = (1-\frac{3q}{2})(1-\frac{q}{2}) \geq 0$ for $q \leq \frac{2}{3}$.

Now let $t \geq 2$. From $K_t(1) = (1-\frac{q}{2})K_{t-1}(1) + \frac{q}{2}K_{t-1}(2)$ and the same identity with $t-1$,

$$K_{t-1}(1)^2 - K_{t-2}(1)K_t(1) = \frac{q}{2}[K_{t-1}(1)K_{t-2}(2) - K_{t-2}(1)K_{t-1}(2)] = \frac{q}{2}M_{t-1}(1, 2) \geq 0,$$

that is, $f(t)^2 \geq f(t-1)f(t+1)$. The support statement is in Theorem 2.2 together with the existence of a path of every length $t \geq N-1$ (rest at site 1). A log-concave sequence whose support is an interval has nonincreasing ratios $f(t+1)/f(t)$ on its support and is therefore unimodal; moreover $f(k)^2 \geq f(k-j)f(k+j)$ for all $j \geq 0$.

(2) Let $A = Q - I$. For an integer $M \geq 3q/2$ the walk with move probability $q/M \leq \frac{2}{3}$ has transition matrix $I + A/M$ and PMF $f_{q/M}(k) = \frac{q}{2M}[(I + A/M)^{k-1}]_{1,n}$. If $k_M/M \rightarrow t$ then $(I + A/M)^{k_M-1} \rightarrow e^{tA}$, so $Mf_{q/M}(k_M) \rightarrow g(t)$. Given $0 < t_1 < t_2$ take $k_1 = \lceil Mt_1 \rceil$, $j = \lceil M(t_2 - t_1)/2 \rceil$; by (1), $f_{q/M}(k_1 + j)^2 \geq f_{q/M}(k_1)f_{q/M}(k_1 + 2j)$, and letting $M \rightarrow \infty$ gives $g(\frac{t_1+t_2}{2})^2 \geq g(t_1)g(t_2)$. A continuous midpoint-log-concave function is log-concave. $\square$

Remark 3.7. The Claim in the proof holds for every $q \leq \frac{2}{3}$ and all $x < y$: the ratio $f_x(t+1)/f_x(t)$ is nonincreasing in the start site x (monotone likelihood ratio in the distance to the target). Numerically, log-concavity of f from the reflecting end holds exactly for $q \leq q_{lc}(N)$, where $q_{lc}(N)$ is the root of $e_2(\lambda_1, \dots, \lambda_n) = 0$ (the first inequality $f(N)^2 \geq f(N-1)f(N+1)$ is the binding one); $q_{lc}(3) = 0.7639$, $q_{lc}(4) = 0.7847$, $q_{lc}(5) = \frac{4}{5}$, $q_{lc}(N) \uparrow 1$ (Numerical observation, script 02).

3.5 Monotonicity of unimodality in q

Lemma 3.8 (Slowing down preserves unimodality). *Fix N and x_0 . If f_{q_0} is unimodal for some $q_0 \in (0, 1]$, then f_q is unimodal for every $q \in (0, q_0)$. Consequently $\{q \in (0, 1] : f_q \text{ unimodal}\}$ is an interval $(0, q^*(N, x_0)]$.*

Proof. Let $p = q/q_0 \in (0, 1)$. Run the q_0 -walk only at the times of an independent Bernoulli(p) sequence and rest otherwise. The resulting chain moves left/right with probability $pq_0/2 = q/2$ each (cancelled moves at site 1 remain rests), so it is the q -walk, and $T_q = \xi_1 + \dots + \xi_{T_{q_0}}$ with i.i.d. geometric(p) variables ξ_i on $\{1, 2, \dots\}$ independent of T_{q_0} . Hence, with $a_k := f_{q_0}(k)$,

$$f_q(t) = \sum_{k \geq 1} a_k K(k, t), \quad K(k, t) := \binom{t-1}{k-1} p^k (1-p)^{t-k} \quad (k, t \geq 1).$$

Pascal's rule gives $K(k, t+1) = (1-p)K(k, t) + pK(k-1, t)$ with $K(0, t) := 0$, hence

$$f_q(t+1) - f_q(t) = p \sum_{k \geq 1} (a_{k+1} - a_k) K(k, t) =: p e(t).$$

By unimodality of a there is $k_0 \geq 0$ with $b_k := a_{k+1} - a_k \geq 0$ for $k \leq k_0$ and $b_k \leq 0$ for $k > k_0$. We show: if $e(t_1) < 0$ and $t_2 > t_1$ then $e(t_2) \leq 0$; this gives the unimodality of f_q . Since $K(k, t_1) = 0$ for $k > t_1$, $e(t_1) < 0$ forces $k_0 < t_1$ (otherwise $e(t_1)$ is a sum of nonnegative terms); if $k_0 = 0$ all $b_k \leq 0$ and $e \leq 0$ everywhere. For $1 \leq k \leq t_1$,

$$\frac{K(k, t_2)}{K(k, t_1)} = (1-p)^{t_2-t_1} \prod_{i=0}^{k-2} \frac{t_2-1-i}{t_1-1-i}$$

is nondecreasing in k . Let γ be its value at $k = k_0$. Then $K(k, t_2) \leq \gamma K(k, t_1)$ for $k \leq k_0$ (where $b_k \geq 0$) and $K(k, t_2) \geq \gamma K(k, t_1)$ for $k > k_0$ (where $b_k \leq 0$; for $k > t_1$ the right side is 0). Therefore $e(t_2) = \sum_k b_k K(k, t_2) \leq \gamma \sum_k b_k K(k, t_1) = \gamma e(t_1) < 0$. The last assertion follows because unimodality is preserved under the pointwise limit $q \uparrow q^*$. $\square$

3.6 What fails for q close to 1

Proposition 3.9 (Sharp obstruction at the first three values). *Let $x_0 = 1$.*

1. For $N \geq 4$ and every $q \in (0, 1]$: $f(N+1) > f(N)$.
2. $f(N) < f(N-1)$ if and only if $q > 1 - \frac{1}{2N-3}$.
3. Hence for $N \geq 4$ and $q > 1 - \frac{1}{2N-3}$ the PMF is not unimodal ($f(N-1) > f(N) < f(N+1)$), so $q^*(N, 1) \leq 1 - \frac{1}{2N-3}$. In particular for every $q \in (\frac{4}{5}, 1]$ the PMF with $N = 4$ is not unimodal, and for $q = 1$ the PMF is not unimodal for any $N \geq 4$.
4. For $N \in \{2, 3\}$ the PMF is unimodal for every $q \in (0, 1]$. For $N = 4$ it is unimodal if and only if $q \leq \frac{4}{5}$; at $q = \frac{4}{5}$, $f(3) = f(4) = \frac{8}{125} < f(5) = \frac{208}{3125}$, the mode is $t = 5$.

Proof. By Theorem 2.3, $f(n+k) = (q/2)^n h_k(\lambda)$ with $h_0 = 1$, $h_1 = \text{tr } Q = n - (n - \frac{1}{2})q$ and $h_2 = \frac{1}{2}(h_1^2 + \text{tr } Q^2)$, $\text{tr } Q^2 = \sum_{i,j} Q_{ij}^2 = (1 - \frac{q}{2})^2 + (n-1)(1-q)^2 + (n-1)\frac{q^2}{2}$. A direct expansion gives

$$2(h_2 - h_1) = n(n-1) - n(2n-1)q + (n^2 + \frac{n}{2} - 1)q^2,$$

a quadratic in q with positive leading coefficient and discriminant $n(-2n^2 + 7n - 4) < 0$ for $n \geq 3$. This proves (1). (2) is $h_1 < 1 \iff q > (n-1)/(n-\frac{1}{2})$. (3) follows. (4): for $N = 2$, f is geometric. For $N = 3$ and $q = 1$, $e_1 = \lambda_1 + \lambda_2 = \frac{1}{2}$, $e_2 = \lambda_1 \lambda_2 = \det Q = -\frac{1}{4}$, so $h_{k+1} = \frac{1}{2}h_k + \frac{1}{4}h_{k-1}$, $h_0 = 1$, $h_1 = \frac{1}{2}$, i.e. $h_k = F_{k+1}/2^k$ with the Fibonacci numbers $F_1 = F_2 = 1$; $h_{k+1} \leq h_k$ is $F_{k+2} \leq 2F_{k+1}$, which holds. So f is nonincreasing on its support at $q = 1$, hence unimodal, and Lemma 3.8 gives all $q < 1$. For $N = 4$, $q = \frac{4}{5}$ the values $f(3) = f(4) = (\frac{2}{5})^3$ and $f(5) = (\frac{2}{5})^3 h_2 = \frac{208}{3125}$ follow from the formulas above, and beyond $t = 5$ the sequence decreases (Theorem 3.15(2)), so f is unimodal at $q = \frac{4}{5}$, hence for $q \leq \frac{4}{5}$ by Lemma 3.8; for $q > \frac{4}{5}$ use (3). $\square$

Theorem 3.10 ($q = 1$: each parity class is unimodal). *Let $q = 1$, $x_0 \in \{1, \dots, N-1\}$, $d = N - x_0$ and $d' = 2N - 1 - d = N + x_0 - 1$. Then $f(t) = 0$ for $t < d$, and also for $t < d'$ with $t \equiv d' \pmod{2}$. Each of the two sequences*

$$i \mapsto f(d+2i) \quad \text{and} \quad i \mapsto f(d'+2i), \quad i \geq 0,$$

is positive and strictly increasing up to an index i^ , satisfies $f(\cdot + 2i^*) \geq f(\cdot + 2i^* + 2)$ there, and is strictly decreasing afterwards. Moreover, with $\vartheta_k = k\pi/(2N-1)$, for $D \in \{d, d'\}$ and $t \equiv D \pmod{2}$,*

$$f(t) = \frac{2}{2N-1} \sum_{k=1}^{N-1} \sin \vartheta_k \sin(D\vartheta_k) \cos^{t-1} \vartheta_k. \quad (6)$$

Proof. Let $W = 2N - 1$. For $m \leq N/2$ put $k = 2m - 1$ (odd, $k \leq N - 1$): $\theta_m = \vartheta_k$, $\lambda_m = \cos \vartheta_k$. For $m > N/2$ put $k = 2(N - m)$ (even, $2 \leq k \leq N - 1$): $\theta_m = \pi - \vartheta_k$, $\lambda_m = -\cos \vartheta_k$, $\sin \theta_m = \sin \vartheta_k$ and $\sin(d\theta_m) = (-1)^{d+1} \sin(d\vartheta_k)$. This is a bijection $m \leftrightarrow k \in \{1, \dots, N - 1\}$, and (2) becomes

$$\frac{W}{2}f(t) = \sum_{k \text{ odd}} \sin \vartheta_k \sin(d\vartheta_k) \cos^{t-1} \vartheta_k + (-1)^{d+t} \sum_{k \text{ even}} \sin \vartheta_k \sin(d\vartheta_k) \cos^{t-1} \vartheta_k.$$

For $t \equiv d$ this is (6) with $D = d$. For $t \equiv d + 1 \equiv d'$ the sign is $(-1)^{k+1}$ for all k , and $(-1)^{k+1} \sin(d\vartheta_k) = \sin(k\pi - d\vartheta_k) = \sin(d'\vartheta_k)$ because $k\pi = W\vartheta_k$; this is (6) with $D = d'$.

Vanishing: a path from x_0 to N of length t has r right steps, l left steps and s cancelled steps (rests at site 1), $r - l = d$, $r + l + s = t$. If $t \equiv d' \equiv d + 1$ then s is odd, so the path visits site 1; then $l \geq x_0 - 1$, $r = d + l \geq N - 1$ and $t \geq (N - 1) + (x_0 - 1) + 1 = d'$.

Fix $D \in \{d, d'\}$ and let $\Phi_D(t)$ be the right-hand side of (6) times $W/2$, for real t . It is an exponential sum with exponents $\ln \cos \vartheta_k$, strictly decreasing in k since $0 < \vartheta_k < \pi/2$, and by Lemma 3.2 (with $\Theta = \pi/2$) its coefficients have at most $V_D := \lfloor (D - 1)/2 \rfloor$ sign changes. The function $\Psi_D(t) = \Phi_D(t + 2) - \Phi_D(t)$ has coefficients $-\sin^3 \vartheta_k \sin(D\vartheta_k)$, with the same number of sign changes. Φ_D vanishes at the V_D integers $t = D - 2i \geq 1$, $i \geq 1$; hence Ψ_D vanishes at the $\max(V_D - 1, 0)$ integers $t = D - 2i \geq 1$, $i \geq 2$. If $V_D = 0$, $\Psi_D < 0$ everywhere and the class sequence is strictly decreasing. If $V_D \geq 1$, Lemma 3.1 leaves room for at most one further real zero of Ψ_D ; since $\Psi_D(D - 2) = \Phi_D(D) = \frac{W}{2}f(D) > 0$ and $f(D + 2i) \rightarrow 0$, there is exactly one zero $t_0 > D - 2$, $\Psi_D > 0$ on $[D - 2, t_0)$ and $\Psi_D < 0$ on (t_0, ∞) . $\square$

Remark 3.11. Unfolding the reflection identifies the class $t \equiv d$ with exit of the simple random walk from $(1 - N, N)$ through N and the class $t \equiv d'$ with exit through $1 - N$. At $q = 1$ the two classes have different envelopes (their ratio is $1 + O(1/N)$), which is why the raw PMF oscillates.

3.7 Beyond $q = \frac{2}{3}$: eventual total positivity

For $q > \frac{2}{3}$ the matrix Q is not TP_2 (the minor $Q_{xx}Q_{x+1,x+1} - Q_{x,x+1}Q_{x+1,x} = (1 - q)^2 - \frac{q^2}{4}$ is negative), and the induction in the proof of Theorem 3.6 breaks down at its first step. What that proof really needs, however, is the total positivity of the powers Q^t for the relevant t , and high powers of Q are TP_2 again. For a matrix A and indices $x < y$, $z < w$ write $A[x, y|z, w] := A_{xz}A_{yw} - A_{xw}A_{yz}$; a matrix with nonnegative entries is called TP_2 if all these minors are nonnegative. We keep the notation $K_t(x) = (Q^t)_{x,n}$ and $M_t(x, y) = K_t(x)K_{t-1}(y) - K_{t-1}(x)K_t(y)$ of the proof of Theorem 3.6; recall $f(t) = \frac{q}{2}K_{t-1}(1)$ and, for $t \geq 1$,

$$f(t + 1)^2 - f(t)f(t + 2) = \left(\frac{q}{2}\right)^3 M_t(1, 2). \quad (7)$$

Lemma 3.12 (Products). *1. For $n \times n$ matrices A, B and $x < y$, $z < w$: $(AB)[x, y|z, w] = \sum_{u < v} A[x, y|u, v] B[u, v|z, w]$. Hence a product of TP_2 matrices is TP_2 .*

2. For $t \geq 1$ and $x < y$: $M_t(x, y) = \sum_{u < v} (Q^{t-1})[x, y|u, v] M_1(u, v)$, and $M_1(u, v) = Q_{u,n}\delta_{v,n} \geq 0$. Consequently, if Q^k is TP_2 , then $M_{k+1}(x, y) \geq 0$ for all $x < y$.

Proof. (1) $(AB)[x, y|z, w] = \sum_{u, v} A_{xu}A_{yv}\beta(u, v)$ with $\beta(u, v) := B_{uz}B_{vw} - B_{uw}B_{vz} = -\beta(v, u)$; grouping the terms (u, v) and (v, u) gives $\sum_{u < v} (A_{xu}A_{yv} - A_{xv}A_{yu})\beta(u, v)$ (the Cauchy–Binet formula for minors of order two). (2) Since $K_t = Q^{t-1}K_1$ and $K_{t-1} = Q^{t-1}K_0$, $M_t(x, y) = \sum_{u, v} (Q^{t-1})_{xu}(Q^{t-1})_{yv} [K_1(u)K_0(v) - K_0(u)K_1(v)]$, the bracket is antisymmetric and equals $M_1(u, v)$ for $u < v$; and $K_0 = e_n$ gives $M_1(u, v) = K_1(u)\delta_{v,n} = Q_{u,n}\delta_{v,n}$ for $u < v \leq n$. $\square$

Lemma 3.13 (Contiguous minors suffice). *Let $k \geq 1$ and let P be an $n \times n$ matrix with $P_{xz} > 0$ if $|x - z| \leq k$ and $P_{xz} = 0$ otherwise. If $P[x, x + 1|z, z + 1] \geq 0$ for all x, z with $x + 1 - k \leq z$ and $z + 1 \leq x + k$, then P is TP_2 .*

Proof. Let $x < y$, $z < w$. If $P_{xw}P_{yz} = 0$ then $P[x, y|z, w] = P_{xz}P_{yw} \geq 0$. Otherwise $|x - w| \leq k$ and $|y - z| \leq k$, hence $y - k \leq z < w \leq x + k$. For all rows $r \in [x, y]$ and columns $c \in [z, w]$ we then have $r - c \leq y - z \leq k$ and $c - r \leq w - x \leq k$, so $P_{rc} > 0$; and for $x \leq r < y$, $z \leq c < w$ the pair (r, c) satisfies $r + 1 - k \leq y - k \leq c$ and $c + 1 \leq w \leq r + k$, so $P_{r,c}P_{r+1,c+1} \geq P_{r,c+1}P_{r+1,c}$ by hypothesis, i.e. $P_{r+1,c+1}/P_{r+1,c} \geq P_{r,c+1}/P_{r,c}$. Multiplying these inequalities over $c = z, \dots, w - 1$ gives $P_{r+1,w}/P_{r+1,z} \geq P_{r,w}/P_{r,z}$, and chaining over $r = x, \dots, y - 1$ gives $P_{yw}/P_{yz} \geq P_{xw}/P_{xz}$, that is $P[x, y|z, w] \geq 0$. $\square$

For $0 < q < 1$ the support of Q^k is exactly the band $|x - z| \leq k$ (a path may rest), so Lemma 3.13 applies to $P = Q^k$.

Lemma 3.14 (Locality). *Write Q_N for the matrix (1) on $n = N - 1$ sites. Let $k \geq 1$ and $N \geq N_0$ with $N_0 - 1 \geq 4k$. Then every minor $Q_N^k[x, y|z, w]$ ($x < y$, $z < w$) is either nonnegative because $(Q_N^k)_{xw}(Q_N^k)_{yz} = 0$, or equal to some minor $Q_{N_0}^k[x', y'|z', w']$. In particular, if $Q_{N_0}^k$ is TP_2 , then Q_N^k is TP_2 for every $N \geq N_0$.*

Proof. $(Q_N^k)_{x,z}$ is the sum over the paths $x = s_0, s_1, \dots, s_k = z$ in $\{1, \dots, n\}$ with $|s_{i+1} - s_i| \leq 1$ of the products of the weights $\frac{q}{2}$ (a move), $1 - q$ (a rest at a site ≥ 2) and $1 - \frac{q}{2}$ (a rest at site 1). A path starting at x stays in $[x - k, x + k]$. Hence: (a) if $x \leq n - k$, the entry does not depend on N (the absorbing end is not felt): $(Q_N^k)_{x,z} = W_k(x, z)$; (b) if $x \geq k + 1$, no path rests at site 1 and the entry depends only on the distances to the absorbing end: $(Q_N^k)_{x,z} = V_k(n - x, n - z)$; (c) if $k + 1 \leq x \leq n - k$, $(Q_N^k)_{x,z} = p_k(z - x)$ with the k -step kernel p_k of the lazy walk on $\mathbb{Z}$.

Let $(Q_N^k)_{xw}(Q_N^k)_{yz} \neq 0$; then (bandwidth k) $y - k \leq z < w \leq x + k$, in particular $y - x \leq 2k - 1$, and $n_0 := N_0 - 1 \geq 4k$, $n \geq n_0$. *Case $y \leq n - k$, $x \leq k$.* Then $y \leq 3k - 1 \leq n_0 - k$, both rows are given by (a), and the minor equals $Q_{N_0}^k[x, y|z, w]$. *Case $y \leq n - k$, $x \geq k + 1$.* Both rows are given by (c), so the minor depends only on $(y - x, z - x, w - x)$ and equals $Q_{N_0}^k[k + 1, k + 1 + y - x | k + 1 + z - x, k + 1 + w - x]$; here $k + 1 + y - x \leq 3k \leq n_0 - k$ and $1 \leq k + 1 + z - x$, $k + 1 + w - x \leq 2k + 1 \leq n_0$. *Case $y \geq n - k + 1$.* Then $x \geq y - 2k + 1 \geq n - 3k + 2 \geq k + 1$, both rows are given by (b), the minor depends only on $(n - x, n - y, n - z, n - w)$ and equals the minor of $Q_{N_0}^k$ at the indices shifted by $n_0 - n$; the shifted row index $x - (n - n_0) \geq n_0 - 3k + 2 \geq k + 1$ and the shifted column index $z - (n - n_0) \geq n_0 - 2k + 1 \geq 1$ are admissible. $\square$

Theorem 3.15 ($q = \frac{4}{5}$: eventual total positivity, log-concavity, unimodality; exact CA). 1.

For $q = \frac{4}{5}$ and every $N \geq 4$ the matrix Q^k is TP_2 for all $k \geq 10$. (For large N it is not TP_2 for $k \leq 7$ and for $k = 9$; e.g. $N = 80$.)

2. For $q = \frac{4}{5}$ the PMF f from the reflecting end is log-concave if $N = 2$ or $N \geq 5$. For $N \in \{3, 4\}$ it is unimodal but not log-concave.

3. For every $N \geq 2$ and every $q \in (0, \frac{4}{5}]$ the PMF f from the reflecting end is unimodal. The bound $\frac{4}{5}$ is optimal for a statement valid for all N (Proposition 3.9(4), $N = 4$).

Proof. (1) Script 31_tp2_powers.py computes the integer matrices A^k , $A := 10Q$ ($A_{11} = 6$, $A_{jj} = 2$, $A_{j,j\pm 1} = 4$), for $1 \leq k \leq 19$ and $3 \leq N \leq 80$ in exact integer arithmetic and verifies, for $4 \leq N \leq 80$ and $10 \leq k \leq 19$, that all contiguous minors $A^k[x, x + 1|z, z + 1]$ are nonnegative (712 830 minors; for safety the script also tests all 96 377 478 minors $A^k[x, y|z, w]$ with $y - k \leq z < w \leq x + k$ directly). By Lemma 3.13, Q^k is TP_2 for these N and k ; by Lemma 3.14 with $N_0 = 80$ ($N_0 - 1 = 79 \geq 4 \cdot 19$) the same holds for all $N \geq 80$. Every integer $k \geq 20$ is a sum of integers from $\{10, \dots, 19\}$, and products of TP_2 matrices are TP_2 (Lemma 3.12(1)). The negative minors for small k are listed in the output of the script.

(2) $N = 2$: f is geometric. Let $N \geq 5$. By (1) and Lemma 3.12(2), $M_t(1, 2) \geq 0$ for all $t \geq 11$. For $1 \leq t \leq 10$: if $N \geq 14$ then $K_s(2) = 0$ for $s \leq 10 < n - 2$, so $M_t(1, 2) = 0$; for $5 \leq N \leq 13$ the script computes the ten integers exactly and finds them nonnegative. By (7), $f(t + 1)^2 \geq f(t)f(t + 2)$ for all $t \geq 1$, and the support of f is $\{t \geq N - 1\}$. For $N = 3$,

$M_2(1, 2) < 0$ and for $N = 4$, $M_3(1, 2) < 0$ (script), so f is not log-concave. $N = 3$ is unimodal by Proposition 3.9(4). For $N = 4$ the script finds the signs of $f(t+1) - f(t)$, $t = 1, \dots, 12$, to be $0, +, 0, +, -, -, -, -, -, -, -$ (exact), and $M_t(1, 2) \geq 0$ for $t \geq 11$ shows that the ratios $f(t+1)/f(t)$ are nonincreasing for $t \geq 11$; since $f(13) < f(12)$, f decreases from $t = 5$ on.

(3) For $q = \frac{4}{5}$ this is (2); for $q < \frac{4}{5}$ apply Lemma 3.8. $\square$

Theorem 3.16 (Beyond $\frac{4}{5}$; exact CA). 1. For $q = \frac{17}{20}$ and $N \geq 5$, Q^k is TP_2 for all $k \geq 18$. The PMF f is log-concave iff $N = 2$ or $N \geq 12$, and unimodal iff $N \notin \{4, 5\}$.

2. For $q = \frac{9}{10}$ and $N \geq 7$, Q^k is TP_2 for all $k \geq 40$. The PMF f is log-concave iff $N = 2$ or $N \geq 33$, and unimodal iff $N \notin \{4, 5, 6, 7, 8\}$.

3. Consequently f_q is unimodal for all $q \leq \frac{17}{20}$ if $N \geq 6$, and for all $q \leq \frac{9}{10}$ if $N \geq 9$; in the notation of Lemma 3.8: $q^*(N, 1) \geq \frac{4}{5}$ for all N , $q^*(N, 1) \geq \frac{17}{20}$ for $N \geq 6$, $q^*(N, 1) \geq \frac{9}{10}$ for $N \geq 9$.

Proof. The same script with $(q, k_*, N_0) = (\frac{17}{20}, 18, 141)$ and $(\frac{9}{10}, 40, 320)$: for $3 \leq N \leq N_0$ and $k_* \leq k \leq 2k_* - 1$ all contiguous minors of A^k ($A = 40Q$, resp. $20Q$) are computed exactly (7 460 898, resp. 196 413 420 minors); they are nonnegative for $N \geq 5$, resp. $N \geq 7$. Lemmas 3.13, 3.14 ($N_0 - 1 \geq 4(2k_* - 1)$) and 3.12 give the TP_2 statements, hence $M_t \geq 0$ for $t \geq k_* + 1$. For each $N \leq N_0$ the script computes exactly the signs of $M_t(1, 2)$, $t \leq k_*$, and of the increments $f(t+1) - f(t)$, $t \leq k_* + 2$. f is log-concave iff all these $M_t(1, 2)$ are ≥ 0 (for $N > N_0$ they vanish). Since the ratios $f(t+1)/f(t)$ are nonincreasing for $t \geq k_* + 1$, f is unimodal iff the exact sign sequence of the increments up to $t = k_* + 2$ contains no $-$ followed by a later $+$. This decides all $N \geq 5$ ($N \geq 7$ for $q = \frac{9}{10}$); for the remaining $N \in \{4, 5, 6\}$ at $q = \frac{9}{10}$ and $N = 4$ at $q = \frac{17}{20}$ the sign sequence itself contains $-$, $+$ (not unimodal; also Proposition 3.9(3) for $N \leq 6$, resp. $N = 4$), and $N \in \{2, 3\}$ are covered by Proposition 3.9(4). (3) follows from Lemma 3.8. (Outputs: `data/31_tp2_powers_q*.json`; brute-force confirmation for $N \leq 40$ by separately written code: script 33.) $\square$

Remark 3.17 (The mechanism, and a question). (a) $M_t(x, y) \geq 0$ for all $x < y$ means that $x \mapsto f_x(t+1)/f_x(t)$ is nonincreasing in the start x , or equivalently that the two walks started at the sites $n-1$ and n next to the target are ordered in the monotone likelihood ratio order at time $t-1$: $x \mapsto (Q^{t-1})_{n-1, x} / (Q^{t-1})_{n, x}$ is nonincreasing. For $q > \frac{2}{3}$ this fails at early times because of parity oscillations, but only in a bounded neighbourhood of the target: for $q = \frac{4}{5}$ exactly 16 triples $(n-x, n-y, t)$ have $M_t(x, y) < 0$, all with $n-x \leq 6$ and $t \leq 10$, the same for every $12 \leq N \leq 20$ (script 30, $t \leq N^2$; for $t \geq 11$ there are none for any $N \geq 4$ by Theorem 3.15(1)). Starting from the reflecting end one never sees them when $N \geq 5$. (b) The smallest k_* such that Q^k is TP_2 for all $k \geq k_*$ and all large N is $k_* = 2, 4, 10, 18, 40$ for $q = \frac{7}{10}, \frac{3}{4}, \frac{4}{5}, \frac{17}{20}, \frac{9}{10}$ (exact computations; for $q \in \{\frac{7}{10}, \frac{3}{4}\}$ the TP_2 property holds for every $N \geq 3$, and the proof of Theorem 3.15(2) then shows that f is log-concave for every $N \geq 2$, see the logs of script 31); heuristically $k_* \approx \frac{q^2}{2(1-q)^2}$, from the first two coefficients of $(b + az + bz^2)^k$. *Question:* is Q^k eventually TP_2 for every $q < 1$, uniformly in N ? This would give: for every $q < 1$, f is log-concave for all $N \geq N_1(q)$.

Proposition 3.18 (Table of modes for $q = \frac{4}{5}$; CA). For $q = \frac{4}{5}$ and every $N \in \{2, 3, \dots, 600\}$ the maximiser of f is unique; the table of modes is `data/08_modes_q45.jsonl`. (The computation also proves unimodality for these N a second time, by a method that does not use Theorem 3.15.)

Proof. Script `08_unimodality_q45.py` computes $r_t = (Q^t)_{1, n}$ in IEEE double precision. All quantities are nonnegative and each entry of $Q^{t+1}e$ is a sum of at most three products of a rounded constant and a nonnegative number, so (round to nearest; no underflow, see below) the computed value equals $r_t(1 + \vartheta_t)$ with $|\vartheta_t| \leq (1 + 2^{-53})^{4t} - 1 < 5t 2^{-53}$ (one rounded constant, one multiplication and two additions per term and step; sums of nonnegative numbers do not amplify relative errors). The sign of $f(t+1) - f(t)$ is accepted only if the computed lower bar of one value

exceeds the computed upper bar of the other. These bars, $\text{fl}(\hat{r}_t \cdot (1 \mp 5t2^{-53}))$, are themselves formed in floating point, which is harmless: with $u = 2^{-53}$ and $P_t := (1+u)^{4t} - 1 \leq 4tu + 16t^2u^2$, the factor $1 - 5tu$ is an exact floating-point number, $1 + 5tu$ is exact for even t and correct to within u for odd t , the product is rounded with relative error at most u , and the comparison itself is exact. So the computed lower bar is at most $r_t(1 + P_t)(1 - 5tu)(1 + u) = r_t(1 - (t-1)u + O(t^2u^2))$ and the computed upper bar at least $r_t(1 - P_t)(1 + 5tu - u)(1 - u) = r_t(1 + (t-2)u + O(t^2u^2))$ for odd t (and $r_t(1 + (t-1)u + O(t^2u^2))$ for even t). Since all indices used satisfy $t \leq t_{\text{end}} + 2 \leq 1.1 \times 10^6$ (`data/08_modes_q45.jsonl`, field `t_end`), the second-order terms, at most $50t^2u^2 < 10^{-18}$ in absolute value, are far below u , and the bars are rigorous for every $t \geq 2$. For $t \leq 1$ the compared values are exact zeros or differ by a relative amount of order 1 (e.g. $r_0 = 1$, $r_1 = 1 - \frac{q}{2}$ for $N = 2$), far beyond any rounding. The remaining indices (435 in all, for 210 values of N : the exact tie $N = 4$, $t = 3$, and, for $N \geq 251$, indices within distance 3 of the mode, where the PMF is too flat for double precision) are decided in 200-bit ball arithmetic from (2), and the exact tie $f(3) = f(4)$ for $N = 4$ by exact rational arithmetic.

No underflow. For $1 \leq j \leq n$ and $t \geq j - 1$ the Chapman–Kolmogorov equation (all terms nonnegative) gives $(Q^t)_{1,j} \geq (Q^{t-j+1})_{1,1}(Q^{j-1})_{1,j} = (q/2)^{j-1}(Q^{t-j+1})_{1,1}$, while $(Q^t)_{1,j} = 0$ for $t < j - 1$ (and is computed as an exact zero). Moreover $(Q^s)_{1,1} \geq (1 - \frac{q}{2})\lambda_1^s\varphi_1(1)^2$ for all $s \geq 0$, where $\varphi_1(1)^2 = v_1(1)^2/\|v_1\|^2 = 4\cos^2(\theta_1/2)/(2N-1)$: for even s by the spectral decomposition $(Q^s)_{1,1} = \sum_m \lambda_m^s \varphi_m(1)^2 \geq \lambda_1^s \varphi_1(1)^2$ (Lemma 2.1), and for odd s because $(Q^s)_{1,1} \geq Q_{11}(Q^{s-1})_{1,1}$ and $\lambda_1 < 1$. Hence every nonzero entry of $Q^t e_1$ with $t \leq t_{\text{end}} + 2$ is at least $(\frac{q}{2})^{N-2}(1 - \frac{q}{2})\lambda_1^{t_{\text{end}}+2}\varphi_1(1)^2$; script 24 certifies that this number exceeds 10^{-242} for $q = \frac{4}{5}$ and every $N \leq 600$. So all intermediate products exceed 10^{-243} , far above the underflow threshold 2^{-1022} , and the error model applies. For $t \geq t_{\text{end}}(N)$ the increment is negative because in $f(t+1) - f(t) = -\frac{2q}{2N-1} \sum_m c_m \nu_m \lambda_m^{t-1}$ the term $m = 1$ dominates: $(\max_{m \geq 2} |\lambda_m|/\lambda_1)^{t-1} < c_1 \nu_1 / \sum_{m \geq 2} |c_m| \nu_m$, with t_{end} computed in ball arithmetic. The script verifies that the certified signs are nonnegative up to the mode and nonpositive afterwards. $\square$

Remark 3.19 (Cross-check with exact-arithmetic certificates). The exact-arithmetic certificates of the companion note R5 (`data/msc_rigorous_certified`; exact multi-limb integer arithmetic, a different method, written separately within the same project and run on the same machine) establish the same statement for $q = \frac{4}{5}$, $2 \leq N \leq 600$, and in addition for $N \in \{800, 1000\}$, in agreement with Theorem 3.15. Script 22 compares the two sets of certificates: all 598 modes for $3 \leq N \leq 600$ agree, and so do the unimodality flags (both find the single plateau $N = 4$).

Numerical observation 3.20. Floating-point bisection (script 02) gives $q^*(N, 1) = 1 - \frac{1}{2N-3}$ for $N = 4$, $N = 20$ and $22 \leq N \leq 80$, and a strictly smaller value for $5 \leq N \leq 19$ and $N = 21$ (e.g. $q^*(5, 1) \approx 0.8354$, $q^*(10, 1) \approx 0.9209$); in all cases $q^*(N, 1) \geq \frac{4}{5}$, with equality only for $N = 4$ (the inequality is Theorem 3.15(3); the values are consistent with Theorem 3.16).

4 The continuum limit and the constant c

4.1 The limiting density

For $u > 0$ and integers $j \geq 0$ put

$$G^{(j)}(u) := \sum_{k \text{ odd}} \sigma_k 2y_k (-2y_k^2)^j e^{-2uy_k^2}, \quad G(u) := G^{(0)}(u) = \frac{\pi}{2} \sum_{k \text{ odd}} \sigma_k k e^{-k^2\pi^2 u/8} \quad (8)$$

($y_k = k\pi/4$, $\sigma_k = (-1)^{(k-1)/2}$). The series converge absolutely and uniformly on $[u_0, \infty)$ for every $u_0 > 0$, together with all termwise derivatives; hence G is real-analytic on $(0, \infty)$ and $G^{(j)}$ is its j -th derivative.

Lemma 4.1 (Tails). *Let $a > 0$, $p \geq 0$ and let $K \geq 1$ be odd with $r := (1 + \frac{2}{K})^p e^{-a\pi^2(K+1)/2} < 1$. Then*

$$\sum_{k \geq K, k \text{ odd}} y_k^p e^{-2ay_k^2} \leq \frac{y_K^p e^{-2ay_K^2}}{1-r}.$$

Proof. With $t_k = y_k^p e^{-2ay_k^2}$ and $y_{k+2}^2 - y_k^2 = \pi^2(k+1)/4$, we get $t_{k+2}/t_k = (1 + \frac{2}{k})^p e^{-a\pi^2(k+1)/2} \leq r$ for $k \geq K$; sum the geometric series. $\square$

Recall $g(s) = g_1(s)$, the continuous-time density from the reflecting end, and define the *scaled lattice density*

$$G_N(u) := \frac{L^2}{q} g\left(\frac{L^2 u}{q}\right) = \sum_{\substack{k \text{ odd} \\ k \leq 2N-3}} \sigma_k 2y_k A_0(\zeta_k) e^{-2uy_k^2 \beta(\zeta_k)}, \quad x_k = \frac{y_k}{L}, \quad \zeta_k = x_k^2, \quad (9)$$

where, for $\zeta \geq 0$ and $x = \sqrt{\zeta}$,

$$S(\zeta) := \frac{\sin x}{x}, \quad C(\zeta) := \cos^2 x = 1 - \zeta S(\zeta)^2, \quad \beta := S^2, \quad A_j := S^{2j+1} C. \quad (10)$$

Indeed, in (3) with $x_0 = 1$, $k = 2m - 1$: $\theta_m = 2x_k$, $\sin \theta_m \cos(\theta_m/2) = 2 \sin x_k \cos^2 x_k$, $L \sin x_k = y_k S(\zeta_k)$ and $\nu_m L^2 u/q = 2uL^2 \sin^2 x_k = 2uy_k^2 \beta(\zeta_k)$. G_N does not depend on q , G_N is a probability density with mean $1 - \frac{1}{4L^2}$ (Corollary 2.5), and

$$G_N^{(j)}(u) = \sum_{k \text{ odd}, k \leq 2N-3} \sigma_k 2y_k (-2y_k^2)^j A_j(\zeta_k) e^{-2uy_k^2 \beta(\zeta_k)}. \quad (11)$$

Theorem 4.2 (Convergence). *For every $u_0 > 0$ and $j \geq 0$, $\sup_{u \geq u_0} |G_N^{(j)}(u) - G^{(j)}(u)| \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. For $0 < x \leq \pi/2$ we have $\frac{2}{\pi} \leq S \leq 1$ and $0 \leq C \leq 1$, so the k -th term of (11) is bounded in absolute value by $b_k := 2^{j+1} y_k^{2j+1} e^{-8u_0 y_k^2 / \pi^2}$ for all N and all $u \geq u_0$, and so is the k -th term of (8); $\sum_k b_k < \infty$. For fixed k , $\zeta_k \rightarrow 0$ as $N \rightarrow \infty$, $A_j(\zeta_k) \rightarrow 1$, $\beta(\zeta_k) \rightarrow 1$, and $\sup_{u \geq u_0} |A_j(\zeta_k) e^{-2uy_k^2 \beta(\zeta_k)} - e^{-2uy_k^2}| \leq |A_j(\zeta_k) - 1| + 2y_k^2(1 - \beta(\zeta_k)) \sup_{u \geq u_0} u e^{-2uy_k^2 \beta(\zeta_k)} \rightarrow 0$ (mean value theorem in the exponent). Given $\varepsilon > 0$ choose K with $\sum_{k > K} b_k < \varepsilon$; the finitely many terms $k \leq K$ converge uniformly. Hence the claim (Tannery's theorem). An explicit rate is given in Proposition 5.4. $\square$

Theorem 4.3 (The limiting density has a unique maximiser). *1. $G > 0$ on $(0, \infty)$ and $\ln G$ is concave.*

2. G' has exactly one zero c in $(0, \infty)$; $G' > 0$ on $(0, c)$ and $G' < 0$ on (c, ∞) . In particular c is the unique maximiser of G , and $\frac{3}{10} < c < \frac{2}{5}$.

3. $G''(c) < 0$.

Proof. (1) By Theorem 3.6(2) (and $g > 0$ on $(0, \infty)$, Theorem 3.3), each G_N , $N \geq 3$, is positive and log-concave on $(0, \infty)$: $G_N(\frac{u_1+u_2}{2})^2 \geq G_N(u_1)G_N(u_2)$. By Theorem 4.2 the same inequality holds for G , and $G \geq 0$. G is real-analytic and not identically zero, since $e^{\pi^2 u/8} G(u) \rightarrow \frac{\pi}{2}$ as $u \rightarrow \infty$. Suppose $G(u_0) = 0$ for some $u_0 > 0$. Zeros of G are isolated, so there is $\delta \in (0, u_0)$ with $G(u_0 \pm \delta) \neq 0$, i.e. $G(u_0 \pm \delta) > 0$, and then $0 = G(u_0)^2 \geq G(u_0 - \delta)G(u_0 + \delta) > 0$, a contradiction. Thus $G > 0$, and a positive continuous midpoint-log-concave function is log-concave.

(2) Since $\ln G$ is concave and differentiable, G'/G is nonincreasing; therefore $\{G' = 0\}$ is an interval. As G' is real-analytic and not identically zero, this interval is empty or a single point. It is non-empty: ball arithmetic gives $G'(\frac{3}{10}) > 0 > G'(\frac{2}{5})$ (script 03; the series (8) is summed up to an odd K and the remainder is bounded by Lemma 4.1). The sign statements follow from the monotonicity of G'/G .

(3) By ball arithmetic, $G'' < 0$ on the enclosure of c in Theorem 4.6 (script 03). $\square$

Remark 4.4 (A second proof of log-concavity). Part (1) can also be obtained without the discrete TP_2 argument: by Theorem 2.3, G_N is the density of a finite sum of independent exponential variables, exponential densities are log-concave, and the convolution of log-concave densities is log-concave (in dimension one due to I. J. Schoenberg (1951); see A. Saumard and J. A. Wellner, *Log-concavity and strong log-concavity: a review*, Statist. Surveys **8** (2014) 45–114, Proposition 3.5 in the numbering of arXiv:1404.5886v1, where the result is derived from Prékopa’s theorem on marginals of log-concave functions and its history is given). We do not use this route; the proof given above is self-contained.

Remark 4.5 (Probabilistic meaning; why $c \approx \frac{1}{3}$). $\int_0^\infty e^{-su} G(u) du = 1/\cosh \sqrt{2s} = \prod_{k \text{ odd}} (1 + s/(2y_k^2))^{-1}$: G is the density of the first time a standard Brownian motion started at 0 leaves $(-1, 1)$ (equivalently, reflected Brownian motion reaches 1), which is the law of $\sum_{k \text{ odd}} \xi_k/(2y_k^2)$ with i.i.d. standard exponential ξ_k (the limit of Theorem 2.3). By Poisson summation (method of images) $G(u) = \sum_{n \in \mathbb{Z}} (-1)^n \frac{2n+1}{\sqrt{2\pi u^3}} e^{-(2n+1)^2/(2u)}$; the two leading images $n \in \{0, -1\}$ give the Lévy density $2(2\pi u^3)^{-1/2} e^{-1/(2u)}$, whose maximiser is exactly $\frac{1}{3}$, and the next images are of relative size $3e^{-4/u} \approx 2 \times 10^{-5}$ at $u = \frac{1}{3}$. This is why $c = \frac{1}{3} - 4.9068 \dots \times 10^{-5}$. (Both identities were checked numerically, script 09; they are not used in the proofs.)

Theorem 4.6 (Enclosure of the constants; CA). *With*

$$c_0 = 0.33328426549494789874852691654344242109703655907113188526100593708624 \dots$$

(the 80-digit decimal stored in `data/03_constants.json`) we have $|c - c_0| < 10^{-70}$, and

$$\begin{aligned} G(c) &= 0.92506494218414348715254834 \dots, & G''(c) &= -12.514127592957827401576602 \dots, \\ G'''(c) &= 149.52919423275263698621782 \dots, & G''''(c) &= -1523.6016850748786418965293 \dots, \end{aligned}$$

$$\begin{aligned} \kappa &:= \frac{3}{4} + \frac{c G'''(c)}{6 G''(c)} = 0.086273779378239017181791 \dots, \\ \rho &:= -\frac{c G'''(c)}{2 G''(c)} = 1.9911786618652829484546 \dots, \end{aligned}$$

all digits shown being correct; note $\kappa = \frac{3}{4} - \frac{\rho}{3}$.

Proof. Script `03_constant_enclosure.py` certifies in 400-bit ball arithmetic (Arb) that $G'(c_0 - 10^{-70}) > 0 > G'(c_0 + 10^{-70})$; by Theorem 4.3(2) the unique zero c of G' lies between. The other quantities are evaluated on the ball $[c_0 \pm 10^{-70}]$ with rigorous tails (Lemma 4.1); their radii are below 10^{-56} . $\square$

5 The lattice expansion and the continuous-time mode

5.1 Elementary bounds

Lemma 5.1. *Let S, C, β, A_j be as in (10); primes denote $d/d\zeta$.*

1. For $\zeta \in [0, \pi^2]$: $0 \leq S \leq 1$, $-\frac{1}{6} \leq S' \leq 0$, $0 \leq S'' \leq \frac{1}{60}$.
2. For $\zeta \in [0, \pi^2/4]$: $S \geq \frac{2}{\pi}$, $0 \leq C \leq 1$, $-1 \leq C' \leq 0$, $0 \leq C'' \leq \frac{2}{3}$, $-\frac{1}{3} \leq \beta' \leq 0$, $0 \leq \beta'' \leq \frac{4}{45}$.
3. For $\zeta \in [0, \pi^2/4]$ and $j \in \{0, 1\}$: $0 \leq A_j \leq 1$, $-\alpha_{1,j} \leq A'_j \leq 0$, $0 \leq A''_j \leq \alpha_{2,j}$, where $\alpha_{1,j} = \frac{2j+7}{6}$, $\alpha_{2,0} = \frac{61}{60}$, $\alpha_{2,1} = \frac{113}{60}$; moreover $A_j(0) = 1$, $A'_j(0) = -\frac{2j+7}{6}$, $\beta(0) = 1$, $\beta'(0) = -\frac{1}{3}$.

Proof. (1) With $x = \sqrt{\zeta} \in [0, \pi]$ and $d/d\zeta = (2x)^{-1}d/dx$,

$$S' = -\frac{\varphi(x)}{2x^3}, \quad \varphi(x) := \sin x - x \cos x; \quad S'' = \frac{\psi(x)}{4x^5}, \quad \psi(x) := 3 \sin x - 3x \cos x - x^2 \sin x.$$

Now $\varphi(0) = 0$, $\varphi'(x) = x \sin x \in [0, x^2]$, so $0 \leq \varphi \leq x^3/3$; and $\psi(0) = 0$, $\psi'(x) = x\varphi(x) \in [0, x^4/3]$, so $0 \leq \psi \leq x^5/15$. This gives the bounds on S', S'' ; $0 \leq S \leq 1$ is clear.

(2) S is nonincreasing by (1), so $S \geq S(\pi^2/4) = 2/\pi$. Since $C(\zeta) = \frac{1}{2}(1 + \cos(2\sqrt{\zeta}))$ and $\frac{d}{d\eta} \cos \sqrt{\eta} = -\frac{1}{2}S(\eta)$, we get $C'(\zeta) = -S(4\zeta)$ and $C''(\zeta) = -4S'(4\zeta)$ with $4\zeta \in [0, \pi^2]$; apply (1). Finally $\beta' = 2SS'$ and $\beta'' = 2S'^2 + 2SS'' \leq \frac{2}{36} + \frac{2}{60} = \frac{4}{45}$.

(3) $A'_j = (2j+1)S^{2j}S'C + S^{2j+1}C' \in [-\frac{2j+1}{6} - 1, 0]$ and

$$A''_j = (2j+1)(2j)S^{2j-1}S''C + (2j+1)S^{2j}S''C + 2(2j+1)S^{2j}S'C' + S^{2j+1}C'',$$

a sum of four nonnegative terms bounded by $\frac{(2j+1)2j}{36} + \frac{2j+1}{60} + \frac{2j+1}{3} + \frac{2}{3}$. The values at 0 follow from $S = 1 - \frac{\zeta}{6} + \dots$, $C = 1 - \zeta + \dots$. $\square$

For the discrete-time walk we need $\Lambda(w) := -\ln(1-w)/w = \sum_{i \geq 0} w^i/(i+1)$, $0 \leq w < 1$ ($\Lambda(0) = 1$), and

$$B(\zeta; q) := \beta(\zeta) \Lambda(w(\zeta)), \quad w(\zeta) := 2q \zeta \beta(\zeta) = 2q(1 - C(\zeta)) = 2q \sin^2 \sqrt{\zeta}, \quad (12)$$

so that $\lambda = 1 - w$ satisfies $\lambda^{t-1} = \exp(-(t-1)w\Lambda(w))$ whenever $w < 1$. In continuous time we set $B := \beta$ (formally $q = 0$).

Lemma 5.2. *Let $0 < q \leq \bar{q} \leq 1$, $\bar{w} := \bar{q}/2$ and $0 \leq \zeta \leq \pi^2/36$ (the low range). Then $0 \leq w \leq \bar{w}$, $B \geq \beta \geq \beta_* := 9/\pi^2$, $B(0) = 1$, $B'(0) = q - \frac{1}{3}$, and*

$$|B'| \leq b_1 := \max\{\frac{1}{3}\Lambda(\bar{w}), 2\bar{q}\Lambda'(\bar{w})\}, \quad |B''| \leq b_2 := \max\{\frac{4}{45}\Lambda(\bar{w}) + 4\bar{q}^2\Lambda''(\bar{w}), \frac{8}{3}\bar{q}\Lambda'(\bar{w})\},$$

where $\Lambda'(w) = (\frac{w}{1-w} + \ln(1-w))/w^2$ and $\Lambda''(w) = \frac{1}{w(1-w)^2} - \frac{2}{w^2(1-w)} - \frac{2\ln(1-w)}{w^3}$. For $\bar{q} = \frac{4}{5}$: $b_1 < 1.559$, $b_2 < 5.425$; for $\bar{q} = 1$: $b_1 < 2.455$, $b_2 < 12.49$. In continuous time ($B = \beta$) the same holds with $b_1 = \frac{1}{3}$, $b_2 = \frac{4}{45}$ and $B'(0) = -\frac{1}{3}$. In all cases $|B'(0)| \leq b_0 := \max\{\frac{1}{3}, \bar{q} - \frac{1}{3}\}$.

Proof. On the low range $\sin^2 \sqrt{\zeta} \leq \frac{1}{4}$, so $w \leq q/2 \leq \bar{w} \leq \frac{1}{2}$; β is nonincreasing, so $\beta \geq \beta(\pi^2/36) = (\frac{1/2}{\pi/6})^2 = 9/\pi^2$; $\Lambda \geq 1$. The power series of $\Lambda, \Lambda', \Lambda''$ have nonnegative coefficients, hence these functions are nonnegative and nondecreasing on $[0, \bar{w}]$, with the stated closed forms. By Lemma 5.1, $w' = -2qC' \in [0, 2q]$ and $w'' = -2qC'' \in [-\frac{4}{3}q, 0]$. Then

$$\begin{aligned} B' &= \beta' \Lambda + \beta \Lambda' w' \in [-\frac{1}{3}\Lambda(\bar{w}), 2\bar{q}\Lambda'(\bar{w})], \\ B'' &= \underbrace{\beta'' \Lambda}_{\in [0, \frac{4}{45}\Lambda(\bar{w})]} + \underbrace{2\beta' \Lambda' w'}_{\in [-\frac{4}{3}\bar{q}\Lambda'(\bar{w}), 0]} + \underbrace{\beta \Lambda'' w'^2}_{\in [0, 4\bar{q}^2\Lambda''(\bar{w})]} + \underbrace{\beta \Lambda' w''}_{\in [-\frac{4}{3}\bar{q}\Lambda'(\bar{w}), 0]}. \end{aligned}$$

At $\zeta = 0$: $\Lambda(0) = 1$, $\Lambda'(0) = \frac{1}{2}$, $w'(0) = 2q$, so $B'(0) = -\frac{1}{3} + q$. $\square$

Lemma 5.3. *For $j \in \{0, 1\}$, $y > 0$, $v > 0$ put $a := 2vy^2$ and $F_j(\zeta) := A_j(\zeta) e^{-aB(\zeta; q)}$. Then $F_j(0) = e^{-a}$, $F'_j(0) = (-\frac{2j+1}{6} + a(\frac{1}{3} - q))e^{-a}$ and, on the low range,*

$$F''_j = [A''_j - 2aA'_jB' + A_j(a^2B'^2 - aB'')]e^{-aB}, \quad |F''_j| \leq [\alpha_{2,j} + 2a\alpha_{1,j}b_1 + a^2b_1^2 + ab_2]e^{-a\beta_*}.$$

Proof. Differentiate twice and use Lemmas 5.1 and 5.2. $\square$

5.2 The expansion

For the discrete-time walk from the reflecting end and integers $t \geq 1$ define, with $k = 2m - 1$ and $\lambda_m = 1 - 2q \sin^2 x_k$,

$$\mathfrak{S}_j(t) := \sum_{m=1}^{N-1} \sigma_k 2y_k (-2y_k^2)^j A_j(\zeta_k) \lambda_m^{t-1}, \quad \frac{L^2}{q} f(t) = \mathfrak{S}_0(t), \quad \frac{L^4}{q^2} (f(t+1) - f(t)) = \mathfrak{S}_1(t). \quad (13)$$

(The two identities follow from (2) exactly as (9) followed from (3), using $\lambda_m - 1 = -\frac{q}{L^2} 2y_k^2 \beta(\zeta_k)$.) In continuous time we write $\mathfrak{S}_j := G_N^{(j)}(v)$. Put

$$H_j^{(q)}(v) := \frac{7+2j}{12} G^{(j+1)}(v) + \frac{v}{6} (1-3q) G^{(j+2)}(v) \quad (q = 0 \text{ in continuous time}). \quad (14)$$

Proposition 5.4 (Lattice expansion with certified remainder). *Let $j \in \{0, 1\}$, $\bar{q} \in [0, 1]$, $0 < v_- \leq v_+$, and put*

$$\chi(q) := \frac{1}{2} \quad (0 \leq q \leq \frac{4}{5}), \quad \chi(q) := \min\{\frac{1}{2}, -\ln(2q-1)/q\} \quad (\frac{4}{5} < q < 1), \quad (15)$$

a nonincreasing function of q . Let $L_0 \geq 3$ with $L_0^2 > (2j+6)/(2\chi(\bar{q})v_-)$, and let k_0 be odd with $k_0 \leq \frac{2}{3}L_0$. Let N be such that $L = N - \frac{1}{2} \geq L_0$, and

- *either (continuous time, $\bar{q} = 0$) $v \in [v_-, v_+]$ and $\mathfrak{S}_j = G_N^{(j)}(v)$,*
- *or (discrete time) $q \in (0, \bar{q}]$, $t \geq 1$ an integer with $v := q(t-1)/L^2 \in [v_-, v_+]$, and $\mathfrak{S}_j = \mathfrak{S}_j(t)$.*

Then

$$\mathfrak{S}_j = G^{(j)}(v) + \frac{H_j^{(q)}(v)}{L^2} + \frac{R}{L^4}, \quad R = \sum_{\substack{k \leq k_0 \\ k \text{ odd}}} \sigma_k (-2)^j y_k^{2j+5} F_j''(\xi_k; y_k, v, q) + R', \quad (16)$$

for some $\xi_k \in (0, y_k^2/L^2) \subset [0, y_k^2/L_0^2]$, where $|R'| \leq B_{\text{low}} + B_{\text{high}} + B_{\text{tail}}$,

$$\begin{aligned} B_{\text{low}} &:= \sum_{k > k_0, k \text{ odd}} 2^j y_k^{2j+5} e^{-2v-y_k^2 \beta_*} [\alpha_{2,j} + 2a_k \alpha_{1,j} b_1 + a_k^2 b_1^2 + a_k b_2], \quad a_k := 2v + y_k^2, \\ B_{\text{high}} &:= 2^{j+1} \left(\frac{\pi}{2}\right)^{2j+1} L_0^{2j+6} e^{-\chi(\bar{q})v-L_0^2}, \\ B_{\text{tail}} &:= \sum_{k > \frac{2}{3}L_0, k \text{ odd}} 2^{j+1} y_k^{2j+5} e^{-2v-y_k^2} \left[\left(\frac{6}{\pi}\right)^4 + \left(\frac{6}{\pi}\right)^2 \left(\frac{2j+7}{6} + 2v + b_0 y_k^2\right) \right], \end{aligned}$$

with the constants of Lemmas 5.1–5.2 for $\bar{q}$. For an individual pair (N, q) one may replace B_{high} by $B_{\text{high}}(L, q) := 2^{j+1} \left(\frac{\pi}{2}\right)^{2j+1} L^{2j+6} e^{-\chi(q)v-L^2}$ (no condition linking L_0 and χ is then needed).

Proof. Call a lattice mode $k = 2m - 1$ *low* if $x_k \leq \pi/6$, i.e. $k \leq \frac{2}{3}L$, and *high* otherwise; all $k \leq k_0$ are low.

Low modes. Here $\zeta_k \leq \pi^2/36$ and $w = 2q \sin^2 x_k \leq q/2 < 1$, so $\lambda_m > 0$ and $\lambda_m^{t-1} = \exp(-(t-1)w\lambda(w)) = \exp(-2vy_k^2 B(\zeta_k; q))$, because $(t-1)w = \frac{vL^2}{q} 2q \sin^2 x_k = 2vy_k^2 \beta(\zeta_k)$. Thus the k -th term of $\mathfrak{S}_j$ is $\sigma_k 2(-2)^j y_k^{2j+1} F_j(\zeta_k)$ with F_j from Lemma 5.3 (also in continuous time). By Taylor's formula with Lagrange remainder, $F_j(\zeta_k) = F_j(0) + \zeta_k F_j'(0) + \frac{1}{2} \zeta_k^2 F_j''(\xi_k)$ with $\xi_k \in (0, \zeta_k)$, and $\zeta_k = y_k^2/L^2$.

Bookkeeping. Summed over all odd $k \geq 1$, the first two Taylor terms give

$$\sum_k \sigma_k 2(-2)^j y_k^{2j+1} e^{-2vy_k^2} = G^{(j)}(v), \quad \sum_k \sigma_k 2(-2)^j y_k^{2j+3} \left(-\frac{2j+7}{6} + 2vy_k^2 \left(\frac{1}{3} - q \right) \right) e^{-2vy_k^2} = H_j^{(q)}(v),$$

because $\sum_k \sigma_k 2(-2)^j y_k^{2j+3} e^{-2vy_k^2} = -\frac{1}{2}G^{(j+1)}$ and $\sum_k \sigma_k 2(-2)^j y_k^{2j+5} e^{-2vy_k^2} = \frac{1}{4}G^{(j+2)}$. Therefore

$$\begin{aligned} L^4 \left(\mathfrak{S}_j - G^{(j)} - \frac{H_j^{(q)}}{L^2} \right) &= \sum_{k \text{ low}} \sigma_k (-2)^j y_k^{2j+5} F_j''(\xi_k) + L^4 \sum_{k \text{ high}} (k\text{-th term of } \mathfrak{S}_j) \\ &\quad - L^4 \sum_{k > \frac{2}{3}L, k \text{ odd}} \sigma_k 2(-2)^j y_k^{2j+1} (F_j(0) + \frac{y_k^2}{L^2} F_j'(0)). \end{aligned}$$

The low modes $k \leq k_0$ give the explicit sum in (16). The low modes $k > k_0$ are bounded by Lemma 5.3 with $a = 2vy_k^2 \leq a_k$ and $e^{-a\beta_*} \leq e^{-2v-y_k^2\beta_*}$; summing over all odd $k > k_0$ gives B_{low} .

High modes. Here $\frac{1}{4} < \sin^2 x_k < 1$ and $|A_j| \leq 1$. In continuous time the exponential factor is $\exp(-2vL^2 \sin^2 x_k) \leq e^{-vL^2/2}$. In discrete time $\lambda_m = 1 - w$ with $w \in (\frac{q}{2}, 2q)$, hence $|\lambda_m| \leq \max\{1 - \frac{q}{2}, 2q - 1\}$. Now $(1 - \frac{q}{2})^{t-1} \leq e^{-q(t-1)/2} = e^{-vL^2/2}$ and, for $q > \frac{1}{2}$, $(2q - 1)^{t-1} = \exp(-vL^2 \cdot (-\ln(2q - 1))/q)$; moreover $2q - 1 \leq 1 - \frac{q}{2}$ iff $q \leq \frac{4}{5}$. Hence $|\lambda_m|^{t-1} \leq e^{-\chi(q)vL^2} \leq e^{-\chi(\bar{q})v-L^2}$ in all cases (in continuous time with $\chi = \frac{1}{2}$). There are fewer than L high modes and $y_k < \pi L/2$, so the high modes contribute at most $2^{j+1}(\pi/2)^{2j+1} L^{2j+6} e^{-\chi v-L^2}$ to $|R'|$; the function $L \mapsto L^{2j+6} e^{-\chi v-L^2}$ is decreasing for $L^2 \geq (2j+6)/(2\chi v_-)$, which gives B_{high} .

Continuum tail. For odd $k > \frac{2}{3}L$ we have $L < 6y_k/\pi$, $|F_j(0)| = e^{-a}$ and $|F_j'(0)| \leq (\frac{2j+7}{6} + ab_0)e^{-a}$ with $a = 2vy_k^2$; hence the last sum is bounded by B_{tail} (the index set $\{k > \frac{2}{3}L\}$ is contained in $\{k > \frac{2}{3}L_0\}$). $\square$

Remark 5.5 (How the proposition is used). The numbers $B_{\text{low}}, B_{\text{high}}, B_{\text{tail}}$ are evaluated rigorously (finitely many terms plus Lemma 4.1); in all applications below they are smaller than 10^{-10} . The explicit sum over $k \leq k_0$ (at most 8 terms) is enclosed in ball arithmetic: F_j'' is evaluated by second-order Taylor-model arithmetic (“jets” (F, F', F'') in ζ) on the ball $Z_k = [0, y_k^2/L_0^2] \ni \xi_k$, with v and q replaced by balls. The resulting ball contains R for *all* $L \geq L_0$ simultaneously. Implementation: `scripts/core.py` (R_enclosure); sanity check against direct 50-digit evaluation of $L^4(\mathfrak{S}_j - G^{(j)} - H_j/L^2)$ for sample N, v, q : `scripts/11_expansion_sanity.py`.

Corollary 5.6 (First-order asymptotics, all derivatives used). *For fixed $u > 0$: $G_N^{(j)}(u) = G^{(j)}(u) + L^{-2}[\frac{7+2j}{12}G^{(j+1)}(u) + \frac{u}{6}G^{(j+2)}(u)] + O(L^{-4})$, $j = 0, 1$, with the explicit O -constant of Proposition 5.4.*

5.3 The mode in continuous time

For $N \geq 3$ let $t_c = t_c(N, q)$ be the unique maximiser of g (Theorem 3.3) and $u_N := qt_c/L^2$, the unique zero of G'_N ; u_N does not depend on q .

Theorem 5.7 (Continuous-time mode; analytic for $N \geq 301$, CA for $3 \leq N \leq 300$). *For every $N \geq 3$,*

$$\left| u_N - c + \frac{\kappa}{L^2} \right| < \frac{1.78}{L^4}, \quad \text{i.e.} \quad \left| t_c - \frac{cL^2 - \kappa}{q} \right| < \frac{1.78}{qL^2}, \quad (17)$$

and 1.78 can be replaced by 1.54 for $N \geq 201$. Moreover $c - 0.3614L^{-2} < u_N < c - \kappa L^{-2}$ for all $N \geq 3$. Finally the limit $\kappa_2 := \lim_{N \rightarrow \infty} L^4(u_N - c + \kappa L^{-2})$ exists and is given, with all derivatives of G taken at c , by

$$\kappa_2 = -\frac{\Gamma_0}{G''(c)}, \quad \Gamma_0 = \frac{1}{2}\kappa^2 G''' - \kappa \left(\frac{11}{12}G''' + \frac{c}{6}G'''' \right) + \frac{113}{480}G''' + \frac{49c}{360}G'''' + \frac{c^2}{72}G^{(5)}; \quad (18)$$

numerically $\Gamma_0 = -19.1577530390322526\dots$ and $\kappa_2 = -1.5308900198375030446\dots$ (all digits shown correct).

Proof. Analytic part. Fix N_0 , $L_0 = N_0 - \frac{1}{2}$, $\varepsilon_0 = L_0^{-2}$, $K > 0$, and let $L \geq L_0$, $\varepsilon = L^{-2}$, $s \in [-K, K]$,

$$u = c - \kappa\varepsilon + s\varepsilon^2 = c - \varepsilon d, \quad d := \kappa - s\varepsilon.$$

By Proposition 5.4 ($j = 1$, continuous time), $G'_N(u) = G'(u) + \varepsilon H_1(u) + \varepsilon^2 R(u)$ with $H_1 = \frac{3}{4}G'' + \frac{u}{6}G'''$. Taylor's formula at c (with $G'(c) = 0$) gives

$$G'(u) = -\varepsilon d G''(c) + \frac{1}{2}\varepsilon^2 d^2 G'''(c) - \frac{1}{6}\varepsilon^3 d^3 G''''(\xi), \quad H_1(u) = H_1(c) - \varepsilon d H'_1(c) + \frac{1}{2}\varepsilon^2 d^2 H''_1(\xi'),$$

with ξ, ξ' between u and c , $H'_1 = \frac{11}{12}G''' + \frac{u}{6}G''''$, $H''_1 = \frac{13}{12}G'''' + \frac{u}{6}G^{(5)}$. Since $H_1(c) = G''(c)(\frac{3}{4} + \frac{c}{6} \frac{G'''(c)}{G''(c)}) = \kappa G''(c)$, the terms of order ε combine to $\varepsilon G''(c)(\kappa - d) = \varepsilon^2 s G''(c)$, and

$$L^4 G'_N(u) = s G''(c) + \Gamma, \quad \Gamma := \frac{1}{2}G'''(c)d^2 - H'_1(c)d + \varepsilon \left[-\frac{1}{6}G''''(\xi)d^3 + \frac{1}{2}H''_1(\xi')d^2 \right] + R(u). \quad (19)$$

Script `12_cont_theorem.py` encloses Γ in ball arithmetic simultaneously for all $\varepsilon \in [0, \varepsilon_0]$, $s \in [-K, K]$ (so d, u, ξ, ξ' range over explicit balls, and R is enclosed as in Remark 5.5) and verifies $\sup |\Gamma| < K|G''(c)|$. Then $G'_N(c - \kappa\varepsilon - K\varepsilon^2) > 0 > G'_N(c - \kappa\varepsilon + K\varepsilon^2)$, and since G'_N has a single zero, positive before and negative after (Theorem 3.3), $|u_N - c + \kappa\varepsilon| < K\varepsilon^2$. The certified values are

N_0	k_0	enclosure of Γ	$B_{\text{low}} + B_{\text{high}} + B_{\text{tail}}$	certified K
21	9	$[-24.56, -13.69]$	$< 5 \times 10^{-11}$	1.975
51	11	$[-20.01, -18.29]$	$< 5 \times 10^{-18}$	1.603
101	13	$[-19.38, -18.94]$	$< 2 \times 10^{-26}$	1.556
201	13	$[-19.22, -19.10]$	$< 2 \times 10^{-26}$	1.54
301	15	$[-19.19, -19.13]$	$< 3 \times 10^{-36}$	1.54
2001	15	$[-19.1584, -19.1571]$	$< 3 \times 10^{-36}$	1.54

(file `data/12_cont_theorem.json`; the Γ column is rounded outward to the digits shown). In every row $\Gamma < 0$, so the zero $s^* = \Gamma/|G''(c)|$ of $sG''(c) + \Gamma$ is negative: $u_N < c - \kappa\varepsilon$. For every $N \geq N_0$ the number $s_N^* := L^4(u_N - c + \kappa L^{-2})$ is such a zero, so it lies in the certified enclosure of $-\Gamma/G''(c)$; for $N_0 = 2001$ the script gives the enclosure $[-1.5309330, -1.5308464]$ (computed directly, not from the rounded table).

Existence and value of κ_2 . Put $\Gamma_N(s) := L^4 G'_N(c - \kappa\varepsilon + s\varepsilon^2) - sG''(c)$, a well-defined function of s for each N ($\varepsilon = L^{-2}$). Since $G'_N(u_N) = 0$ and $|s_N^*| < 1.78$ for all $N \geq 3$ (first part of the proof), $s_N^* = -\Gamma_N(s_N^*)/G''(c)$. Hence it suffices to show that $\Gamma_N(s) \rightarrow \Gamma_0$ as $N \rightarrow \infty$, uniformly for $|s| \leq K := 1.78$; then $|s_N^* - (-\Gamma_0/G''(c))| \leq \sup_{|s| \leq K} |\Gamma_N(s) - \Gamma_0|/|G''(c)| \rightarrow 0$. Write $u = c - \varepsilon d$, $d = \kappa - s\varepsilon$, and decompose, by Proposition 5.4 ($j = 1$, continuous time, $H_1 = \frac{3}{4}G'' + \frac{u}{6}G'''$),

$$\Gamma_N(s) = \underbrace{\varepsilon^{-2}G'(u) + \varepsilon^{-1}H_1(u) - sG''(c)}_{=: A_N(s)} + R_N(u), \quad R_N(u) := L^4(G'_N(u) - G'(u) - \varepsilon H_1(u)).$$

(i) By Taylor's formula at c , exactly as in (19), $A_N(s) = \frac{1}{2}G'''(c)d^2 - H'_1(c)d + \varepsilon[-\frac{1}{6}G''''(\xi)d^3 + \frac{1}{2}H''_1(\xi')d^2]$ with $\xi, \xi' \in [c - \varepsilon(\kappa + K\varepsilon), c]$; as $d \rightarrow \kappa$ uniformly and G'''' , H''_1 are bounded near c , $A_N(s) \rightarrow \frac{1}{2}G'''(c)\kappa^2 - \kappa H'_1(c)$ uniformly, with $H'_1(c) = \frac{11}{12}G'''(c) + \frac{c}{6}G''''(c)$. (ii) In the proof of Proposition 5.4 write the Taylor remainder of each low mode in integral form, $F_1(\zeta_k) - F_1(0) - \zeta_k F'_1(0) = \zeta_k^2 \int_0^1 (1 - \theta) F''_1(\theta \zeta_k) d\theta$. Then

$$R_N(u) = \sum_{k \leq \frac{2}{3}L, k \text{ odd}} \sigma_k(-2)y_k^7 2 \int_0^1 (1 - \theta) F''_1(\theta \zeta_k; y_k, u) d\theta + (\text{high modes}) - (\text{continuum tail}),$$

where, as shown in that proof, $|\text{high modes}| \leq 2^2(\frac{\pi}{2})^3 L^8 e^{-uL^2/2}$ and $|\text{continuum tail}| \leq B_{\text{tail}}(L)$, the expression B_{tail} of Proposition 5.4 with L in place of L_0 . For u in $[c - \varepsilon(\kappa + K\varepsilon), c] \subset$

$[\frac{c}{2}, c]$ (all large N) both tend to 0 uniformly in $|s| \leq K$: the first because $L^8 e^{-cL^2/4} \rightarrow 0$, the second because it is a tail, over $k > \frac{2}{3}L$, of a convergent series with s -independent terms (take $v_- = \frac{c}{2}$, $v_+ = c$). In the low-mode sum, each summand is bounded, uniformly in N , $|s| \leq K$ and θ , by the k -th term of B_{low} (Lemma 5.3 with $v_- = \frac{c}{2}$, $v_+ = c$), which is summable over k ; and for each fixed k , $\theta\zeta_k \rightarrow 0$ and $u \rightarrow c$ uniformly, so by continuity of F_1'' in (ζ, u) the summand tends to $\sigma_k(-2)y_k^7 F_1''(0; y_k, c)$ uniformly. By dominated convergence (Tannery's theorem) $R_N(u) \rightarrow R_0 := \sum_{k \text{ odd}} \sigma_k(-2)y_k^7 F_1''(0; y_k, c)$ uniformly in $|s| \leq K$. (iii) Evaluation of R_0 . With $a = 2cy^2$, $A_1(0) = 1$, $A_1'(0) = -\frac{3}{2}$, $A_1''(0) = \frac{113}{60}$ (from $S^3 C = 1 - \frac{3}{2}\zeta + \frac{113}{120}\zeta^2 + \dots$), $\beta'(0) = -\frac{1}{3}$ and $\beta''(0) = \frac{4}{45}$ (from $\beta = 1 - \frac{\zeta}{3} + \frac{2}{45}\zeta^2 + \dots$), the formula of Lemma 5.3 gives $F_1''(0; y, c) = [\frac{113}{60} - \frac{49}{45}a + \frac{a^2}{9}]e^{-a}$. By (8), $\sum_k \sigma_k(-2)y_k^7 e^{-2cy_k^2} = \frac{1}{8}G'''(c)$, $\sum_k \sigma_k(-2)y_k^9 e^{-2cy_k^2} = -\frac{1}{16}G''''(c)$ and $\sum_k \sigma_k(-2)y_k^{11} e^{-2cy_k^2} = \frac{1}{32}G^{(5)}(c)$, so $R_0 = \frac{113}{480}G'''(c) + \frac{49c}{360}G''''(c) + \frac{c^2}{72}G^{(5)}(c)$. Together with (i) this is Γ_0 of (18), and $\kappa_2 = -\Gamma_0/G''(c)$. The Taylor coefficients are checked symbolically and the right-hand side of (18) is enclosed in 256-bit ball arithmetic (series tails by Lemma 4.1) by script 37; the result $\kappa_2 \in [-1.53089001983750304465868904580 \pm 6 \times 10^{-31}]$ lies inside the enclosure $[-1.5309330, -1.5308464]$ given by the row $N_0 = 2001$, as it must. As a numerical illustration (not used), $L^2(s_N^* - \kappa_2) = -4.74925\dots$ for $N = 501, \dots, 8001$.

Computer-assisted part. For each $3 \leq N \leq 300$, script 13_cont_small_N.py evaluates the finite sum (11) ($j = 1$) in 160-bit ball arithmetic and certifies $G'_N(c - \kappa L^{-2} - 1.78L^{-4}) > 0 > G'_N(c - \kappa L^{-2} + 1.78L^{-4})$, as well as $G'_N(c - 0.3614L^{-2}) > 0 > G'_N(c - \kappa L^{-2})$. For $N \geq 301$ the bound $u_N > c - 0.3614L^{-2}$ follows from (17). (The largest value of $L^4|u_N - c + \kappa L^{-2}|$ is $1.77465\dots$, attained at $N = 4$.) $\square$

Corollary 5.8 (Mode over mean in continuous time). *For $N \geq 3$, with $\mathbb{E}T = (L^2 - \frac{1}{4})/q$,*

$$\frac{t_c}{\mathbb{E}T} = \frac{u_N}{1 - \frac{1}{4L^2}} = c + \frac{\frac{c}{4} - \kappa}{L^2} + \frac{\vartheta}{L^4}, \quad \frac{c}{4} - \kappa = -0.0029527130\dots, \quad |\vartheta| \leq 1.86.$$

In particular $|t_c/\mathbb{E}T - c| \leq 0.0030L^{-2} + 1.86L^{-4}$: the ratio mode/mean converges to $c = 0.3332842654949\dots$ at rate $O(N^{-2})$ with a tiny leading coefficient.

Proof. Write $\delta = \frac{1}{4L^2} \leq \frac{1}{25}$ and $u_N = c - \kappa L^{-2} + \theta_1 L^{-4}$, $|\theta_1| < 1.78$ (Theorem 5.7). Using $(1 - \delta)^{-1} = 1 + \delta + \delta^2(1 - \delta)^{-1}$ and $c\delta = \frac{c}{4}L^{-2}$,

$$\frac{u_N}{1 - \delta} - c - \frac{c/4 - \kappa}{L^2} = \frac{c\delta^2}{1 - \delta} - \frac{\kappa}{L^2} \frac{\delta}{1 - \delta} + \frac{\theta_1}{L^4(1 - \delta)} = \frac{1}{L^4(1 - \delta)} \left[\theta_1 + \frac{c}{16} - \frac{\kappa}{4} \right],$$

and $\frac{25}{24}(1.78 + |\frac{c}{16} - \frac{\kappa}{4}|) < \frac{25}{24} \cdot 1.7808 < 1.86$. $\square$

6 The mode in discrete time for $q \leq \frac{9}{10}$

Throughout this section $x_0 = 1$, $D(t) := f(t+1) - f(t)$, and

$$\tau = \tau(N, q) := \frac{cL^2 - \kappa}{q} + 1 - \rho, \quad E_K := \frac{K}{qL^2} \quad (K > 0). \quad (20)$$

Theorem 6.1 (Sign of the increments near the mode; $N \geq 41$). *Let $N \geq 41$, $q \in (0, \frac{4}{5}]$ and $E = E_{1.68}$. Then*

(A) $D(t) > 0$ for every integer $t \in [\tau - E - 1, \tau - E]$;

(B) $D(t) < 0$ for every integer $t \in [\tau + E, \tau + E + 1]$.

The constant 1.68 can be replaced by 1.63 for $N \geq 51$, by 1.58 for $N \geq 101$ and by 1.56 for $N \geq 601$. Moreover (A) and (B) hold with $E = E_{1.61}$ for all $N \geq 61$ and all $q \in (0, \frac{9}{10}]$.

Proof. Let $L \geq L_0$, $\varepsilon = L^{-2} \leq \varepsilon_0 = L_0^{-2}$. For an integer t write $t = \tau + \sigma$, $s := q\sigma L^2$ and $v := q(t-1)/L^2$. By (20), $q(\tau-1) = cL^2 - \kappa - q\rho$, so

$$v = c - \varepsilon d, \quad d := \kappa + q\rho - s\varepsilon.$$

By (13) and Proposition 5.4 ($j = 1$, discrete time, $\bar{q} = \frac{4}{5}$), $\frac{L^4}{q^2}D(t) = \mathfrak{S}_1(t) = G'(v) + \varepsilon H(v) + \varepsilon^2 R$ with $H = H_1^{(q)} = \frac{3}{4}G'' + \frac{v}{6}(1-3q)G'''$. Exactly as in the proof of Theorem 5.7, using now $H(c) = G''(c)(\frac{3}{4} + \frac{\varepsilon}{6}(1-3q)\frac{G'''(c)}{G''(c)}) = (\kappa + q\rho)G''(c)$,

$$\frac{L^8}{q^2}D(t) = \underbrace{sG''(c) + \frac{1}{2}G'''(c)d^2 - H'(c)d}_{=: M(s)} + \underbrace{\varepsilon[-\frac{1}{6}G''''(\xi)d^3 + \frac{1}{2}H''(\xi')d^2]}_{=: \Xi} + R, \quad (21)$$

with ξ, ξ' between v and c , $H' = \frac{3}{4}G''' + \frac{1-3q}{6}G'''' + \frac{v}{6}(1-3q)G''''$ and $H'' = \frac{3}{4}G'''' + \frac{1-3q}{3}G'''' + \frac{v}{6}(1-3q)G^{(5)}$. On the window (A), $\sigma \in [-E-1, -E]$, i.e. $s \in [-K - qL^2, -K]$ and $d \in \kappa + q\rho + [K\varepsilon, K\varepsilon + q]$; on (B), $s \in [K, K + qL^2]$ and $d \in \kappa + q\rho - [K\varepsilon, K\varepsilon + q]$. Since $dd/ds = -\varepsilon$,

$M'(s) = G''(c) - \varepsilon(G'''(c)d - H'(c)) < 0$ if $|G''(c)| > \varepsilon_0 \sup |G'''(c)d - H'(c)|$ (“slope condition”),

so $M(s) \geq M(-K)$ on (A) and $M(s) \leq M(K)$ on (B). The certificate splits $(0, \frac{4}{5}]$ into 2400 intervals $[q_i, q_{i+1}]$ and, for each of them, encloses in ball arithmetic, simultaneously for all $\varepsilon \in [0, \varepsilon_0]$, all $q \in [q_i, q_{i+1}]$ and all d in $\kappa + q\rho + [-(q + K\varepsilon_0), q + K\varepsilon_0]$ (hence all v, ξ, ξ' in the corresponding ball V): the quantity Ξ (with R enclosed as in Remark 5.5, k_0 as in the table), the slope condition, and

$$M(-K) + \Xi > 0, \quad M(K) + \Xi < 0.$$

This is done in two steps. Script `14_disc_theorem.py` determines, for each interval, the smallest constant K_i that it can certify; the maximum of the K_i over the intervals is listed in the table below (it is attained as $q \rightarrow 0$; the minimum is ≈ 0.27 , near $q = 0.45$). A certificate at K_i covers the windows (A), (B) at K_i , which are not the windows at a larger K ; therefore script `35_disc_uniformK.py` repeats the certification on every interval with the single constant K of the statement (1.68, 1.63, 1.58, 1.56 for $N_0 = 41, 51, 101, 601$): no interval fails. The constants are (files `data/14_disc_N0_*.json` and `data/35_disc_uniformK.json`):

N_0	k_0	$\max_i K_i, q \in (0, \frac{4}{5}]$	$\max_i K_i, q \in (0, \frac{2}{3}]$	uniform K certified on all intervals
41	11	1.6733	1.6590	1.68
51	11	1.6222	1.6222	1.63
101	13	1.5732	1.5732	1.58
601	15	1.5572	1.5572	1.56

This proves (A) and (B) for $q \leq \frac{4}{5}$. For $q \leq \frac{9}{10}$ the same two scripts are run with $\bar{q} = \frac{9}{10}$ (Proposition 5.4 with $\chi(\frac{9}{10}) = -\frac{10}{9} \ln \frac{4}{5} = 0.2479 \dots$) and 2700 intervals in q : $\max_i K_i = 1.6023$ for $N_0 = 61$ ($k_0 = 15$), 1.5732 for $N_0 = 101$, 1.5572 for $N_0 = 601$ (files `data/14_disc_N0_*.q9_10.json`), and the uniform constants 1.61, 1.58, 1.56 are certified on every interval (script 35). $\square$

Proposition 6.2 (Small N ; CA). *Let $E = E_{1.78}$, $t_- := \lfloor \tau - E \rfloor$, $t_+ := \lceil \tau + E \rceil$, and let (N, q) satisfy one of*

- (a) $3 \leq N \leq 40$ and $q \in (0, \frac{4}{5}]$;
- (b) $7 \leq N \leq 9$ and $q \in [\frac{4}{5}, \frac{17}{20}]$;
- (c) $10 \leq N \leq 60$ and $q \in [\frac{4}{5}, \frac{9}{10}]$.

Then $t_+ \geq N - 1$, $D(t_+) < 0$, and either $t_- \leq N - 2$ or $D(t_-) > 0$, with the single exception $N = 4$, $q = \frac{4}{5}$, $t_- = 3$, where $D(3) = 0$.

Proof. Script `15_disc_small_N.py`, 128-bit ball arithmetic.

Part 1: $0 < q \leq \frac{1}{2}$. All λ_m are positive, and by the proof of Theorem 3.5 the real-variable function $\Psi(t) = \Phi(t+1) - \Phi(t)$ (which equals a positive multiple of $D(t)$ at integers) has exactly one zero t_0 in $(N-2, \infty)$, is positive on $[N-2, t_0)$ and negative on (t_0, ∞) . In the scaled variable $v = q(t-1)/L^2$ (now t real) we have $\frac{L^4}{q^2} \cdot \frac{2q}{2N-1} \Psi(t) = \mathfrak{D}(v; q)$, $\mathfrak{D}(v; q) := \sum_k \sigma_k 2y_k (-2y_k^2) A_1(\zeta_k) \exp(-2vy_k^2 \beta(\zeta_k) \Lambda(2q \sin^2 x_k))$ (the sum over all lattice modes $k = 2m-1$, $m \leq N-1$; $2q \sin^2 x_k < 1$). The script certifies, by adaptive bisection of $q \in [0, \frac{1}{2}]$ into balls, that $\mathfrak{D}(v_+(q); q) < 0$ and $\tau + E > N-2$, and that $\mathfrak{D}(v_-(q); q) > 0$ or $\tau - E \leq N-2$, where $v_{\mp}(q) = c - (\kappa + q\rho)L^{-2} \mp 1.78L^{-4}$ are the values of v at $t = \tau \mp E$ (Λ is enclosed on a ball $[w_1, w_2]$ by $[\Lambda(w_1), \Lambda(w_2)]$, monotonicity). Hence $\tau - E < t_0 < \tau + E$, so $D(t_+) < 0$ and, if $t_- \geq N-1$, $D(t_-) > 0$.

Part 2: $\frac{1}{2} \leq q \leq \frac{4}{5}$. Put $q_t^{\mp} := (cL^2 - \kappa \mp 1.78L^{-2})/(t-1+\rho)$; then $t_- = t \iff q \in (q_{t+1}^-, q_t^-]$ and $t_+ = t \iff q \in [q_t^+, q_{t-1}^+)$. For every integer $t \geq N-1$ for which the interval $I = [q_{t+1}^-, q_t^-] \cap [\frac{1}{2}, \frac{4}{5}]$ is non-empty the script certifies $\partial_q \mathfrak{S}_1(t) < 0$ on I (one ball evaluation of $\sum_k c_k (t-1) \lambda_k^{t-2} (-2 \sin^2 x_k)$) and $\mathfrak{S}_1(t) > 0$ at the right end point of I ; similarly on $I = [q_t^+, q_{t-1}^+] \cap [\frac{1}{2}, \frac{4}{5}]$, $\partial_q \mathfrak{S}_1(t) < 0$ and $\mathfrak{S}_1(t) < 0$ at the left end point (no such t is smaller than $N-1$). The only case not of this form is $t = N-1$ on windows of type t_- , where $D(N-1) = (q/2)^{N-1}(h_1-1)$ has the sign of $1 - \frac{1}{2N-3} - q$ (Proposition 3.9(2)); for $3 \leq N \leq 40$ this is positive on the relevant q -interval except for $N=4$, where $D(3) \geq 0$ with equality exactly at $q = \frac{4}{5}$. All 38 values of N pass (`data/15_disc_small_N.jsonl`; 782 windows for $N=40$).

Part 3: cases (b), (c). Script `32_disc_small_N_ext.py` applies the method of Part 2 with $[\frac{1}{2}, \frac{4}{5}]$ replaced by $[\frac{4}{5}, \frac{17}{20}]$ ($N=7, 8, 9$), resp. $[\frac{4}{5}, \frac{9}{10}]$ ($10 \leq N \leq 60$); for integer t the formula $\mathfrak{S}_1(t) = \sum_k c_k \lambda_k^{t-1}$ is valid for eigenvalues of either sign. All 54 cases pass (6752 windows; `data/32_disc_small_N_ext.jsonl`). The restrictions on N cannot simply be dropped: the same computation fails for $N=6$ (at $t=11$, q slightly below 0.829) and for $N=9$ (at $t=26$, q slightly below 0.888), where the mode leaves the window (22); e.g. for $N=6$, $q=0.828$ the mode is $t^* = 11 < \tau - E = 11.0097$, and for $N=9$, $q=0.887$ it is $t^* = 26 < \tau - E = 26.031$ (exact rational computation). $\square$

Theorem 6.3 (Mode in discrete time). *Let (N, q) satisfy one of*

- (a) $N \geq 3$ and $q \in (0, \frac{4}{5}]$;
- (b) $N \geq 7$ and $q \in (0, \frac{17}{20}]$;
- (c) $N \geq 10$ and $q \in (0, \frac{9}{10}]$.

Then f is unimodal and every maximiser t^ of f satisfies*

$$\tau - E < t^* < \tau + 1 + E, \quad \tau = \frac{cL^2 - \kappa}{q} + 1 - \rho, \quad E = \frac{1.78}{qL^2} \quad (22)$$

(with 1.68 instead of 1.78 for $N \geq 41$, $q \leq \frac{4}{5}$, and with 1.61 for $N \geq 61$). In particular there are at most $\lfloor 1 + 2E \rfloor + 1$ maximisers, and exactly one or two adjacent ones as soon as $qL^2 > 3.56$.

Proof. f is unimodal by Theorem 3.15(3) in case (a) and by Theorem 3.16(3) in cases (b), (c). Let $K := 1.61$ if $N \geq 61$; $K := 1.68$ if $41 \leq N \leq 60$ and $q \leq \frac{4}{5}$; $K := 1.78$ otherwise; and in this proof let $E := E_K \leq E_{1.78}$, $t_- := \lfloor \tau - E \rfloor$, $t_+ := \lceil \tau + E \rceil$. It suffices to prove $\tau - E < t^* < \tau + 1 + E$ for this E . We have $D(t_+) < 0$, and $t_- \leq N-2$ or $D(t_-) > 0$ (apart from the exception $N=4$, $q = \frac{4}{5}$): for $N \geq 61$, and for $41 \leq N \leq 60$ with $q \leq \frac{4}{5}$, by Theorem 6.1, because t_- is an integer of the window (A) and t_+ an integer of the window (B); for $41 \leq N \leq 60$ with $q > \frac{4}{5}$ by Proposition 6.2(c); for $N \leq 40$ by Proposition 6.2. Now $D(t_+) < 0$ means $f(t_+ + 1) < f(t_+)$; if some maximiser satisfied $t^* \geq t_+ + 1$, unimodality would force $f(t_+) \leq f(t_+ + 1)$. Hence $t^* \leq t_+ < \tau + E + 1$. If $t_- \leq N-2$ then $t^* \geq N-1 > t_-$ trivially (f vanishes below

$N - 1$). Otherwise $D(t_-) > 0$, i.e. $f(t_- + 1) > f(t_-)$, and a maximiser $t^* \leq t_-$ would force $f(t_-) \geq f(t_- + 1)$; hence $t^* \geq t_- + 1 > \tau - E$. (The per-interval constants $K_i \leq K$ of script 14 would also suffice here: by unimodality, $D(\lfloor \tau - E_{K_i} \rfloor) > 0$ gives $t^* > \tau - E_{K_i} \geq \tau - E_K$, and $D(\lceil \tau + E_{K_i} \rceil) < 0$ gives $t^* < \tau + E_{K_i} + 1 \leq \tau + E_K + 1$.) In the exceptional case $N = 4$, $q = \frac{4}{5}$ the unique maximiser is $t^* = 5$ (Proposition 3.9(4)), while $\tau = 4.0043\dots$ and $E = 0.1816\dots$, so (22) holds. The count: the open interval (22) has length $1 + 2E$, which is smaller than 2 when $qL^2 > 3.56$, and then it contains at most two integers, necessarily adjacent. $\square$

Corollary 6.4 (Explicit finite- N bounds). *Under the hypotheses of Theorem 6.3, every mode t^* satisfies*

$$\left| t^* - \frac{cL^2 - \kappa}{q} + \rho - \frac{3}{2} \right| < \frac{1}{2} + \frac{1.78}{qL^2}, \quad (23)$$

$$-\frac{\kappa + (\rho - 1)q}{L^2} - \frac{1.78}{L^4} < \frac{qt^*}{L^2} - c < -\frac{\kappa - (2 - \rho)q}{L^2} + \frac{1.78}{L^4}, \quad (24)$$

with $\kappa + (\rho - 1)q = 0.08627\dots + 0.99117\dots q$ and $\kappa - (2 - \rho)q = 0.08627\dots - 0.00882\dots q > 0$. Consequently, for all $N \geq 3$,

$$\left| \frac{qt^*}{(N - \frac{1}{2})^2} - c \right| < \frac{0.0863 + 0.9912q}{(N - \frac{1}{2})^2} + \frac{1.78}{(N - \frac{1}{2})^4} \leq \frac{0.372 + 0.992q}{(N - \frac{1}{2})^2} \leq \frac{1.44(0.372 + 0.992q)}{N^2}, \quad (25)$$

$qt^*/L^2 < c$ as soon as $N \geq 6$, and for the ratio mode/mean, with $\mathbb{E}T = N(N - 1)/q = (L^2 - \frac{1}{4})/q$,

$$\frac{t^*}{\mathbb{E}T} = c + \frac{\frac{c}{4} - \kappa - q(\rho - \frac{3}{2}) + \frac{q}{2}\vartheta_1}{L^2} + \frac{\vartheta_2}{L^4}, \quad |\vartheta_1| < 1, \quad |\vartheta_2| \leq 2.2, \quad (26)$$

where $\frac{c}{4} - \kappa = -0.00295\dots$ and $\rho - \frac{3}{2} = 0.49117\dots$. In particular $|t^*/\mathbb{E}T - c| \leq (0.003 + 0.992q)L^{-2} + 2.2L^{-4}$. The exponent 2 in (25) is optimal: by the right-hand inequality in (24), for all $N \geq 6$,

$$c - \frac{qt^*}{L^2} > \frac{\kappa - (2 - \rho)q}{L^2} - \frac{1.78}{L^4} \geq \frac{0.0774}{L^2} - \frac{1.78}{L^4} > 0.$$

(The interval (24) has length $q/L^2 + 3.56/L^4$; the term q/L^2 reflects the integrality of t^* , which (23) determines up to at most two adjacent candidates.) Finally, the mode is given by an exact formula except near ties: if the distance from τ to the nearest integer is at least $E = 1.78/(qL^2)$, then the maximiser is unique and

$$t^* = \lceil \tau \rceil = \left\lceil \frac{c(N - \frac{1}{2})^2 - \kappa}{q} + 1 - \rho \right\rceil; \quad (27)$$

in the remaining cases, if $E < \frac{1}{2}$ (i.e. $qL^2 > 3.56$), all modes lie in $\{\lceil \tau \rceil - 1, \lceil \tau \rceil\}$ or all lie in $\{\lceil \tau \rceil, \lceil \tau \rceil + 1\}$.

Proof. (23) is (22). Multiplying $\tau - E < t^* < \tau + 1 + E$ by q/L^2 gives (24). For (25): $2 - \rho > 0$, so the right side of (24) is at most $-(\kappa - 0.0089)L^{-2} + 1.78L^{-4}$, which is negative for $L^2 > 1.78/0.0774$, i.e. $N \geq 6$, and in absolute value smaller than the left side; and $L^{-4} \leq L^{-2}/6.25$, $L \geq \frac{5}{6}N$ for $N \geq 3$ (the middle inequality of (25) needs $1.78/L^2 \leq 0.372 - 0.0863$, i.e. $L^2 \geq 6.23$, and $L^2 = 6.25$ at $N = 3$; the last inequality is an equality at $N = 3$, where $L = \frac{5}{6}N$). For (26) write $qt^*/L^2 = c - (\kappa + q(\rho - \frac{3}{2}))L^{-2} + \frac{q}{2}\vartheta_1L^{-2} + \vartheta_3L^{-4}$ with $|\vartheta_1| < 1$, $|\vartheta_3| \leq 1.78$ (this is (24)), divide by $1 - \delta$, $\delta = \frac{1}{4L^2} \leq \frac{1}{25}$, and expand as in Corollary 5.8: the L^{-4} coefficient is $(1 - \delta)^{-1}[\vartheta_3 + \frac{c}{16} - \frac{1}{4}(\kappa + q(\rho - \frac{3}{2}) - \frac{q}{2}\vartheta_1)]$, of modulus at most $\frac{25}{24}(1.78 + 0.0209 + \frac{1}{4}(0.0863 + 0.9912)) < 2.2$. For (27): with $m = \lceil \tau \rceil \in [\tau, \tau + 1)$ and $E < 1$, the integers in the open interval $(\tau - E, \tau + 1 + E)$ are m , and in addition $m - 1$ if $\tau - (m - 1) < E$, and $m + 1$ if $m - \tau < E$. If $\text{dist}(\tau, \mathbb{Z}) \geq E$ (which forces $E \leq \frac{1}{2}$) only m remains, and it is then the unique maximiser by Theorem 6.3. If $E < \frac{1}{2}$ the two conditions exclude each other, because $(\tau - (m - 1)) + (m - \tau) = 1 > 2E$. $\square$

Remark 6.5. For $q = \frac{4}{5}$, (27) reads $t^* = \lceil \frac{5}{4}cL^2 - 1.09902\dots \rceil$, and for $q = \frac{1}{2}$, $t^* = \lceil 2cL^2 - 1.16372\dots \rceil$. These are the empirical rules $t^* = \lfloor cL^2/q - 0.099 \rfloor$ ($q = \frac{4}{5}$) and $t^* = \lfloor cL^2/q - 0.164 \rfloor$ ($q = \frac{1}{2}$) found and certified case by case for $10 \leq N \leq 600$, resp. $10 \leq N \leq 300$, by the exact-arithmetic certificates of R5; here they follow from a theorem valid for all N , but only for the N with $\text{dist}(\tau, \mathbb{Z}) \geq E$; for the other N the theorem leaves two adjacent candidates, and the rules rest on the certificates alone. (Script 22: on those certified data the hypothesis $\text{dist}(\tau, \mathbb{Z}) \geq E$ holds in 596 of 600 cases for $q = \frac{4}{5}$ and in 289 of 298 cases for $q = \frac{1}{2}$, and (27) gives the certified mode in each of them.)

7 The simple random walk $q = 1$ and the range $\frac{9}{10} < q < 1$

7.1 Parity classes at $q = 1$: notation and expansions

In this section $x_0 = 1$. For $q = 1$ we use the class representation of Theorem 3.10: with $W = 2N - 1 = 2L$, $D \in \{N - 1, N\}$ and $\Phi_D(t) := \sum_{k=1}^{N-1} \sin \vartheta_k \sin(D\vartheta_k) \cos^{t-1} \vartheta_k$ (real $t \geq 1$; all $\cos \vartheta_k > 0$) we have $f(t) = \Phi_D(t)/L$ for integers $t \equiv D \pmod{2}$. The class $D = N - 1$ is called *near* (upper signs below), the class $D = N$ *far* (lower signs). Put, for integers $k \geq 1$,

$$\hat{y}_k := \frac{k\pi}{2}, \quad x_k := \frac{\hat{y}_k}{L} = \vartheta_k, \quad \zeta_k := x_k^2, \quad \tau_k := (-1)^{k/2} \text{ (} k \text{ even)}, \quad \eta := \frac{1}{L}, \quad v := \frac{t-1}{L^2},$$

(so $\hat{y}_k = 2y_k$ for odd k), and for $0 \leq \zeta < \pi^2/4$

$$c_2(\zeta) := \cos \frac{\sqrt{\zeta}}{2}, \quad S_4(\zeta) := S(\zeta/4), \quad M(\zeta) := -\frac{2 \ln \cos \sqrt{\zeta}}{\zeta} \quad (M(0) = 1),$$

so that $\cos^{t-1} x_k = \exp(-\frac{1}{2}v\hat{y}_k^2 M(\zeta_k))$. Finally

$$\mathcal{E}^{(j)}(v) := \sum_{k \text{ even}, k \geq 2} \tau_k \frac{\hat{y}_k^2}{2} \left(-\frac{\hat{y}_k^2}{2}\right)^j e^{-v\hat{y}_k^2/2}, \quad \mathcal{E} := \mathcal{E}^{(0)} = \sum_{k \text{ even}} (-1)^{k/2} \frac{k^2 \pi^2}{8} e^{-k^2 \pi^2 v/8}. \quad (28)$$

Lemma 7.1 (Class sums). *For real $t \geq 1$,*

$$L \Phi_D(t) = P(t) \mp \eta E(t), \quad L^3(\Phi_D(t+2) - \Phi_D(t)) = P_1(t) \mp \eta E_1(t),$$

$$\begin{aligned} P &:= \sum_{k \text{ odd}} \sigma_k \hat{y}_k (Sc_2)(\zeta_k) \cos^{t-1} x_k, & E &:= \sum_{k \text{ even}} \tau_k \frac{\hat{y}_k^2}{2} (SS_4)(\zeta_k) \cos^{t-1} x_k, \\ P_1 &:= - \sum_{k \text{ odd}} \sigma_k \hat{y}_k^3 (S^3 c_2)(\zeta_k) \cos^{t-1} x_k, & E_1 &:= - \sum_{k \text{ even}} \tau_k \frac{\hat{y}_k^4}{2} (S^3 S_4)(\zeta_k) \cos^{t-1} x_k, \end{aligned}$$

the sums running over $1 \leq k \leq N - 1$. Thus $L^2 f(t) = P \mp \eta E$ and $L^4(f(t+2) - f(t)) = P_1 \mp \eta E_1$ for integers $t \equiv D$.

Proof. $D = L \mp \frac{1}{2}$, so $D\vartheta_k = \hat{y}_k \mp \frac{x_k}{2}$ and $\sin(D\vartheta_k) = \sin \hat{y}_k \cos \frac{x_k}{2} \mp \cos \hat{y}_k \sin \frac{x_k}{2}$, where $\sin \hat{y}_k = \sigma_k$, $\cos \hat{y}_k = 0$ for odd k and $\sin \hat{y}_k = 0$, $\cos \hat{y}_k = \tau_k$ for even k . Use $L \sin x_k = \hat{y}_k S(\zeta_k)$, $\sin \frac{x_k}{2} = \frac{\hat{y}_k}{2L} S_4(\zeta_k)$ and $\cos^{t+1} x - \cos^{t-1} x = -\sin^2 x \cos^{t-1} x = -L^{-2} \hat{y}^2 S^2 \cos^{t-1} x$. $\square$

Lemma 7.2 (Bounds on the low range $\zeta \leq \pi^2/9$). *The power series of M at 0 has nonnegative coefficients and radius of convergence $\pi^2/4$; hence M, M', M'' are nonnegative and nondecreasing on $[0, \pi^2/4)$. With $x = \sqrt{\zeta}$, $M' = \frac{x \tan x + 2 \ln \cos x}{x^4}$ and $M'' = \frac{x^2 \sec^2 x - 5x \tan x - 8 \ln \cos x}{2x^6}$, so on $[0, \pi^2/9]$*

$$M \geq 1, \quad 0 \leq M' \leq m_1 := M'(\frac{\pi^2}{9}) = 0.35548\dots, \quad 0 \leq M'' \leq m_2 := M''(\frac{\pi^2}{9}) = 0.32707\dots$$

Moreover, on $[0, \pi^2/4]$: $-\frac{1}{8} \leq c'_2 \leq 0$, $0 \leq c''_2 \leq \frac{1}{192}$, $-\frac{1}{24} \leq S'_4 \leq 0$, and

$$\begin{aligned} |(Sc_2)'| &\leq \frac{7}{24}, \quad |(SS_4)'| \leq \frac{5}{24}, \quad |(S^3 S_4)'| \leq \frac{13}{24}, \quad -\frac{5}{8} \leq (S^3 c_2)' \leq 0, \quad 0 \leq (S^3 c_2)'' \leq \hat{\alpha}_2 := \frac{111}{320}; \\ (S^3 c_2)'(0) &= -\frac{5}{8}, \quad M'(0) = \frac{1}{6}; \text{ all four functions } Sc_2, SS_4, S^3 S_4, S^3 c_2 \text{ take values in } [0, 1]. \end{aligned}$$

Proof. $\tan$ has nonnegative Maclaurin coefficients (from $\tan' = 1 + \tan^2$ by induction), hence so has $-\ln \cos x = \int_0^x \tan$, and $M(\zeta) = -2 \ln \cos(\sqrt{\zeta})/\zeta$ is a power series in ζ with nonnegative coefficients. The formulas for M', M'' follow from $d/d\zeta = (2x)^{-1}d/dx$. As in Lemma 5.1, $c_2(\zeta) = \cos \sqrt{\zeta/4}$ gives $c_2' = -\frac{1}{8}S(\zeta/4)$ and $c_2'' = -\frac{1}{32}S'(\zeta/4)$; $S_4' = \frac{1}{4}S'(\zeta/4)$. The product rule and Lemma 5.1(1) give the remaining bounds, e.g. $(S^3 c_2)'' = (6SS'^2 + 3S^2 S'')c_2 + 6S^2 S' c_2' + S^3 c_2'' \leq \frac{1}{6} + \frac{1}{20} + \frac{1}{8} + \frac{1}{192} = \frac{111}{320}$, all terms being nonnegative. $\square$

Lemma 7.3 (Expansions for $q = 1$). *Let $0 < v_- \leq v \leq v_+$, $L \geq L_0 \geq 3$, $L_0^2 > 8/(2v_- \ln 2)$, and $t = 1 + vL^2$ (real). Then*

$$\begin{aligned} P &= G(v) + \eta^2 R_{P0}, & E &= \mathcal{E}(v) + \eta^2 R_{E0}, \\ P_1 &= 2G'(v) + \eta^2 \left[\frac{5}{2}G''(v) - \frac{2v}{3}G'''(v) \right] + \eta^4 R_P, & E_1 &= 2\mathcal{E}'(v) + \eta^2 R_E, \end{aligned}$$

with $|R_{P0}| \leq \rho_{P0}$, $|R_{E0}| \leq \rho_{E0}$, $|R_P| \leq \rho_P$, $|R_E| \leq \rho_E$, where, writing $a_k^\pm := \frac{1}{2}v_\pm \hat{y}_k^2$ and $\sum'$ for sums over $k > \frac{2}{3}L_0$,

$$\begin{aligned} \rho_P &:= \frac{1}{2} \sum_{k \text{ odd}} \hat{y}_k^7 [\hat{\alpha}_2 + a_k^+ (\frac{5}{4}m_1 + m_2) + (a_k^+ m_1)^2] e^{-a_k^-} + \left(\frac{\pi}{2}\right)^3 L_0^8 2^{-v-L_0^2} \\ &\quad + \sum'_{k \text{ odd}} \hat{y}_k^7 \left[\left(\frac{3}{\pi}\right)^4 + \left(\frac{3}{\pi}\right)^2 \left(\frac{5}{8} + \frac{a_k^+}{6}\right) \right] e^{-a_k^-}, \\ \rho_E &:= \frac{1}{2} \sum_{k \text{ even}} \hat{y}_k^6 \left[\frac{13}{24} + a_k^+ m_1 \right] e^{-a_k^-} + \frac{1}{2} \left(\frac{\pi}{2}\right)^4 L_0^7 2^{-v-L_0^2} + \frac{1}{2} \left(\frac{3}{\pi}\right)^2 \sum'_{k \text{ even}} \hat{y}_k^6 e^{-a_k^-}, \\ \rho_{P0} &:= \sum_{k \text{ odd}} \hat{y}_k^3 \left[\frac{7}{24} + a_k^+ m_1 \right] e^{-a_k^-} + \frac{\pi}{2} L_0^4 2^{-v-L_0^2} + \left(\frac{3}{\pi}\right)^2 \sum'_{k \text{ odd}} \hat{y}_k^3 e^{-a_k^-}, \\ \rho_{E0} &:= \frac{1}{2} \sum_{k \text{ even}} \hat{y}_k^4 \left[\frac{5}{24} + a_k^+ m_1 \right] e^{-a_k^-} + \frac{1}{2} \left(\frac{\pi}{2}\right)^2 L_0^5 2^{-v-L_0^2} + \frac{1}{2} \left(\frac{3}{\pi}\right)^2 \sum'_{k \text{ even}} \hat{y}_k^4 e^{-a_k^-}. \end{aligned}$$

Proof. The proof is that of Proposition 5.4, with the split at $x_k = \pi/3$ (low modes: $k \leq \frac{2}{3}L$, $\cos x_k \geq \frac{1}{2}$, $\zeta_k \leq \pi^2/9$) and without the computer-assisted part. On a low mode the summand is $\pm \hat{y}_k^p \hat{A}(\zeta_k) \exp(-aM(\zeta_k))$ with $a = \frac{1}{2}v\hat{y}_k^2$ and $\hat{A} \in \{Sc_2, SS_4, S^3 c_2, S^3 S_4\}$. Put $F = \hat{A}e^{-aM}$; by Lemma 7.2, $|F'| \leq (|\hat{A}'| + am_1)e^{-a}$ and $|F''| \leq (\hat{\alpha}_2 + 2a \cdot \frac{5}{8}m_1 + a^2 m_1^2 + am_2)e^{-a}$ (for $\hat{A} = S^3 c_2$) on the low range. For P, E, E_1 use $F(\zeta_k) = F(0) + \zeta_k F'(\xi_k)$; for P_1 use $F(\zeta_k) = F(0) + \zeta_k F'(0) + \frac{1}{2}\zeta_k^2 F''(\xi_k)$ with $F(0) = e^{-a}$, $F'(0) = (-\frac{5}{8} - \frac{a}{6})e^{-a}$. Summing the Taylor polynomials over all k of the relevant parity gives $G, \mathcal{E}, 2\mathcal{E}'$ (by (8), (28) and $\hat{y}_k = 2y_k$) and, for P_1 ,

$$- \sum_{k \text{ odd}} \sigma_k \hat{y}_k^3 e^{-a} + \eta^2 \sum_{k \text{ odd}} \sigma_k \hat{y}_k^5 \left(\frac{5}{8} + \frac{v\hat{y}_k^2}{12} \right) e^{-a} = 2G' + \eta^2 \left[\frac{5}{8} \cdot 4G'' + \frac{v}{12} \cdot (-8G''') \right].$$

The three parts of each ρ bound, in this order, the Taylor remainders of the low modes (summed over all k of the parity), the high lattice modes ($|\cos x_k|^{t-1} \leq 2^{-vL^2}$, fewer than L of them, $\hat{y}_k < \pi L/2$; the bound is decreasing in L for $L^2 > 8/(2v_- \ln 2)$), and the Taylor polynomials for $k > \frac{2}{3}L$ (where $L < 3\hat{y}_k/\pi$). $\square$

7.2 The mode at $q = 1$

Define the constants (enclosed in ball arithmetic, script 16; all digits correct)

$$\mathcal{E}(c) = -0.92536002838\dots, \quad \mathcal{E}'(c) = 4.1610548333\dots, \quad \mathcal{E}''(c) = -12.547560414\dots,$$

$$b := -\frac{\mathcal{E}'(c)}{G''(c)} = 0.33250858299642786\dots, \tag{29}$$

$$s_2 := -\frac{\Gamma_0}{G''(c)} = -2.2503062886311552\dots, \quad \Gamma_0 := \frac{1}{2}G'''(c)b^2 + b\mathcal{E}''(c) + \frac{5}{4}G''(c) - \frac{c}{3}G'''(c),$$

$$\tau_1 := cL^2 - bL + 1 + s_2.$$

Theorem 7.4 (Mode of the simple random walk). *Let $q = 1$, $x_0 = 1$, $N \geq 3$. Every global maximiser t^* of f satisfies*

1. $t^* \equiv N - 1 \pmod{2}$ (the mode lies in the near class), and

2. $\tau_1 - \frac{K}{L} < t^* < \tau_1 + 2 + \frac{K}{L}$ with $K = 10.3$; for $3 \leq N \leq 600$ this holds with $K = 0.031$, and moreover $-0.0032 < t^* - \tau_1 < 1.9986$ there.

Consequently there are at most two maximisers (then t^* and $t^* + 2$; this happens for $N \in \{5, 6, 8, 9\}$ and for no other $N \leq 600$), and

$$\left| \frac{t^*}{L^2} - c + \frac{b}{L} \right| < \frac{1.2504 + K/L}{L^2}, \quad \lim_{N \rightarrow \infty} L \left(c - \frac{t^*}{L^2} \right) = b = 0.33250858 \dots \neq 0; \quad (30)$$

so for $q = 1$ the scaled mode converges to c exactly at rate N^{-1} , not N^{-2} , and $t^*/\mathbb{E}T = c - b/L + O(L^{-2})$.

Proof. Analytic part, $N \geq 601$. Let $L \geq L_0 = 600.5$, $\eta \leq \eta_0 = 1/L_0$. By Lemmas 7.1 and 7.3, for real t and $v = (t - 1)/L^2$ in a window $[v_-, v_+] \ni c$,

$$\frac{1}{2}L^3\Psi_D(t) = G'(v) \mp \eta \mathcal{E}'(v) + \eta^2 J(v) + \eta^3 r, \quad J := \frac{5}{4}G'' - \frac{v}{3}G''', \quad |r| \leq \frac{1}{2}(\rho_E + \eta_0 \rho_P), \quad (31)$$

where $\Psi_D(t) = \Phi_D(t + 2) - \Phi_D(t)$. Write $v = c \mp b\eta + s\eta^2$ and $\delta := \mp b + s\eta$, so $v - c = \eta\delta$. Taylor's formula at c gives $G'(v) = G'(c) + G''(c)\eta\delta + \frac{1}{2}G'''(c)\eta^2\delta^2 + \frac{1}{6}G''''(c)\eta^3\delta^3$, $\mathcal{E}'(v) = \mathcal{E}'(c) + \mathcal{E}''(c)\eta\delta + \frac{1}{2}\mathcal{E}'''(c)\eta^2\delta^2$, $J(v) = J(c) + J'(\xi'')\eta\delta$. The terms of order η are $\eta(G''(c)\delta \mp \mathcal{E}'(c)) = \eta^2 s G''(c)$ by the definition of b , and collecting the rest,

$$\begin{aligned} \frac{1}{2}L^5\Psi_D(t) &= (s - s_2)G''(c) + \eta\Gamma_1, \\ \Gamma_1 &:= \frac{1}{2}G'''(c)(\mp 2bs + s^2\eta) \mp \mathcal{E}''(c)s + \frac{1}{6}G''''(c)\delta^3 \mp \frac{1}{2}\mathcal{E}'''(c)\delta^2 + J'(\xi'')\delta + r, \end{aligned} \quad (32)$$

because $\frac{1}{2}G'''(c)\delta^2 \mp \mathcal{E}''(c)\delta + J(c) = \Gamma_0 + \eta[\frac{1}{2}G'''(c)(\mp 2bs + s^2\eta) \mp \mathcal{E}''(c)s]$ and $\Gamma_0 = -s_2 G''(c)$. Script `16.q1.theorem.py` encloses Γ_1 for both classes, all $\eta \in [0, \eta_0]$ and all $s \in [s_2 - K_1\eta_0, s_2 + K_1\eta_0]$ (the window $[v_-, v_+]$ is chosen to contain all occurring v , here $[0.33272, 0.33385]$; $\rho_P \leq 4190$, $\rho_E \leq 231$, $|r| \leq 118.7$) and certifies

$$\sup |\Gamma_1| < K_1 |G''(c)| \quad \text{with } K_1 = 10.21.$$

Hence $\Psi_D > 0$ at $s = s_2 - K_1\eta$ and $\Psi_D < 0$ at $s = s_2 + K_1\eta$. By Theorem 3.10 (proof), Ψ_D has exactly one zero t_0^D in $(D - 2, \infty)$, is positive on $[D - 2, t_0^D)$ and negative on (t_0^D, ∞) ; since $\tau_1 - K_1\eta > N - 2$ (checked in the script), we conclude

$$|t_0^{N-1} - (cL^2 - bL + 1 + s_2)| < \frac{K_1}{L}, \quad |t_0^N - (cL^2 + bL + 1 + s_2)| < \frac{K_1}{L}.$$

Within the class D , $f(t + 2) - f(t) = \Psi_D(t)/L$ is positive for class points $t < t_0^D$ and negative for $t > t_0^D$; therefore every maximiser of f over the class lies in $[t_0^D, t_0^D + 2]$.

The near class wins. Let t_n, t_f be class maximisers, $v_n = c + \eta\delta_n$, $v_f = c + \eta\delta_f$, with $\delta_n = -b + s_n\eta$, $\delta_f = b + s_f\eta$ and $s_n, s_f \in [s_2 - K_1\eta_0, s_2 + 2 + K_1\eta_0]$. By Lemmas 7.1–7.3 and Taylor's formula ($G'(c) = 0$),

$$\frac{L^2(f(t_n) - f(t_f))}{\eta} = -2\mathcal{E}(c) + \eta \left[\frac{1}{2}G''(\xi_1)\delta_n^2 - \frac{1}{2}G''(\xi_2)\delta_f^2 - \mathcal{E}'(\xi_3)\delta_n - \mathcal{E}'(\xi_4)\delta_f + \theta \right]$$

with $|\theta| \leq 2(\rho_{P0} + \eta_0\rho_{E0})$, and the script certifies that the right-hand side lies in $[1.832, 1.869]$ for all $L \geq 600.5$ (it is close to $-2\mathcal{E}(c) = 1.8507 \dots > 0$). Hence every global maximiser is a near-class maximiser, $t^* \in [t_0^{N-1}, t_0^{N-1} + 2]$, which gives (1) and (2) with $K = K_1$ for $N \geq 601$.

Computer-assisted part, $3 \leq N \leq 600$. Script `17_q1_small_N.py`: for $N \leq 14$ the PMF is computed exactly (integer path counts), all maximisers are listed, and the monotone tail is guaranteed by Theorem 3.10. For $15 \leq N \leq 600$ the script certifies in 256-bit ball arithmetic, for each class, $\Psi_D(t_D - 2) > 0 > \Psi_D(t_D)$ at a class point t_D (so t_D is the unique class maximiser) and $\Phi_{N-1}(t_{N-1}) > \Phi_N(t_N)$. In all cases the maximisers are in the near class and $-0.0032 < t^* - \tau_1 < 1.9986$, $L \cdot \max\{\tau_1 - t^*, t^* - \tau_1 - 2, 0\} \leq 0.0309$ (`data/17_q1_small_N.json`). The exact ties $f(t^*) = f(t^* + 2)$ occur for $N = 5, 6, 8, 9$ (modes $\{4, 6\}, \{7, 9\}, \{15, 17\}, \{20, 22\}$).

Consequences. (1),(2) give $1 + s_2 - K/L < t^* - cL^2 + bL < 3 + s_2 + K/L$ and $\max(|1 + s_2|, |3 + s_2|) = 1.2503 \dots$. Two maximisers in the near class differ by 2 and must both lie in $[t_0, t_0 + 2]$. $\square$

Remark 7.5. (a) The first-order shift $-bL$ has a simple meaning: unfolding the reflection, the near class is the exit of the simple random walk from an interval through the end at distance $L - \frac{1}{2}$, the far class through the end at distance $L + \frac{1}{2}$; to first order the near class has the larger and earlier peak. For $q < 1$ the two classes are mixed by the holding times and the shift disappears (Theorem 7.7). (b) The analytic constant $K_1 = 10.21$ is far from optimal (we used absolute values for R_E, R_P); the certified data suggest that (2) holds with $K \approx 0.03$ for all N (numerical observation).

7.3 Global maximisers for $\frac{2}{3} < q < 1$

For $q > \frac{9}{10}$ we have no unimodality theorem (Theorems 3.15, 3.16 stop at $\frac{9}{10}$), and for $q > 1 - \frac{1}{2N-3}$ the PMF is *not* unimodal (Proposition 3.9). The following results locate all global maximisers without using unimodality; they are needed for $\frac{9}{10} < q < 1$ (and give a second proof of the localisation for $\frac{2}{3} < q \leq \frac{9}{10}$, $N \geq 601$). The next lemma controls the far left tail, where the oscillations live, by the $q = 1$ walk.

Lemma 7.6 (Comparison with the simple random walk). *For $q \in (0, 1)$, $N \geq 2$ and $t \geq 1$: $f_q(t) = q \mathbb{E}[f_1(1 + \text{Bin}(t-1, q))] \leq q \max_{1 \leq k \leq t} f_1(k)$.*

Proof. By the proof of Lemma 3.8 with $q_0 = 1$, $p = q$: $f_q(t) = \sum_{k=1}^t f_1(k) \binom{t-1}{k-1} q^k (1-q)^{t-k}$, and $\sum_{k=1}^t \binom{t-1}{k-1} q^k (1-q)^{t-k} = q$. $\square$

Theorem 7.7 (All global maximisers, $q < 1$, $N \geq 601$). *Let $N \geq 601$, $q \in (0, 1)$, and assume*

$$\mathfrak{h}(q, L) := 4\left(\frac{\pi}{2}\right)^3 L^8 \exp(-0.15 \chi(q) L^2) \leq 10^{-12}, \quad (33)$$

with χ from (15). Then, with $\tau = (cL^2 - \kappa)/q + 1 - \rho$ and $E = 1.56/(qL^2)$:

1. *$D(t) > 0$ for all integers t with $q(t-1)/L^2 \geq 0.15$ and $t \leq \tau - E$; $D(t) < 0$ for all integers $t \geq \tau + E$.*
2. *If $q \geq \frac{2}{3}$: $f(t) < \max f$ for all t with $q(t-1)/L^2 < 0.15$.*
3. *Every global maximiser t^* of f satisfies $\tau - E < t^* < \tau + 1 + E$; there are at most two, and they are adjacent.*

Condition (33) holds for all $N \geq 601$ if $q \leq 0.999$, for all $N \geq 2001$ if $q \leq 0.9999$, and in general whenever $\min\{\frac{1}{2}, 2(1-q)\} L^2 \geq (8 \ln L + 30.4)/0.15$.

Proof. For $q \leq \frac{2}{3}$ the PMF is log-concave, so the ratios $f(t+1)/f(t)$ are nonincreasing on the support, and (1), (3) follow from Theorem 6.1 (with $K = 1.56$, $N \geq 601$) as in the proof of Theorem 6.3. So let $q \in [\frac{2}{3}, 1)$. Put $\varepsilon = L^{-2} \leq \varepsilon_0 = 600.5^{-2}$. For the given pair (N, q) , Proposition 5.4 holds with B_{high} replaced by $B_{\text{high}}(L, q) \leq \mathfrak{h}(q, L) \leq 10^{-12}$ whenever $v_- \geq 0.15$, $j \leq 1$ (“tolerance mode” of the scripts: `core.HIGH_TOL`); the other parts of the enclosure of R are uniform in $L \geq 600.5$ and q in a ball, with $\bar{q} = 1$ in Lemma 5.2.

(1), near c . Script 14 (run with $N_0 = 601$, $q \in (0, 1]$ split into 3000 balls, tolerance mode) certifies (A) and (B) of Theorem 6.1 on each q -interval with its own constant $K_i \leq 1.5572$ (file `14_disc_N0_601_q1_tol.json`), and script 35 certifies them on every interval with the single constant $K = 1.56$ used in the statement (file `35_disc_uniformK.json`).

(1), $|v - c| \leq \delta_0 := 0.01$. Write $v = c - d\varepsilon$. By Proposition 5.4 and Taylor's formula ($G'(v) = -G''(c)d\varepsilon + \frac{1}{2}G'''(\xi)d^2\varepsilon^2$, $H(v) = H(c) - H'(\xi')d\varepsilon$, $H(c) = (\kappa + q\rho)G''(c)$),

$$\frac{\mathfrak{S}_1(t)}{\varepsilon} = |G''(c)| (d - \kappa - q\rho) + d\varepsilon \left[\frac{1}{2}G'''(\xi)d - H'(\xi') \right] + \varepsilon R.$$

Let $M_3 := \sup_{|v-c| \leq \delta_0} |G'''| \leq 170.4$, $M_{H'} := \sup |H'| \leq 315.2$, $R_{\max} := \sup |R| \leq 406$ (suprema over $|v-c| \leq \delta_0$, $q \in [\frac{2}{3}, 1]$, $L \geq 600.5$) and $m := |G''(c)| - \frac{1}{2}M_3\delta_0 - \varepsilon_0 M_{H'} \geq 11.66$. If $d \geq 0$, $|d\varepsilon| \leq \delta_0$: $\mathfrak{S}_1/\varepsilon \geq md - |G''(c)|(\kappa + q\rho) - \varepsilon_0 R_{\max} > 0$ as soon as $d > d_1(q) := (|G''(c)|(\kappa + q\rho) + \varepsilon_0 R_{\max})/m$, and the script checks $d_1(q) < \kappa + q\rho + q$ on 64 subintervals of $[\frac{2}{3}, 1]$. Since window (A) of Theorem 6.1 is $d \in \kappa + q\rho + [K\varepsilon, K\varepsilon + q]$, this gives $D(t) > 0$ for all integers $t \leq \tau - E$ with $v \geq c - \delta_0$. If $0 \leq d \leq \kappa + q\rho - q$: $\mathfrak{S}_1/\varepsilon \leq -|G''(c)|q + d\varepsilon_0(\frac{1}{2}M_3d + M_{H'}) + \varepsilon_0 R_{\max} < 0$; if $d < 0$, $|d\varepsilon| \leq \delta_0$: $\mathfrak{S}_1/\varepsilon \leq -m|d| - |G''(c)|(\kappa + q\rho) + \varepsilon_0 R_{\max} < 0$; together with window (B) this gives $D(t) < 0$ for all integers $t \geq \tau + E$ with $v \leq c + \delta_0$.

(1), $v \in [0.15, c - \delta_0]$ and $v \in [c + \delta_0, 0.5]$. Each interval is divided into 400 pieces V_i ; on each the script encloses $G'(V_i) + \varepsilon H(V_i) + \varepsilon^2 R(V_i)$ for all $\varepsilon \in [0, \varepsilon_0]$, $q \in [\frac{2}{3}, 1]$ and finds it ≥ 0.1319 , resp. ≤ -0.1172 .

(1), $v \geq v_b := 0.5$. In $\mathfrak{S}_1(t) = \sum_k c_k \lambda_k^{t-1}$, $c_k = -\sigma_k 4y_k^3 A_1(\zeta_k)$, the first term is $-4y_1^3 A_1(\zeta_1) \lambda_1^{t-1} < 0$. For a low mode $k \geq 3$, $|c_k \lambda_k^{t-1}| \leq 4y_1^3 \lambda_1^{t-1} \cdot (y_k/y_1)^3 \exp(-2v(y_k^2 B(\zeta_k) - y_1^2 B(\zeta_1)))$ with $B(\zeta_k) \geq \beta_*$ and $B(\zeta_1) \leq \Lambda(w_1) \leq (1 - w_1)^{-1}$, $w_1 \leq 2q\zeta_1$; the exponent is negative and decreasing in v , so the last factor is at most its value at v_b . For a high mode the corresponding factor is at most $(\pi L/(2y_1))^3 \exp(-v_b(\chi L^2 - 2y_1^2(1 - w_1)^{-1}))$, and there are fewer than L of them; this total is $\leq \mathfrak{h}(q, L)$. The script certifies $\sum_{k \geq 3 \text{ odd}} (y_k/y_1)^3 e^{-2v_b(\beta_* y_k^2 - (1 - w_1)^{-1} y_1^2)} + 10^{-12} \leq 0.3169 < 0.9999 \leq A_1(\zeta_1)$. Hence $\mathfrak{S}_1(t) < 0$.

(2). Let t be an integer with $q(t-1)/L^2 < 0.15$; then $t \leq K_a := \lceil 0.225L^2 \rceil + 1$ because $q \geq \frac{2}{3}$. By the proof of Theorem 7.4, both parity classes of f_1 are increasing up to $t_0^D > cL^2 - bL + 1 + s_2 - K_1/L > K_a + 2$ (checked at L_0 ; the difference increases with L). Hence $\max_{k \leq t} f_1(k) \leq \max\{f_1(K_a - 1), f_1(K_a)\}$, and for $k \in \{K_a - 1, K_a\}$, $w := (k-1)/L^2 \in [0.225 - \varepsilon_0, 0.225 + \varepsilon_0]$ (here $v = w$, $q = 1$); by Lemmas 7.1, 7.3 and since G is increasing on $(0, c)$,

$$L^2 f_1(k) \leq G(0.225 + \varepsilon_0) + \eta_0 \sup |\mathcal{E}(w)| + \eta_0^2 (\rho_{P0} + \eta_0 \rho_{E0}) \leq 0.8126.$$

On the other hand, for the integer $t_1 := \lfloor \tau - E \rfloor + 1$ we have $v_1 = q(t_1 - 1)/L^2 \in [c - 4\varepsilon_0, c + \varepsilon_0]$ and, by Proposition 5.4 with $j = 0$, $\frac{L^2}{q} f_q(t_1) = G(v_1) + \varepsilon H_0^{(q)}(v_1) + \varepsilon^2 R \geq 0.9250$. By Lemma 7.6, $f_q(t) \leq q \cdot 0.8126 L^{-2} < q \cdot 0.9250 L^{-2} \leq f_q(t_1)$.

(3). By (1), on $\{t : q(t-1)/L^2 \geq 0.15\}$ the PMF increases strictly up to $\lfloor \tau - E \rfloor + 1$ and decreases strictly from $\lceil \tau + E \rceil$ on; with (2), all global maximisers lie in $[\lfloor \tau - E \rfloor + 1, \lceil \tau + E \rceil] \subset (\tau - E, \tau + E + 1)$, an interval of length $1 + 2E < 2$.

All certificates: `scripts/18_global.py`, outputs `data/18_global_N0_601_q1.1_tol.json` (and, with the explicit B_{high} instead of the hypothesis (33), `18_global_N0_601_q999_1000.json`, `18_global_N0_2001_q9999_10000.json`). For the last statement: $\chi(q) \geq \min\{\frac{1}{2}, 2(1-q)\}$ because $-\ln(1-x) \geq x$ with $x = 2(1-q)$, and (33) is $0.15\chi L^2 \geq 8 \ln L + \ln(4(\pi/2)^3) + 12 \ln 10 = 8 \ln L + 30.37 \dots$; for $q \leq 0.999$ and $L = 600.5$: $0.15\chi L^2 \geq 108.39 > 81.56 \geq 8 \ln L + 30.38$, and for $q \leq 0.9999$, $L = 2000.5$: $0.15\chi L^2 \geq 120.08 > 91.19$; in both cases the left side grows faster than the right for larger L . $\square$

Corollary 7.8 (Limit theorem for every q). *Let $q \in (0, 1]$ be fixed, $x_0 = 1$, and let t_N^* be any global maximiser of $f = f_{N,q}$. Then*

$$\frac{q t_N^*}{(N - \frac{1}{2})^2} \longrightarrow c \quad \text{and} \quad \frac{t_N^*}{\mathbb{E}T} \longrightarrow c = 0.33328426549494789874 \dots$$

as $N \rightarrow \infty$; the error is $O(N^{-2})$ for $q < 1$ (Corollary 6.4, which by Theorem 7.7 holds with 1.56 for all $N \geq N_0(q)$) and $-b/L + O(N^{-2})$ for $q = 1$ (Theorem 7.4). The same holds for the continuous-time walk for every q (Corollary 5.8).

Remark 7.9 (The two-pole closed form does not apply in $d = 1$). The closed-form approximation of the computations of Sections 5–6 of the article (Section 6 of the article; `data/msc_modes`), $\text{mode} \approx \mathbb{E}T \cdot \ln(2dX)/(X + 2d - 1)$ with $X = \mu_1 \mathbb{E}T$ and $\mu_1 = \frac{q}{d}(1 - \cos \frac{\pi}{N})$, was obtained from the two slowest poles of the renewal ratio and is designed for $d \geq 2$, where the target is small. For $d = 1$ one has $X = (1 - \cos \frac{\pi}{N})N(N - 1) \rightarrow \frac{\pi^2}{2}$, so that formula would give $\text{mode}/\mathbb{E}T \rightarrow \ln(\pi^2)/(1 + \frac{\pi^2}{2}) = 0.385768\dots$, whereas by Corollary 7.8 the true limit is $c = 0.333284\dots$: in one dimension the closed form overestimates the mode by a factor tending to $1.15747\dots$ (in agreement with the certified overestimates of 15.8–22.5% for $10 \leq N \leq 600$, $q = \frac{4}{5}$, in R5). In $d = 1$ all poles ν_m are of the same order L^{-2} and all of them contribute at the mode; the correct statement is Theorem 6.3.

Remark 7.10 (The crossover). The hypothesis (33) is essentially $(1 - q)L^2 \gtrsim 27 \ln L + 100$. It is not an artefact: for $1 - q \lesssim L^{-2}$ the holding times do not mix the parity classes before the mode, and the mode follows the $q = 1$ law of Theorem 7.4 (e.g. $N = 10$, $q = 0.99$: $t^* = 27$, the $q = 1$ mode, while $\tau = 29.3$). The transition between the two laws, i.e. the regime $(1 - q)L^2 = O(\ln L)$, is not covered by our theorems (Section 9).

8 General start

8.1 The limit density and the limit theorem

Let now $x_0 \in \{1, \dots, N - 1\}$ be arbitrary, $d = N - x_0$, and $\xi_N := (x_0 - \frac{1}{2})/L \in (0, 1)$, the relative position of the start measured from the reflecting wall. By (3) and $\sin(d\theta_m) = (-1)^{m+1} \cos(\theta_m(x_0 - \frac{1}{2}))$,

$$G_{N, x_0}(u) := \frac{L^2}{q} g_{x_0}\left(\frac{L^2 u}{q}\right) = \sum_{k \text{ odd}, k \leq 2N-3} \sigma_k 2y_k S(\zeta_k) \cos x_k \cos(2y_k \xi_N) e^{-2uy_k^2 \beta(\zeta_k)}, \quad (34)$$

and, for $0 \leq \xi < 1$, the candidate limit is

$$G_\xi(u) := \sum_{k \text{ odd}} \sigma_k 2y_k \cos(2y_k \xi) e^{-2uy_k^2} = \frac{\pi}{2} \sum_{k \text{ odd}} \sigma_k k \cos \frac{k\pi\xi}{2} e^{-k^2\pi^2 u/8}, \quad G_0 = G. \quad (35)$$

Lemma 8.1 (Image form). *For $u > 0$ and $0 \leq \xi < 1$, with $h_a(u) := \frac{a}{\sqrt{2\pi}u^3} e^{-a^2/(2u)}$ ($a \in \mathbb{R}$),*

$$G_\xi(u) = \sum_{n \in \mathbb{Z}} [h_{4n+1-\xi}(u) + h_{4n+1+\xi}(u)].$$

In particular there is $C_\xi < \infty$ with $|G_\xi(u)| \leq C_\xi u^{-3/2} e^{-(1-\xi)^2/(2u)}$ for $0 < u \leq 1$, so that $G_\xi(u) = o(u^p)$ as $u \downarrow 0$ for every p ; and $G_\xi(u) e^{\pi^2 u/8} \rightarrow \frac{\pi}{2} \cos \frac{\pi\xi}{2} > 0$ as $u \rightarrow \infty$.

Proof. Let $p_u(z) = (2\pi u)^{-1/2} e^{-z^2/(2u)}$, whose Fourier transform is $\int p_u(z) e^{-2\pi i \omega z} dz = e^{-2\pi^2 \omega^2 u}$. The Poisson summation formula for the Schwartz function $z \mapsto p_u(a + 4z)$ (E. M. Stein and R. Shakarchi, *Fourier Analysis: an Introduction*, Princeton Univ. Press 2003, Chapter 5, Theorem 3.1) gives $\sum_{n \in \mathbb{Z}} p_u(a + 4n) = \frac{1}{4} \sum_{m \in \mathbb{Z}} e^{-\pi^2 m^2 u/8} e^{i\pi m a/2}$. Both sides may be differentiated termwise in a ; since $-\partial_a p_u(a) = h_a(u)$,

$$\sum_{n \in \mathbb{Z}} h_{a+4n}(u) = \frac{\pi}{4} \sum_{m \geq 1} m \sin \frac{\pi m a}{2} e^{-\pi^2 m^2 u/8}.$$

Adding this for $a = 1 - \xi$ and $a = 1 + \xi$ and using $\sin \frac{\pi m(1-\xi)}{2} + \sin \frac{\pi m(1+\xi)}{2} = 2 \sin \frac{\pi m}{2} \cos \frac{\pi m \xi}{2}$, which vanishes for even m and equals $2\sigma_m \cos \frac{\pi m \xi}{2}$ for odd m , gives (35). All indices $a = 4n + 1 \mp \xi$ satisfy $|a| \geq a_0 := 1 - \xi > 0$, and for $0 < u \leq 1$ we have $e^{-a^2/(2u)} \leq e^{-a_0^2/(2u)} e^{-(a^2 - a_0^2)/2}$; hence $\sum_a |h_a(u)| \leq (2\pi u^3)^{-1/2} e^{-a_0^2/(2u)} \sum_a |a| e^{-(a^2 - a_0^2)/2}$, which is the stated bound with $C_\xi = (2\pi)^{-1/2} \sum_a |a| e^{-(a^2 - a_0^2)/2} < \infty$. The behaviour at infinity is read off from (35). (Numerical check of the identity to 60 digits: script 19.) $\square$

Lemma 8.2 (Zero counts on the lattice). *Let $x_0 \in \{1, \dots, N-1\}$, $d = N - x_0$ and $0 \leq i \leq d-1$. Then the i -th derivative $g_{x_0}^{(i)}$ has at most i zeros in $(0, \infty)$, counted with multiplicity. The same holds for $G_{N, x_0}^{(i)}$.*

Proof. By (3), $g^{(i)}(t) = \frac{2q}{2N-1} \sum_m c_m (-\nu_m)^i e^{-\nu_m t}$ with $c_m = \sin \theta_m \sin(d\theta_m)$; the coefficient sequence $((-1)^i \nu_m^i c_m)_m$ has the same number $V \leq d-1$ of sign changes as $(c_m)_m$ (Lemma 3.2). By Theorem 2.2, $g^{(l)}(0) = 0$ for $l < d-1$ and $g^{(d-1)}(0) \neq 0$, so $g^{(i)}$ has a zero of multiplicity exactly $d-1-i$ at $t = 0$. By Lemma 3.1 the number of real zeros of $g^{(i)}$ other than 0, counted with multiplicity, is at most $V - (d-1-i) \leq i$. G_{N, x_0} is g_{x_0} up to positive scalings of the variable and of the function. $\square$

Theorem 8.3 (The limit density for a general start). *Let $0 \leq \xi < 1$.*

1. $G_\xi > 0$ on $(0, \infty)$, G_ξ is real-analytic, and there is a unique $c(\xi) \in (0, \infty)$ such that G_ξ is nondecreasing on $(0, c(\xi)]$ and nonincreasing on $[c(\xi), \infty)$, with $G_\xi(u) < G_\xi(c(\xi))$ for all $u \neq c(\xi)$. Moreover $c(0) = c$.
2. G'_ξ has exactly one zero in $(0, \infty)$, namely $c(\xi)$; $G'_\xi > 0$ on $(0, c(\xi))$ and $G'_\xi < 0$ on $(c(\xi), \infty)$.
3. $G''_\xi(c(\xi)) < 0$. More precisely there are $0 < a_1 < c(\xi) < a_2$ such that $G''_\xi \geq 0$ on $(0, a_1] \cup [a_2, \infty)$ and $G''_\xi \leq 0$ on $[a_1, a_2]$ (G_ξ has exactly two inflection points in the sense of sign changes of G''_ξ).
4. The function $\xi \mapsto c(\xi)$ is real-analytic on $[0, 1)$ (it extends to an even real-analytic function on $(-1, 1)$).

Proof. (1) Choose $x_0 = x_0(N)$ with $\xi_N \rightarrow \xi$ and $d = N - x_0 \geq 2$. Exactly as in Theorem 4.2 (dominated convergence; $|S \cos x_k \cos(2y_k \xi_N)| \leq 1$), $G_{N, x_0} \rightarrow G_\xi$ uniformly on $[u_0, \infty)$ for every $u_0 > 0$. By Theorem 3.3, $G_{N, x_0} > 0$ is increasing on $(0, u_N)$ and decreasing on $[u_N, \infty)$ for some $u_N > 0$. Pass to a subsequence with $u_N \rightarrow u^* \in [0, \infty]$. If $u < u' < u^*$ then $u' < u_N$ eventually, so $G_{N, x_0}(u) \leq G_{N, x_0}(u')$ and in the limit $G_\xi(u) \leq G_\xi(u')$; likewise G_ξ is nonincreasing on (u^*, ∞) ; and $G_\xi \geq 0$. If $u^* = \infty$, G_ξ would be nondecreasing on $(0, \infty)$ with limit 0 at infinity, hence $G_\xi \equiv 0$; if $u^* = 0$, G_ξ would be nonincreasing with limit 0 at 0^+ (Lemma 8.1), hence again $G_\xi \equiv 0$; both contradict the last statement of Lemma 8.1. So $u^* \in (0, \infty)$ and G_ξ is unimodal. If $G_\xi(u_1) = 0$ for some $u_1 > 0$, unimodality and $G_\xi \geq 0$ force $G_\xi = 0$ on $(0, u_1]$ or on $[u_1, \infty)$, which is impossible for a non-zero real-analytic function; so $G_\xi > 0$. The set of maximisers of a unimodal function is an interval, and it is a single point $c(\xi)$ because a non-constant real-analytic function is not constant on an interval; then $G_\xi(u) < G_\xi(c(\xi))$ for $u \neq c(\xi)$, and monotonicity on both sides of $c(\xi)$ follows from unimodality. $c(0) = c$ by Theorem 4.3.

(2), (3). Write $G = G_\xi$ and let $x_0(N)$ be as in (1); then $d = N - x_0 \rightarrow \infty$. Differentiating (34) and (35) termwise i times multiplies the k -th terms by $(-2y_k^2 \beta(\zeta_k))^i$ and $(-2y_k^2)^i$, and the dominated-convergence argument of Theorem 4.2 (majorant $2^{i+1} y_k^{2i+1} e^{-8u_0 y_k^2 / \pi^2}$) gives $G_{N, x_0}^{(i)} \rightarrow G^{(i)}$ uniformly on $[u_0, \infty)$ for every $i \geq 0$ and $u_0 > 0$. Say that a function w on $(0, \infty)$ has at least r sign changes if there are points $u_0 < u_1 < \dots < u_r$ with $w(u_l)w(u_{l+1}) < 0$ for all l . If $G^{(i)}$ had at least $i+1$ sign changes, then for all large N the function $G_{N, x_0}^{(i)}$ would have the same strict

signs at the points $u_0, \dots, u_{i+1}$, hence at least $i + 1$ zeros in $(0, \infty)$, contradicting Lemma 8.2 (note $i \leq d - 1$ for large N). Thus

$$G^{(i)} \text{ has at most } i \text{ sign changes on } (0, \infty), \quad i = 0, 1, 2, 3. \quad (36)$$

Each $G^{(i)}$, $i \leq 3$, is real-analytic and not identically zero (otherwise G would be a polynomial of degree $< i$, which is incompatible with $G > 0$ and $G(u) \rightarrow 0$ as $u \rightarrow \infty$). Hence its zeros are isolated, and by (36) the half-line splits into at most $i + 1$ intervals on each of which $G^{(i)}$ has a weak sign (≥ 0 or ≤ 0 , with isolated zeros); in particular $G^{(i)}$ has a weak sign near 0 and near ∞ . We also use $G^{(i)}(u) \rightarrow 0$ as $u \rightarrow \infty$ (from (35)) and $G(u) = o(u^2)$ as $u \downarrow 0$ (Lemma 8.1).

G'' near 0 and near ∞ . If $G'' \leq 0$ on some $(0, \delta)$, then G is concave there and, since $G(0^+) = 0$, $G(u) \geq (u/\delta')G(\delta')$ for $0 < u < \delta' < \delta$, contradicting $G(u) = o(u)$. Hence $G'' \geq 0$ near 0, and $G''(p_1) > 0$ for some p_1 arbitrarily close to 0. If $G'' \leq 0$ on some (M, ∞) , then G' is nonincreasing there; by (1), $G' \leq 0$ on $(c(\xi), \infty)$ and $G'(u_1) < 0$ for some $u_1 > \max(M, c(\xi))$, so $G' \leq G'(u_1) < 0$ on (u_1, ∞) and $G \rightarrow -\infty$, which is absurd. Hence $G'' \geq 0$ near ∞ and $G''(p_4) > 0$ for some arbitrarily large p_4 . Finally, by (1) there are $p < c(\xi) < p'$ with $G'(p) > 0 > G'(p')$, and the mean value theorem gives $\eta \in (p, p')$ with $G''(\eta) < 0$. So G'' has at least two sign changes ($+$, $-$, $+$ at $p_1 < \eta < p_4$), and by (36) exactly two: there are $a_1 < a_2$ with $G'' \geq 0$ on $(0, a_1]$, $G'' \leq 0$ on $[a_1, a_2]$, $G'' \geq 0$ on $[a_2, \infty)$, the zeros being isolated; by continuity $G''(a_1) = G''(a_2) = 0$.

The zero of G' . G' is strictly increasing on $(0, a_1]$, strictly decreasing on $[a_1, a_2]$ and strictly increasing on $[a_2, \infty)$. Since $G' \geq 0$ near 0 by (1), $G' > 0$ on $(0, a_1]$. Since $G'(u) \rightarrow 0$ as $u \rightarrow \infty$, $G' < 0$ on $[a_2, \infty)$. On $[a_1, a_2]$, G' decreases strictly from a positive to a negative value, so it has exactly one zero there, and none elsewhere. This zero is the maximiser $c(\xi)$ of (1), $a_1 < c(\xi) < a_2$, and the signs of G' are as stated. This proves (2) and the second assertion of (3).

Non-degeneracy. Suppose $G''(c(\xi)) = 0$. We exhibit six points at which G''' has strictly alternating signs. (i) There is $q_0 < p_1$ with $G'''(q_0) > 0$: otherwise G'' would be nonincreasing on $(0, p_1)$, so $G'' \geq G''(p_1) > 0$ there, and (as $G' \geq 0$ near 0) $G(u) \geq \frac{1}{2}G''(p_1)u^2$ for small u , contradicting $G(u) = o(u^2)$. (ii) Since $G''(p_1) > 0 = G''(a_1)$, $G''' < 0$ somewhere in (p_1, a_1) . (iii) On $[a_1, c(\xi)]$, $G'' \leq 0$ vanishes at both ends and is negative at some interior point p_2 ; hence $G''' > 0$ somewhere in $(p_2, c(\xi))$. (iv) Likewise on $[c(\xi), a_2]$ there is p_3 with $G''(p_3) < 0$, hence $G''' < 0$ somewhere in $(c(\xi), p_3)$ and (v) $G''' > 0$ somewhere in (p_3, a_2) . (vi) Since $G''(p_4) > 0$ and $G''(u) \rightarrow 0$ as $u \rightarrow \infty$, $G''' < 0$ somewhere in (p_4, ∞) , where $p_4 > a_2$. These six points are increasing and the signs are $+$, $-$, $+$, $-$, $+$, $-$: five sign changes, contradicting (36) for $i = 3$. Hence $G''(c(\xi)) \neq 0$, and $G''(c(\xi)) < 0$ because $a_1 < c(\xi) < a_2$.

(4) The series $\sum_k \sigma_k 2y_k (-2y_k^2) \cos(2y_k \xi) e^{-2uy_k^2}$ converges locally uniformly for complex (ξ, u) with $\operatorname{Re} u > 0$ (since $|\cos(2y_k \xi)| \leq e^{2y_k |\operatorname{Im} \xi|}$); so $(\xi, u) \mapsto G'_\xi(u)$ is jointly real-analytic on $\mathbb{R} \times (0, \infty)$, and even in ξ . For $|\xi| < 1$, $c(|\xi|)$ is a zero of G'_ξ with $\partial_u G'_\xi(c) = G''_\xi(c) \neq 0$ by (3); by the analytic implicit function theorem there is a real-analytic function $\tilde{c}$ near ξ with $G'_{\xi'}(\tilde{c}(\xi')) = 0$, and by the uniqueness in (2) $\tilde{c}(\xi') = c(|\xi'|)$. $\square$

Remark 8.4. For $\xi = 0$, parts (2) and (3) give a proof of Theorem 4.3(2),(3) which uses neither log-concavity nor any numerical evaluation: the uniqueness of the maximiser of G and its non-degeneracy follow from Descartes' rule on the lattice alone. (The numerical *value* of c of course still requires the enclosure of Theorem 4.6.)

Theorem 8.5 (Limit theorem for a general start). *Let $0 \leq \xi < 1$ and $x_0 = x_0(N)$ with $(x_0 - \frac{1}{2})/(N - \frac{1}{2}) \rightarrow \xi$.*

1. (Continuous time, every $q \in (0, 1]$.) The unique mode t_c of g_{x_0} satisfies $qt_c/L^2 \rightarrow c(\xi)$.
2. (Discrete time, $q \leq \frac{1}{2}$.) Every mode t^* of f_{x_0} satisfies $qt^*/L^2 \rightarrow c(\xi)$.

In both cases mode/ $\mathbb{E}T \rightarrow c(\xi)/(1 - \xi^2)$, since $\mathbb{E}_{x_0} T = (N(N - 1) - x_0(x_0 - 1))/q = L^2(1 - \xi_N^2)/q$.

Proof. (1) Fix $\delta \in (0, c(\xi))$. By Theorem 8.3, $G_\xi(c(\xi) \pm \delta) < G_\xi(c(\xi))$, so for large N , $G_{N,x_0}(c(\xi) \pm \delta) < G_{N,x_0}(c(\xi))$. If the mode u_N of the unimodal function G_{N,x_0} were $\leq c(\xi) - \delta$, then G_{N,x_0} would be nonincreasing on $[c(\xi) - \delta, \infty)$, contradicting $G_{N,x_0}(c(\xi) - \delta) < G_{N,x_0}(c(\xi))$; similarly $u_N < c(\xi) + \delta$. Hence $|u_N - c(\xi)| < \delta$ eventually, and $u_N = qt_c/L^2$.

(2) For $q \leq \frac{1}{2}$ all $\lambda_m \in (0, 1)$ and f_{x_0} is unimodal (Theorem 3.5). As in (13), $\frac{L^2}{q} f_{x_0}(t) = \sum_k \sigma_k 2y_k S(\zeta_k) \cos x_k \cos(2y_k \xi_N) \exp(-2vy_k^2 B(\zeta_k; q))$, $v = q(t-1)/L^2$, with $B = \beta\Lambda(2q \sin^2 x_k) \geq \beta$; the dominated-convergence argument of Theorem 4.2 applies verbatim and gives $\frac{L^2}{q} f_{x_0}(t) - G_\xi(v) \rightarrow 0$ uniformly in $v \geq u_0$. Take integers $t_1 < t_2 < t_3$ with $q(t_i-1)/L^2 \rightarrow c(\xi) - \delta, c(\xi), c(\xi) + \delta$. For large N , $f_{x_0}(t_1) < f_{x_0}(t_2) > f_{x_0}(t_3)$, and by unimodality every mode lies in (t_1, t_3) .

The formula for $\mathbb{E}_{x_0} T$ is Corollary 2.5: $N(N-1) - x_0(x_0-1) = (L^2 - \frac{1}{4}) - ((x_0 - \frac{1}{2})^2 - \frac{1}{4}) = L^2(1 - \xi_N^2)$. $\square$

Proposition 8.6 (Values of $c(\xi)$; CA). *The following enclosures hold (all digits correct; script 19_general_start.py, 600-bit ball arithmetic, certified sign change of G'_ξ on an interval of radius 10^{-40} , which brackets $c(\xi)$ by Theorem 8.3):*

ξ	$c(\xi)$	$(1 - \xi)^2/3$	$G_\xi(c(\xi))$
0	0.33328426549494...	0.33333...	0.9250649...
0.1	0.32286309733114...	0.27	0.9255105...
0.2	0.28518014581572...	0.21333...	0.9340642...
0.3	0.19477604582332...	0.16333...	1.0022603...
0.4	0.12173600568561...	0.12	1.2888334...
0.5	0.08334564332977...	0.08333...	1.8501980...
0.6	0.053333333387474...	0.05333...	2.8908811...
0.7	0.0300000000000000002859...	0.03	5.1393443...
0.8	0.013333333333333...	0.01333...	11.563524...
0.9	0.003333333333333...	0.00333...	46.254098...

Remark 8.7. At the tabulated values $\xi \geq \frac{1}{2}$ the single image $h_{1-\xi}$ dominates and $c(\xi) \approx (1 - \xi)^2/3$ (the mode of the Lévy density of the first passage over the distance $1 - \xi$) to within 1.5×10^{-4} relative error; for small ξ the reflected image $h_{1+\xi}$ matters and $c(\xi)/((1 - \xi)^2/3)$ is not monotone (it is 0.99985, 1.196, 1.337, 1.193, 1.014 at $\xi = 0, 0.1, \dots, 0.4$). For $N = 400$, $q = \frac{1}{2}$, the exact discrete modes satisfy $|qt^*/L^2 - c(\xi_N)| \leq 2.1 \times 10^{-6} \approx 0.33/L^2$ for all twelve tested starts (numerical observation, script 19); this is explained and made rigorous by Theorem 8.12 below. In discrete time with $q > q_N$ and $x_0 \neq 1$, unimodality fails in general (Proposition 8.8); the exact threshold is not known.

Proposition 8.8 (Failure of discrete-time unimodality for starts away from the wall). *Let $N \geq 3$, $2 \leq x_0 \leq N - 1$, $d = N - x_0$ and $q \in (0, 1]$. Then*

$$f(d) = \left(\frac{q}{2}\right)^d, \quad f(d+1) = \left(\frac{q}{2}\right)^d d(1-q), \quad f(d+2) = \left(\frac{q}{2}\right)^d \left[\frac{d(d+1)}{2}(1-q)^2 + \frac{d}{4}q^2\right].$$

Consequently $f(d) > f(d+1) < f(d+2)$, so that f_{x_0} is not unimodal, whenever $q > q_{\text{fail}}(d)$, where

$$q_{\text{fail}}(1) = \frac{4}{5}, \quad q_{\text{fail}}(2) = \frac{4+\sqrt{2}}{7} = 0.77346\dots, \quad q_{\text{fail}}(d) = 1 - \frac{1}{d} \quad (d \geq 3).$$

In the notation of Lemma 3.8, $q_N \leq q^(N, x_0) \leq q_{\text{fail}}(N - x_0)$ for $2 \leq x_0 \leq N - 1$ (the lower bound is Theorem 3.5). For example, $N = 3$, $x_0 = 2$, $q = \frac{9}{10}$ gives $f(1), f(2), f(3) = 0.45, 0.045, 0.095625$, and $N = 5$, $x_0 = 2$, $q = \frac{7}{10}$ gives $f(3), f(4), f(5) = 0.042875, 0.0385875, 0.0389090625$.*

Proof. A path of length t from x_0 to the first visit of N consists of r moves to the right, l moves to the left and s rests, with $r - l = d$ and $r + l + s = t$. For $t = d$ there is one path, of probability

$(\frac{q}{2})^d$. For $t = d + 1$: $r = d$, $s = 1$; the rest is made at one of the d sites $x_0, \dots, N - 1$, all ≥ 2 , so it has probability $1 - q$. For $t = d + 2$: either $s = 2$, $l = 0$, and the two rests are a multiset of size two from the same d sites ($\frac{d(d+1)}{2}$ choices, probability $(1 - q)^2 (\frac{q}{2})^d$ each); or $s = 0$, $l = 1$, $r = d + 1$, and the left move is made from one of the d sites $x_0, \dots, N - 1$ (all ≥ 2 , so it is a genuine move of probability $\frac{q}{2}$), each such path having probability $(\frac{q}{2})^{d+2}$. No other decomposition is possible, which gives the three values. Hence $f(d + 1) < f(d)$ iff $d(1 - q) < 1$, i.e. $q > 1 - \frac{1}{d}$, and with $r := 1 - q$,

$$(\frac{q}{2})^{-d} (f(d + 2) - f(d + 1)) = \frac{d}{4} [(2d + 3)r^2 - 6r + 1].$$

For $d = 1$ the bracket is $(5r - 1)(r - 1)$, positive iff $r < \frac{1}{5}$, and $f(2) < f(1)$ for every $q > 0$. For $d = 2$ the bracket $7r^2 - 6r + 1$ is positive iff $r < \frac{3 - \sqrt{2}}{7}$ or $r > \frac{3 + \sqrt{2}}{7}$; together with $r < \frac{1}{2}$ (from $q > \frac{1}{2}$) this means $q > \frac{4 + \sqrt{2}}{7}$. For $d = 3$ the bracket is $(3r - 1)^2$, positive for $q \neq \frac{2}{3}$, in particular for $q > \frac{2}{3}$. For $d \geq 4$ its discriminant $36 - 4(2d + 3) = 8(3 - d)$ is negative, so $f(d + 2) > f(d + 1)$ for every q . A unimodal sequence cannot have $f(d) > f(d + 1) < f(d + 2)$. Finally $q_N \leq q^*$ by Theorem 3.5, and $q^* \leq q_{\text{fail}}$ because the set of q with unimodal f is the interval $(0, q^*]$ (Lemma 3.8). The three formulas were checked symbolically for $N \leq 9$ and all $x_0 \geq 2$, and the two examples in exact rational arithmetic (script 36). $\square$

Numerical observation 8.9. An exact scan ($3 \leq N \leq 13$, $2 \leq x_0 \leq N - 1$, $q \in \{\frac{3}{5}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{9}{10}, 1\}$, times $t \leq 6N^2$; script 36) finds a certified failure of unimodality (a triple $j < k < l$ with $f(k) < \min\{f(j), f(l)\}$, in exact rational arithmetic) in 171 of the 396 cases. All 165 cases with $q > q_{\text{fail}}(d)$ are among them; the remaining 6 are $d = 5$, $q = \frac{4}{5} = q_{\text{fail}}(5)$, $8 \leq N \leq 13$, with the dip at $t = 10$ instead of $t = d + 1$. So q_{fail} is not the exact threshold. In the remaining 225 cases no failure was found up to $t = 6N^2$ (which, being a finite search, proves nothing).

8.2 Explicit finite- N law for starts in the half next to the wall

We now make Theorem 8.5 quantitative, by the method of Sections 5–6. The key point is to expand around the limit density G_{ξ_N} with the *exact* relative start position $\xi_N = (x_0 - \frac{1}{2})/L$ of the given pair (N, x_0) ; then no drift term appears. Put

$$A_j^{\text{st}}(\zeta) := S(\zeta)^{2j+1} \cos \sqrt{\zeta}, \quad F_j^{\text{st}}(\zeta; y, v, q) := A_j^{\text{st}}(\zeta) e^{-2vy^2 B(\zeta; q)} \quad (j \in \{0, 1\}),$$

with B as in (12) ($B = \beta$ in continuous time), and for real $t \geq 1$ (for integer $t \geq 1$ if $q > \frac{1}{2}$)

$$\mathfrak{S}_j^{\text{st}}(t) := \sum_{m=1}^{N-1} \sigma_k \cos(2y_k \xi_N) 2y_k (-2y_k^2)^j A_j^{\text{st}}(\zeta_k) \lambda_m^{t-1}, \quad k = 2m - 1. \quad (37)$$

Exactly as in (13) (now with $\sin \theta_m \sin(d\theta_m) = \sigma_k 2 \sin x_k \cos x_k \cos(2y_k \xi_N)$),

$$\frac{L^2}{q} f_{x_0}(t) = \mathfrak{S}_0^{\text{st}}(t), \quad \frac{L^4}{q^2} (f_{x_0}(t + 1) - f_{x_0}(t)) = \mathfrak{S}_1^{\text{st}}(t),$$

and in continuous time $G_{N, x_0}^{(j)}(v)$ is the same sum with λ_m^{t-1} replaced by $e^{-2vy_k^2 \beta(\zeta_k)}$; we then write $\mathfrak{S}_j^{\text{st}} := G_{N, x_0}^{(j)}(v)$.

Lemma 8.10. For $\zeta \in [0, \pi^2/4]$ and $j \in \{0, 1\}$: $0 \leq A_j^{\text{st}} \leq 1$, $-\alpha_{1,j}^{\text{st}} \leq (A_j^{\text{st}})' \leq 0$ and $0 \leq (A_j^{\text{st}})'' \leq \alpha_{2,j}^{\text{st}}$, where $\alpha_{1,j}^{\text{st}} = \frac{j+2}{3}$, $\alpha_{2,0}^{\text{st}} = \frac{4}{15}$, $\alpha_{2,1}^{\text{st}} = \frac{4}{5}$; moreover $A_j^{\text{st}}(0) = 1$ and $(A_j^{\text{st}})'(0) = -\frac{j+2}{3}$.

Proof. Let $C_1(\zeta) = \cos \sqrt{\zeta} \in [0, 1]$. As in the proof of Lemma 5.1, $C_1' = -\frac{1}{2}S \in [-\frac{1}{2}, 0]$ and $C_1'' = -\frac{1}{2}S' \in [0, \frac{1}{12}]$. Hence $(A_j^{\text{st}})' = (2j + 1)S^{2j}S'C_1 + S^{2j+1}C_1' \in [-\frac{2j+1}{6} - \frac{1}{2}, 0]$ and $(A_j^{\text{st}})'' = [(2j + 1)2jS^{2j-1}S'^2 + (2j + 1)S^{2j}S'']C_1 + 2(2j + 1)S^{2j}S'C_1' + S^{2j+1}C_1''$ is a sum of nonnegative terms bounded by $\frac{(2j+1)2j}{36} + \frac{2j+1}{60} + \frac{2j+1}{6} + \frac{1}{12}$, which is $\frac{4}{15}$ for $j = 0$ and $\frac{4}{5}$ for $j = 1$. $\square$

Proposition 8.11 (Lattice expansion, general start). *Let $j, \bar{q} \leq \frac{4}{5}$, $v_{\pm}$, L_0 , k_0 , N be as in Proposition 5.4, let $x_0 \in \{1, \dots, N-1\}$, $\xi := \xi_N$, and let either (continuous time, $\bar{q} = 0$) $v \in [v_-, v_+]$, or (discrete time) $q \in (0, \bar{q}]$ and $t \geq 1$ with $v := q(t-1)/L^2 \in [v_-, v_+]$, where t is an integer or, if $\bar{q} \leq \frac{1}{2}$, any real number. Then*

$$\mathfrak{S}_j^{\text{st}} = G_{\xi}^{(j)}(v) + \frac{H_{\xi,j}^{(q)}(v)}{L^2} + \frac{R}{L^4}, \quad H_{\xi,j}^{(q)} := \frac{j+2}{6} G_{\xi}^{(j+1)} + \frac{v}{6} (1-3q) G_{\xi}^{(j+2)} \quad (38)$$

(with $q = 0$ in continuous time), where

$$R = \sum_{k \leq k_0, k \text{ odd}} \sigma_k \cos(2y_k \xi) (-2)^j y_k^{2j+5} (F_j^{\text{st}})''(\tilde{\zeta}_k; y_k, v, q) + R', \quad |R'| \leq B_{\text{low}} + B_{\text{high}} + B_{\text{tail}},$$

with $\tilde{\zeta}_k \in (0, y_k^2/L^2)$ and $B_{\text{low}}, B_{\text{high}}, B_{\text{tail}}$ as in Proposition 5.4, but with $\alpha_{1,j}^{\text{st}}, \alpha_{2,j}^{\text{st}}$ in place of $\alpha_{1,j}, \alpha_{2,j}$ (and $\frac{j+2}{3}$ in place of $\frac{2j+7}{6}$ in B_{tail}).

Proof. The proof of Proposition 5.4 applies word for word: the k -th term of $\mathfrak{S}_j^{\text{st}}$ is $\sigma_k \cos(2y_k \xi) 2(-2)^j y_k^{2j+1} F_j^{\text{st}}(\zeta_k)$ on a low mode, the factor $\cos(2y_k \xi)$ does not depend on ζ and has modulus at most 1, $F_j^{\text{st}}(0) = e^{-a}$ and $(F_j^{\text{st}})'(0) = (-\frac{j+2}{3} + a(\frac{1}{3} - q))e^{-a}$ with $a = 2vy_k^2$, and the Taylor polynomials sum to $G_{\xi}^{(j)}(v)$ and to $\frac{j+2}{6} G_{\xi}^{(j+1)} + \frac{v}{6} (1-3q) G_{\xi}^{(j+2)}$ because $\sum_k \sigma_k \cos(2y_k \xi) 2(-2)^j y_k^{2j+3} e^{-2vy_k^2} = -\frac{1}{2} G_{\xi}^{(j+1)}$ and $\sum_k \sigma_k \cos(2y_k \xi) 2(-2)^j y_k^{2j+5} e^{-2vy_k^2} = \frac{1}{4} G_{\xi}^{(j+2)}$. The bound for $(F_j^{\text{st}})''$ is that of Lemma 5.3 with the constants of Lemma 8.10. For real $t \geq 1$ and $q \leq \frac{1}{2}$ all λ_m are positive, the identity $\lambda_m^{t-1} = \exp(-2vy_k^2 B(\zeta_k; q))$ on low modes and the bound $\lambda_m^{t-1} \leq e^{-vL^2/2}$ on high modes hold as before. $\square$

For $0 \leq \xi < 1$ define, with $c(\xi)$ from Theorem 8.3,

$$\rho(\xi) := -\frac{c(\xi)}{2} \frac{G_{\xi}'''(c(\xi))}{G_{\xi}''(c(\xi))}, \quad \kappa(\xi) := \frac{1}{2} - \frac{\rho(\xi)}{3} = \frac{1}{2} + \frac{c(\xi)}{6} \frac{G_{\xi}'''(c(\xi))}{G_{\xi}''(c(\xi))} \quad (39)$$

(well defined by Theorem 8.3(3); $\rho(0) = \rho$ and $\kappa(0) = \kappa - \frac{1}{4}$ in the notation of Theorem 4.6).

Theorem 8.12 (Explicit finite- N law, general start; analytic for $N \geq 101$, CA for $3 \leq N \leq 100$). *Let $N \geq 3$ and let the start x_0 satisfy $\xi_N = (x_0 - \frac{1}{2})/(N - \frac{1}{2}) \leq \frac{1}{2}$, i.e. $x_0 \leq \frac{2N+1}{4}$. Put $c_N := c(\xi_N)$, $\rho_N := \rho(\xi_N)$, $\kappa_N := \kappa(\xi_N)$.*

1. (Continuous time, every $q \in (0, 1]$.) *The unique mode t_c of g_{x_0} satisfies*

$$\left| \frac{q t_c}{L^2} - c_N + \frac{\kappa_N}{L^2} \right| < \frac{3.5}{L^4}.$$

2. (Discrete time, $q \leq \frac{1}{2}$.) *Every mode t^* of f_{x_0} satisfies*

$$\tau - E < t^* < \tau + 1 + E, \quad \tau := \frac{c_N L^2 - \kappa_N}{q} + 1 - \rho_N, \quad E := \frac{3.5}{q L^2}.$$

Proof. For $0 \leq q \leq \frac{1}{2}$ and real $v > 0$ let

$$\mathfrak{D}(v; q) := \sum_{m=1}^{N-1} \sigma_k \cos(2y_k \xi_N) 2y_k (-2y_k^2) A_1^{\text{st}}(\zeta_k) \exp(-2vy_k^2 B(\zeta_k; q)),$$

where $B(\zeta_k; q) = \beta(\zeta_k) \Lambda(2q \sin^2 x_k)$ (all arguments of Λ are < 1) and $B(\cdot; 0) = \beta$. For $q = 0$ this is $G'_{N, x_0}(v)$, which has exactly one zero $u_N = q t_c / L^2$ in $(0, \infty)$, positive before and negative after

(Theorem 3.3; $d = N - x_0 \geq 2$ because $x_0 \leq \frac{2N+1}{4}$ and $N \geq 3$). For $0 < q \leq \frac{1}{2}$, $\mathfrak{D}(v; q) = \mathfrak{S}_1^{\text{st}}(t)$ at the real time $t = 1 + vL^2/q$ is a positive multiple of the function $\Psi(t)$ in the proof of Theorem 3.5, which has exactly one zero t_0 in $(d-1, \infty)$, is positive on $[d-1, t_0)$ and negative on (t_0, ∞) ; every mode lies in $[t_0, t_0 + 1]$. Put

$$v_{\mp}(q) := c_N - \frac{\kappa_N + q\rho_N}{L^2} \mp \frac{K}{L^4}, \quad K = 3.5,$$

so that $1 + v_{\mp}L^2/q = \tau \mp E$. We claim that for all $q \in [0, \frac{1}{2}]$

$$\mathfrak{D}(v_{-}(q); q) > 0 > \mathfrak{D}(v_{+}(q); q) \quad \text{and, if } q > 0, \quad v_{+}(q)L^2 > q(d-2). \quad (40)$$

Given (40): for $q = 0$, $v_{-} < u_N < v_{+}$, which is (1) (the density g_{x_0} for a given q is a rescaling of the one for $q = 1$). For $q > 0$: $\tau + E > d - 1$ and $\Psi(\tau + E) < 0$ give $\tau + E > t_0$; and $\tau - E < t_0$, either because $\tau - E \leq d - 1 < t_0$ or because $\tau - E > d - 1$ and $\Psi(\tau - E) > 0$. Hence $t^* \in [t_0, t_0 + 1] \subset (\tau - E, \tau + 1 + E)$.

Proof of (40) for $N \geq 101$. Let $L \geq L_0 = 100.5$, $\varepsilon = L^{-2} \leq \varepsilon_0 = L_0^{-2}$, $\xi = \xi_N$, $G = G_{\xi}$, $c = c(\xi)$, $v = c - \varepsilon d$ with $d = \kappa(\xi) + q\rho(\xi) - s\varepsilon$, $s \in [-K, K]$. By Proposition 8.11 ($j = 1$), $\mathfrak{D}(v; q) = G'(v) + \varepsilon H(v) + \varepsilon^2 R$ with $H = \frac{1}{2}G'' + \frac{v}{6}(1 - 3q)G'''$ and $H(c) = G''(c)(\frac{1}{2} + \frac{c}{6}(1 - 3q)\frac{G'''(c)}{G''(c)}) = (\kappa(\xi) + q\rho(\xi))G''(c)$. Taylor's formula at c (where $G'(c) = 0$), exactly as in (19) and (21), gives

$$L^4 \mathfrak{D}(v; q) = sG''(c) + \Gamma, \quad \Gamma = \frac{1}{2}G'''(c)d^2 - H'(c)d + \varepsilon[-\frac{1}{6}G''''(\eta)d^3 + \frac{1}{2}H''(\eta')d^2] + R,$$

with η, η' between v and c , $H' = \frac{1}{2}G''' + \frac{1-3q}{6}G''' + \frac{v}{6}(1 - 3q)G''''$ and $H'' = \frac{1}{2}G'''' + \frac{1-3q}{3}G'''' + \frac{v}{6}(1 - 3q)G^{(5)}$. Script `25_start_theorem.py` covers the rectangle $(\xi, q) \in [0, \frac{1}{2}] \times [0, \frac{1}{2}]$ by 49 238 boxes $\Xi \times Q$ (adaptive bisection, starting from 100 strips in ξ) and certifies for each box, in 128-bit ball arithmetic and simultaneously for all $\xi \in \Xi$, $q \in Q$, $\varepsilon \in [0, \varepsilon_0]$, $s \in [-K, K]$: (i) an enclosure $C \ni c(\xi)$ for all $\xi \in \Xi$, obtained from $G'_{\xi}(C_{\min}) > 0 > G'_{\xi}(C_{\max})$ on Ξ and Theorem 8.3(2); (ii) $G''_{\xi} < 0$ on $C \times \Xi$; (iii) $\sup |\Gamma| < K \inf |G''_{\xi}(c)|$, where $c, G^{(i)}(c), \kappa, \rho, d, v, \eta, \eta'$ are replaced by their balls and R is enclosed as in Remark 5.5 (with $k_0 = 31$ and the weights $\cos(2y_k \Xi)$; module `start.py`); (iv) $v_{\min}L_0 > \frac{1}{2}(1 - \xi_{\min})$, which gives $vL^2 > \frac{1}{2}L(1 - \xi) \geq q(d - 2)$ for all $L \geq L_0$ since $d = L(1 - \xi)$. From (iii), $L^4 \mathfrak{D} = sG''(c) + \Gamma$ is positive at $s = -K$ and negative at $s = K$, which is (40) (file `data/25_start_N0_101_K3p5.json1`; the largest value of $\sup |\Gamma| / \inf |G''(c)|$ over the boxes is $3.4999930 \dots$ (upper end of its ball), the smallest box has ξ -width 3.9×10^{-5} and q -width 9.8×10^{-4}). The closeness of 3.49999 to K is a feature of the adaptive bisection, which subdivides a box until the certified bound drops below K ; it does not indicate that the true quantity comes close to 3.5 for $N \geq 101$ (the largest value found in the numerical tests of script 29, which include small N , is $3.419 \dots$, at $N = 6$, $x_0 = 3$; Remark 8.13).

Proof of (40) for $3 \leq N \leq 100$. Script `26_start_small_N.py` treats the 2499 pairs (N, x_0) one by one in 160-bit ball arithmetic: ξ_N is rational, c_N is enclosed as in (i) (radius $< 10^{-20}$), and the two functions $F_{\mp}(q) = \mathfrak{D}(v_{\mp}(q); q)$ are enclosed on subintervals $Q \ni q_{\text{mid}}$ of $[0, \frac{1}{2}]$ by the mean value form $F(Q) \subset F(q_{\text{mid}}) + F'(Q)(Q - q_{\text{mid}})$, with $F'(q) = \sum_k (\dots) \exp(-ve_k \Lambda(w))(-e_k)[v'(q)\Lambda(w) + vs_k \Lambda'(w)]$, $e_k = 2L^2 \sin^2 x_k$, $s_k = 2 \sin^2 x_k$, $w = qs_k$, $v'(q) = -\rho_N/L^2$, and Λ, Λ' enclosed on $[w_{\min}, w_{\max}]$ by monotonicity (both have nonnegative power series coefficients); adaptive bisection in q (13 115 evaluations in all). All pairs pass (`data/26_start_small_N_K3p5.json1`), including the condition $v_{+}L^2 > q(d - 2)$. $\square$

ξ	$c(\xi)$	$\rho(\xi)$	$\kappa(\xi)$
0	0.333284265494...	1.991178661865...	-0.163726220621...
0.1	0.322863097331...	1.9305185174...	-0.1435061724...
0.2	0.285180145816...	1.7587199904...	-0.0862399968...
0.3	0.194776045823...	2.251816232...	-0.250605410...
0.4	0.121736005686...	2.233308210...	-0.244436070...
0.5	0.083345643329...	2.008874178629...	-0.169624726209...

(Certified values, script 28.)

Remark 8.13. (a) The constant 3.5 is nearly optimal on the range $\xi_N \leq \frac{1}{2}$: for $N = 6$, $x_0 = 3$ one has $L^4(u_N - c_N + \kappa_N/L^2) = -3.419\dots$, and for $\xi_N \rightarrow \frac{1}{2}$, $N \rightarrow \infty$ this quantity tends to ≈ -2.86 (numerical observation, script 29). Numerically it behaves like $-0.75/(1-\xi)^2$, in accordance with the fact that the natural length scale for a start close to the target is the distance $d = (1-\xi)L$; a version of Theorem 8.12 uniform in $\xi < 1$ would have to be formulated in terms of d and is not attempted here. (b) For $x_0 = 1$, $\xi_N = \frac{1}{2L}$ and $c(\xi) = c - \xi^2 + O(\xi^4)$ (indeed $\partial_\xi^2 G_\xi = 2\partial_u G_\xi$, so $c''(0) = -2$), $\kappa(0) = \kappa - \frac{1}{4}$; thus $c_N L^2 - \kappa_N = cL^2 - \kappa + O(L^{-2})$ and Theorem 8.12 is consistent with Theorems 5.7 and 6.3.

Corollary 8.14 (Uniform constants). *In the situation of Theorem 8.12, $1.7206 < \rho_N < 2.5171$ and $-0.3391 < \kappa_N < -0.0735$, and*

$$\begin{aligned} \left| \frac{qt_c}{L^2} - c(\xi_N) \right| &< \frac{0.34}{L^2} + \frac{3.5}{L^4}, & \left| \frac{qt^*}{L^2} - c(\xi_N) \right| &< \frac{0.44}{L^2} + \frac{3.5}{L^4} \quad (q \leq \tfrac{1}{2}), \\ \left| \frac{t_c}{\mathbb{E}_{x_0} T} - \frac{c(\xi_N)}{1 - \xi_N^2} \right| &< \frac{0.46}{L^2} + \frac{4.7}{L^4}, & \left| \frac{t^*}{\mathbb{E}_{x_0} T} - \frac{c(\xi_N)}{1 - \xi_N^2} \right| &< \frac{0.59}{L^2} + \frac{4.7}{L^4} \quad (q \leq \tfrac{1}{2}), \end{aligned}$$

where $\mathbb{E}_{x_0} T = L^2(1 - \xi_N^2)/q$ and $0.1110 < c(\xi)/(1 - \xi^2) < 0.3333$ for $\xi \leq \frac{1}{2}$.

Proof. Script `28_start_constants.py` encloses $c(\xi)$, $\rho(\xi)$, $\kappa(\xi)$ on 4000 balls covering $[0, \frac{1}{2}]$ (ball arithmetic, $c(\xi)$ enclosed as above): $\rho \in [1.72068, 2.51708]$, $\kappa \in [-0.33903, -0.07356]$, $\kappa + \frac{1}{2}(\rho - 1) \in [0.2829, 0.4370]$, $\kappa + \frac{1}{2}(\rho - 2) \in [-0.2171, -0.0630]$, $c(\xi)/(1 - \xi^2) \in [0.11107, 0.33329]$. By Theorem 8.12(2), $-(\kappa_N + q(\rho_N - 1))L^{-2} - 3.5L^{-4} < qt^*/L^2 - c_N < -(\kappa_N + q(\rho_N - 2))L^{-2} + 3.5L^{-4}$; the two coefficients are affine in $q \in [0, \frac{1}{2}]$, so their absolute values are at most $\max\{0.3391, 0.4370, 0.2171\} < 0.44$. The continuous-time bound is (1) with $|\kappa_N| < 0.34$. Dividing by $1 - \xi_N^2 \geq \frac{3}{4}$ gives the last line. $\square$

9 Summary: what is proved, what is computer-assisted, what is open

Proved by hand (complete proofs above). Exact spectral law for every start (Theorem 2.2), product formula and sum-of-geometrics/exponentials representation (Theorem 2.3), moments (Corollary 2.5); strict unimodality in continuous time for every start (Theorem 3.3); strict unimodality in discrete time for every start when $q \leq q_N$ (Theorem 3.5), and its failure for every start $x_0 \geq 2$ when $q > q_{\text{fail}}(N - x_0)$ (Proposition 8.8); log-concavity from the reflecting end for $q \leq \frac{2}{3}$ (Theorem 3.6); monotonicity of unimodality in q (Lemma 3.8); the reduction of unimodality for $q > \frac{2}{3}$ to the total positivity of finitely many powers Q^k on one finite chain (Lemmas 3.12, 3.13, 3.14); failure of unimodality for $q > 1 - \frac{1}{2N-3}$, $N \geq 4$ (Proposition 3.9); parity-class unimodality at $q = 1$ (Theorem 3.10); convergence to the limit density, its strict unimodality and the uniqueness of its maximiser up to one sign evaluation (Theorems 4.2, 4.3); the lattice expansion with explicit remainder (Proposition 5.4, Lemmas 5.1–5.3, 7.1–7.3); the comparison Lemma 7.6; the limit density for a general start, its strict unimodality with a unique non-degenerate critical point, the analyticity of $\xi \mapsto c(\xi)$, and the limit theorem (Lemmas 8.1, 8.2, Theorems 8.3, 8.5).

Computer-assisted (rigorous ball arithmetic or exact arithmetic; scripts and outputs are part of the proof).

statement	script	what is computed
$G'(\frac{3}{10}) > 0 > G'(\frac{2}{5})$, $G''(c) < 0$, 70-digit enclosure of c, κ, ρ (Thms 4.3, 4.6)	03	4 signs, 6 balls
continuous-time mode, $N \geq 301$ (Thm 5.7)	12 (<code>core.py</code>)	enclosure of Γ
value of κ_2 from its closed form (18) (Thm 5.7)	37	one ball
continuous-time mode, $3 \leq N \leq 300$	13	4 signs per N
discrete increments near the mode, $N \geq 41$, $q \leq \frac{4}{5}$; $N \geq 61$, $q \leq \frac{9}{10}$ (Thm 6.1)	14, 35	2400, resp. 2700 q -balls; uniform constants
discrete increments, $3 \leq N \leq 40$, $q \leq \frac{4}{5}$ (Prop. 6.2)	15	bisection in q ; ≤ 782 windows per N
discrete increments, $\frac{4}{5} \leq q \leq \frac{9}{10}$, $N \leq 60$ (Prop. 6.2(b),(c))	32	6752 windows
unimodality: $q \leq \frac{4}{5}$, all N ; $q \leq \frac{17}{20}$, $N \geq 6$; $q \leq \frac{9}{10}$, $N \geq 9$ (Thms 3.15, 3.16)	31	exact integers: contiguous minors of $(2q_{\text{den}}Q)^k$, $k_* \leq k < 2k_*$
table of modes, $q = \frac{4}{5}$, $N \leq 600$ (Prop. 3.18)	08, 24	certified signs of all increments; no-underflow bound
$q = 1$ mode, $N \geq 601$ (Thm 7.4)	16 (<code>q1.py</code>)	enclosure of Γ_1 , class comparison
$q = 1$ mode, $3 \leq N \leq 600$	17	exact / ball
global maximisers $\frac{2}{3} \leq q < 1$, $N \geq 601$, without unimodality (Thm 7.7)	14, 35, 18	3000 q -balls, 800 v -pieces, 4 inequalities
$c(\xi)$ table (Prop. 8.6)	19	sign changes
general start, $\xi_N \leq \frac{1}{2}$, $q \leq \frac{1}{2}$ or continuous time, $N \geq 101$ (Thm 8.12)	25 (<code>start.py</code>)	49 238 boxes in (ξ, q)
general start, $3 \leq N \leq 100$	26	2499 pairs (N, x_0) , bisection in q
ranges of $\rho(\xi), \kappa(\xi)$ (Cor. 8.14)	28	4000 ξ -balls
failure examples for starts $x_0 \geq 2$ (Prop. 8.8; checks only, the proof is by hand)	36	exact rationals, sympy

Sanity checks of every lemma against separately written high-precision or exact computations (same project, same machine): scripts 01 (Section 2), 09 (Sections 3–5), 11 (Proposition 5.4), 21 (Lemmas 4.1, 7.1–7.3); validation of the final theorems against other data: 20 (exact modes of the computations of Sections 5–6 of the article and a fresh grid), 33 (Section 3.7: Cauchy–Binet identity, locality, brute-force confirmation of Theorems 3.15, 3.16 for $N \leq 40$), 34 (a second, separately written implementation of the TP_2 test of Theorem 3.15(1): all 2×2 minors, no band shortcut), 22 (the exact-arithmetic certificates of R5: 600 modes at $q = \frac{4}{5}$, 298 at $q = \frac{1}{2}$, 198 at $q = 1$), 23 (randomised stress test: 450 + 60 pairs (N, q) with $q \leq \frac{4}{5}$, $N \leq 4000$; a fine scan of 400 values of q for each $N \leq 12$; continuous time up to $N = 6000$; $q = 1$ up to $N = 2200$; $\frac{9}{10} < q < 1$; 285 pairs with $\frac{4}{5} < q \leq \frac{9}{10}$), 27 (Section 8: lattice identities, Lemma 8.10, Proposition 8.11, sign patterns of Lemma 8.2 and Theorem 8.3), 29 (Theorem 8.12: 610 discrete and 22 continuous-time cases). All pass, with no violation; outputs in `data/`, logs in `logs/`. The observed extreme values of $t^* - \tau$ in these tests are $-E + 0.00197$ and $1 + E - 0.00305$, so the window (22) is essentially sharp.

Not proved.

1. *Unimodality for $\frac{9}{10} < q < 1$.* The exact threshold $q^*(N, 1)$ is known only through the bounds $\max\{\frac{4}{5}, \frac{17}{20}[N \geq 6], \frac{9}{10}[N \geq 9]\} \leq q^*(N, 1) \leq 1 - \frac{1}{2N-3}$ ($N \geq 4$) and numerically (Observation 3.20). The method of Theorem 3.16 applies to any given rational $q < 1$ (with

$k_* \approx q^2/(2(1-q)^2)$), but we have no proof that Q^k is eventually TP_2 for every $q < 1$ (Remark 3.17).

2. *The crossover:* $\frac{9}{10} < q < 1$ with $N \leq 600$ or $\mathfrak{h}(q, L) > 10^{-12}$, i.e. roughly $(1-q)L^2 \lesssim 27 \ln L + 100$, and the few small N excluded in Theorem 6.3(b),(c): here the mode is in general *not* in the window (22); it interpolates between the laws of Theorems 7.7 and 7.4. No theorem.
3. *Sharp constant for $q = 1$:* $K = 10.3$ is proved, $K \approx 0.03$ is observed for $N \leq 600$.
4. *General start:* explicit finite- N bounds for starts with $\xi_N > \frac{1}{2}$ (they should be formulated in terms of the distance $d = N - x_0$) and, in discrete time, for $q > \frac{1}{2}$; the exact unimodality threshold $q^*(N, x_0)$ for $x_0 \geq 2$, which is only known to satisfy $q_N \leq q^*(N, x_0) \leq q_{\text{fail}}(N - x_0)$ (Theorem 3.5, Proposition 8.8; unimodality does *not* hold in general for $q > q_N$, and q_{fail} is not sharp, Observation 8.9); monotonicity of $\xi \mapsto c(\xi)$ (it is real-analytic and $c(\xi) = c - \xi^2 + O(\xi^4)$, but $c(\xi)/((1-\xi)^2/3)$ is not monotone).